

RUTGERS
Office of Information Technology

COMMUNICATIONS DISTRIBUTION SYSTEM DRAWINGS

OIT Enterprise Infrastructure – Telecommunications Division
6/15/2020

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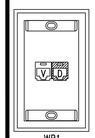
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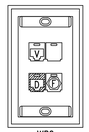
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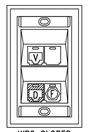
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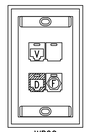

WP1

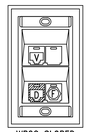

WP1A (STEELCASE)

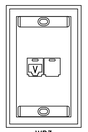

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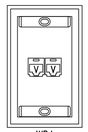

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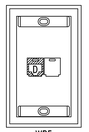

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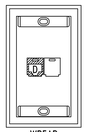

WP2C
INSTALL EXCEPT DO NOT TERMINATE VIDEO IN PODIUM – COIL UP

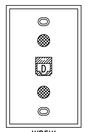

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INSTALL EXCEPT DO NOT TERMINATE VIDEO IN PODIUM – COIL UP

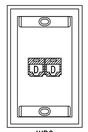

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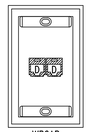

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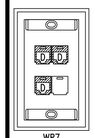

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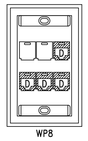

WP5AP

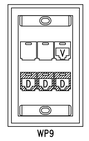

WP5W
STAINLESS STEEL FACE PLATE WITH WALL PHONE SUPPORTS


WP6

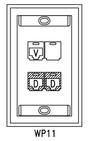

WP6AP


WP7

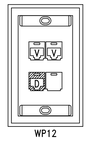

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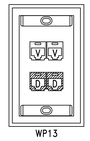

WP9


WP10
STAINLESS STEEL FACE PLATE WITH WALL PHONE SUPPORTS

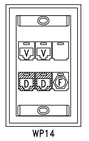

WP11

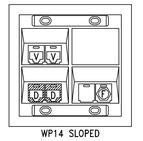

WP11A (STEELCASE)

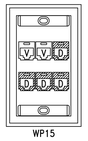

WP12

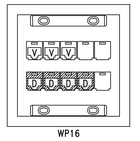

WP13

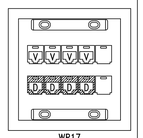

WP13A (STEELCASE)

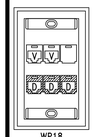

WP14

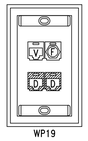

WP14 SLOPED

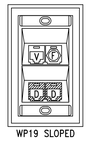

WP15


WP16

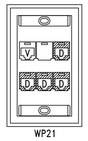

WP17


WP18

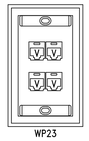

WP19

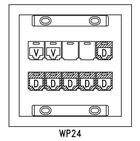

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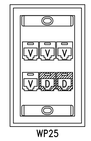

WP20
LEAVE UNTERMINATED @ FP CGP PHONE – BLANK F.P.

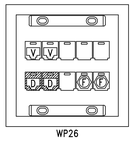

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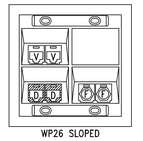

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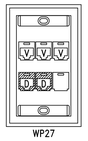

WP23


WP24


WP25


WP26


WP26 SLOPED


WP27

WP28

WP29

WP29 SLOPED

WP30
NO FACEPLATE REQUIRED COIL ABOVE CEILING AND INDICATE ON AS-BUILTS

WP31

WP32
FORWARD & RETURN VIDEO WITH DATA

WP33

WP40
SWIPE CARD BLANK F.P. LEAVE UNTERMINATED @ F.P.

WP41

WP42

WP43

WP44

WP45

WP46

WP47

WP47 SLOPED

WP48

WP48 SLOPED

WP49

WP50

WP51

WP52

WP53

WP54

WP54 SLOPED

WP55

WP57

WP57 SLOPED

WP58

WP58 SLOPED

WP59

WP59 SLOPED

WP60
FORWARD AND RETURN VIDEO (CONTRACTOR COVER PLATE)

WP60
FORWARD & RETURN VIDEO (TERMINATIONS BY RUTGERS)

WP61
RETURN VIDEO (CONTRACTOR COVER PLATE)

WP61
RETURN VIDEO (TERMINATIONS BY RUTGERS)

WP61B
BALANCED RETURN VIDEO WITH STAINLESS STEEL FACE PLATE

WP62
FORWARD VIDEO

WP62 SLOPED
FORWARD VIDEO ONLY

WP64

WP64 SLOPED

WP65

WP66
CAMPUS PHONE

WP67

WP67 SLOPED

WP68

WP68 SLOPED

WP69

WP70

WP70 SLOPED

WP71

WP71 SLOPED

WP72

WP72 SLOPED

WP73

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WP78
BALANCED RETURN VIDEO WITH STAINLESS STEEL FACE PLATE

WP79

WP80

WP80B

WP81

WPFU FUTURE

WP101

WP102

WP103

WP104

WP105

WP106

WP111

WP112

WP113

WP114

WP115

WP116

WP117

WP118

NOTES:

12	ADM		7/22/2020	UPDATE – TELEPRESENCE WALL PLATES AND 2020 SPEC UPDATES
11	ADM		7/15/2020	ADDED TELEPRESENCE WALL PLATES AND 2020 SPEC UPDATES
9	CMW		7/11/16	UPDATE WP 3, 4, 5, 6, & 67. CHANGE FROM 4 TO 2 PORTS. CHANGE WP7 FROM 6 TO 4 PORT
8	ADM		6/18/15	UPDATED
7	SAO		1/09/08	ADDED WP61B
6	SAO		6/11/07	ADDED WP1A, WP2A, WP11A & WP13A
5	SAO		3/22/07	ADDED WP77
4	SAO		3/23/06	ADDED WP76
3	SAO		6/03/03	ADDED WP33
2	SAO		5/06/03	ADDED WP32
1	SAO		3/24/03	REVISED WP60, 61 & 62
ISSUE	BY	CHK	DATE	DESCRIPTION

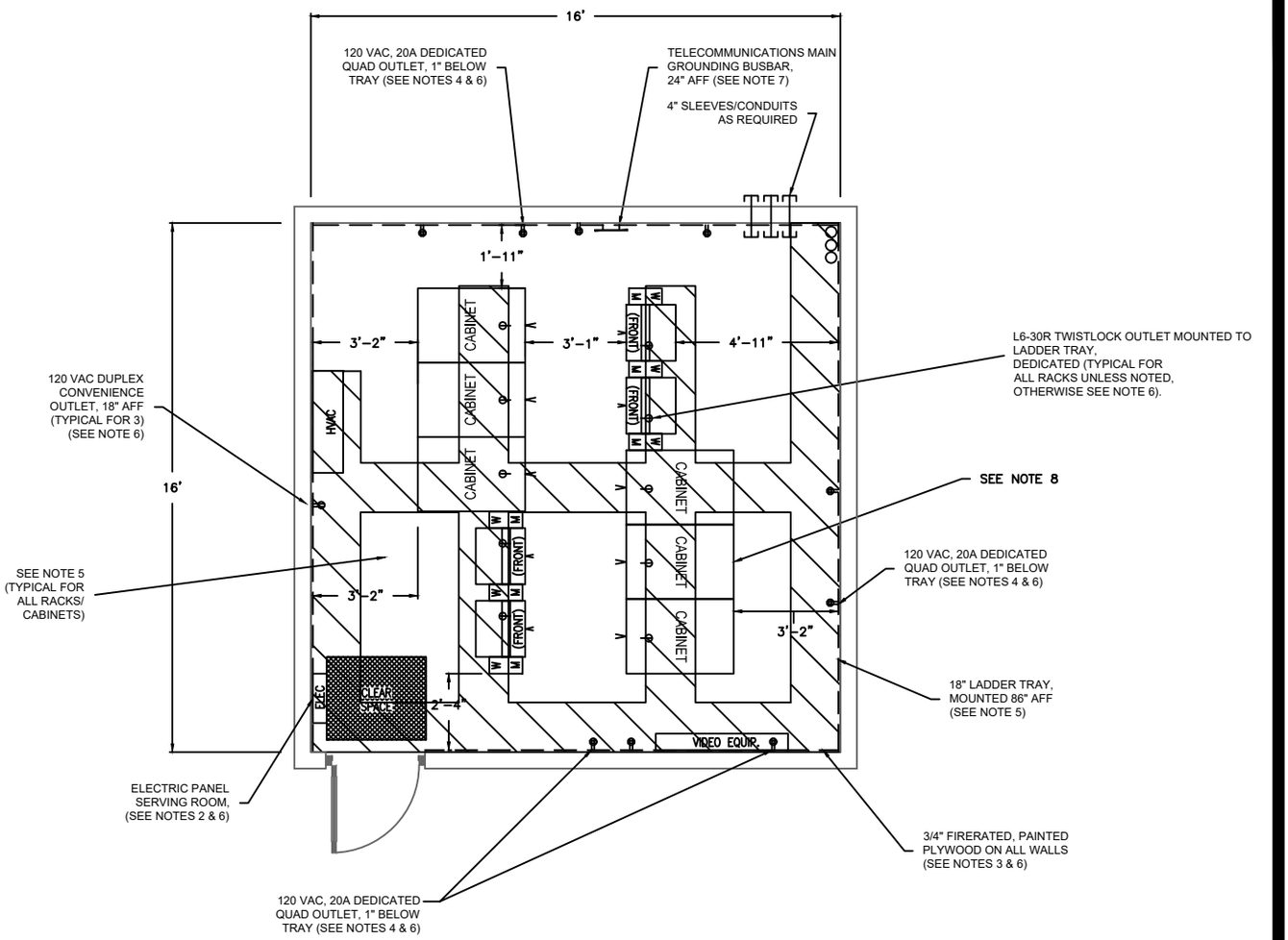
APPROVED FOR RUTGERS–THE STATE UNIVERSITY

THE STATE UNIVERSITY OF NEW JERSEY
RUTGERS
OFFICE OF INFORMATION TECHNOLOGY

PROJECT
NAME OF PROJECT
OR RUNET

DRAWING TITLE
SAMPLE ISP BUILDING
FACE PLATE SYMBOL LIST

DSR NO.	SCALE	DRAWN BY	FILE NO.	DRAWING NO.
	NTS	P.R.		
BLDG. NO. FP	DATE 02/18/02	CHECKED BY P.R.	FACE PLATES	FP-1



NOTES:

1. PROVIDE SERVICE LOOP FOR ALL HORIZONTAL VOICE, DATA, AND VIDEO CABLES NOT TO EXCEED 10 FEET. LOCATION TO BE DETERMINED BY THE RUPM.
2. PROVIDE (3) 30A SPARE CIRCUITS IN ELECTRIC PANEL.
3. 3/4" AC FIRERATED PLYWOOD ON ALL WALLS, PAINTED WITH WHITE FIRE RETARDANT PAINT (DO NOT PAINT PLYWOOD LABEL). MOUNT PLYWOOD VERTICALLY AT 22" AFF WITH STAINLESS STEEL HARDWARE.
4. MOUNT QUAD OUTLETS WITH TOP OF BACKBOX AT 1" BELOW LADDER TRAY OR AT 85" AFF, WHICHEVER IS LOWER. LOCATIONS TO BE COORDINATED WITH RUTGERS UNIVERSITY.
5. BY IT/COMMUNICATIONS CONTRACTOR.
6. BY ARCHITECT/ROOM CONSTRUCTION CONTRACTOR. COORDINATE LOCATION WITH RUTGERS UNIVERSITY.
7. BY IT CONTRACTOR FOR EXISTING ROOMS. BY BUILDING CONTRACTOR FOR NEW ROOMS.
8. CORE AND DISTRIBUTION RACKS MUST BE ADJACENT.

2	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS	4	ADM	DH	4/2/2020	UPDATE
1	ADM		2/10/14	GENERAL REVISION	3	ADM	DH	3/17/2020	UPDATE

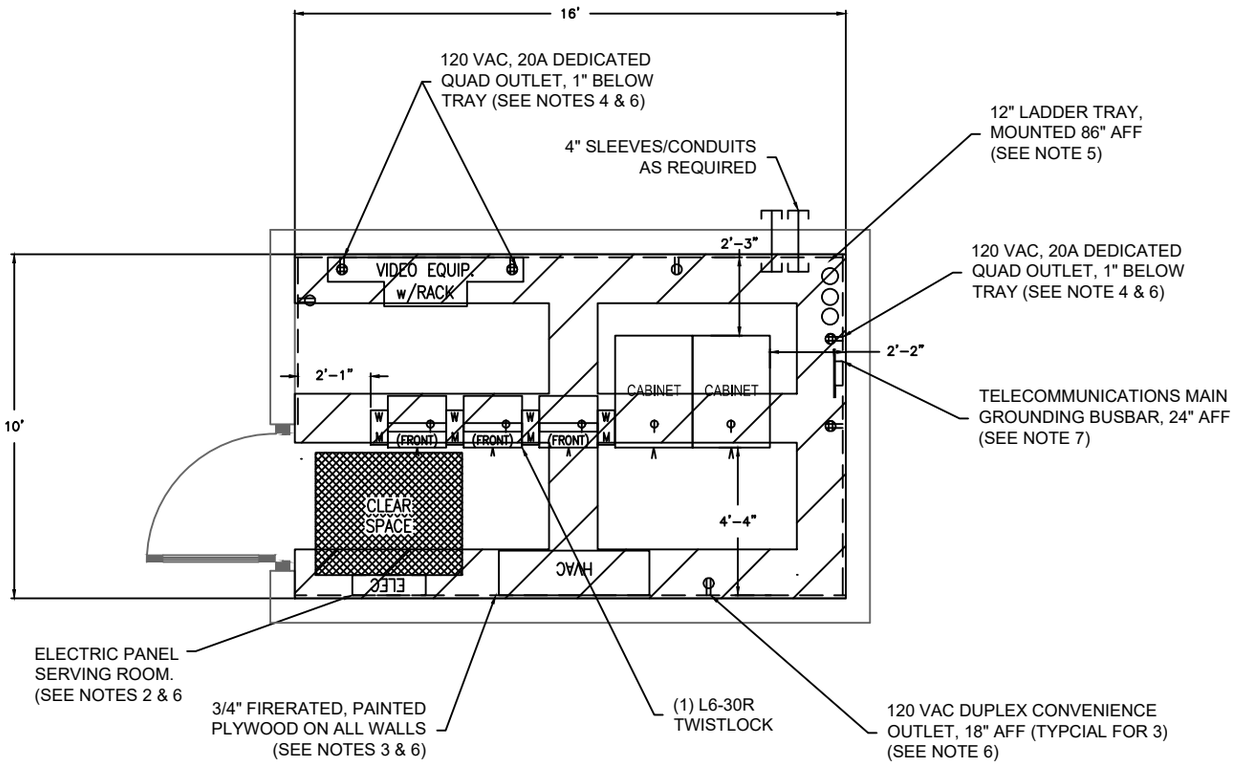
DRAWN BY: SAO
CHECKED BY: WZ
DATE: 3/26/02
DWG. # ER-01

TITLE: TYPICAL LARGE EQUIPMENT ROOM LAYOUT

THE STATE UNIVERSITY OF NEW JERSEY RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

ER-1
5 RACKS (2 ROWS)
(256 FT^2)



NOTES:

1. PROVIDE SERVICE LOOP FOR ALL HORIZONTAL VOICE, DATA, AND VIDEO CABLES NOT TO EXCEED 10 FEET. LOCATION TO BE DETERMINED BY THE RUPM.
2. PROVIDE (3) 30A SPARE CIRCUITS IN ELECTRIC PANEL.
3. 3/4" AC FIRERATED PLYWOOD ON ALL WALLS, PAINTED WITH WHITE FIRE RETARDANT PAINT (DO NOT PAINT PLYWOOD LABEL). MOUNT PLYWOOD VERTICALLY AT 22" AFF WITH STAINLESS STEEL HARDWARE.
4. MOUNT QUAD OUTLETS WITH TOP OF BACKBOX AT 1" BELOW LADDER TRAY OR AT 85" AFF, WHICHEVER IS LOWER. LOCATIONS TO BE COORDINATED WITH RUTGERS UNIVERSITY.
5. BY IT/COMMUNICATIONS CONTRACTOR.
6. BY ARCHITECT/ROOM CONSTRUCTION CONTRACTOR. COORDINATE LOCATION WITH RUTGERS UNIVERSITY.
7. BY IT CONTRACTOR FOR EXISTING ROOMS. BY BUILDING CONTRACTOR FOR NEW ROOMS.

2	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS					
1	ADM		2/10/14	GENERAL REVISION	3	ADM	DH	3/17/2020	UPDATE

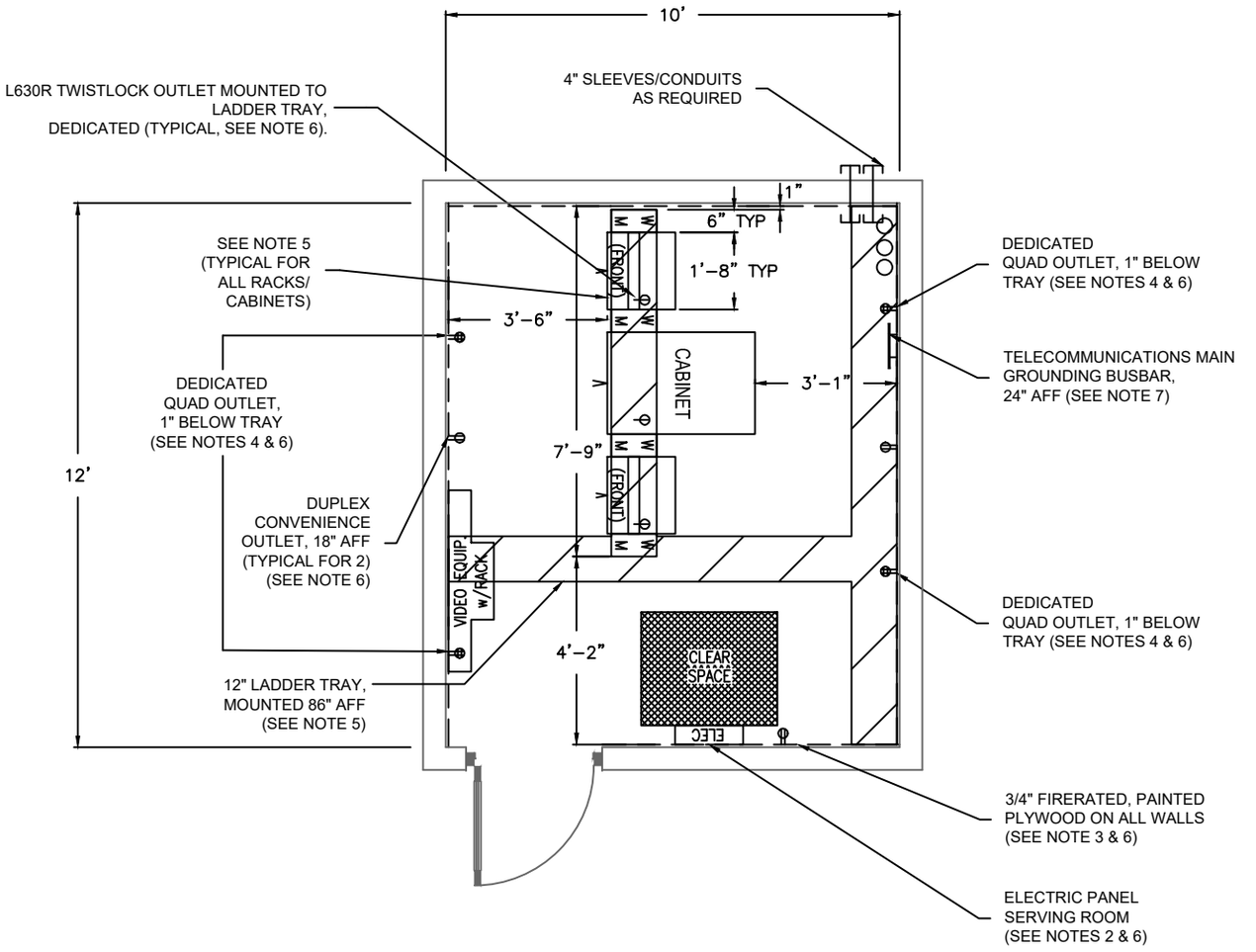
DRAWN BY: SAO
CHECKED BY: WZ
DATE: 3/26/02
DWG. # ER-02

TITLE: TYPICAL MEDIUM EQUIPMENT ROOM LAYOUT

THE STATE UNIVERSITY OF NEW JERSEY RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

ER-2
5 RACKS (160 FT^2)



NOTES:

1. PROVIDE SERVICE LOOP FOR ALL HORIZONTAL VOICE, DATA, AND VIDEO CABLES NOT TO EXCEED 10 FEET. LOCATION TO BE DETERMINED BY THE RUPM.
2. PROVIDE (3) 30A SPARE CIRCUITS IN ELECTRIC PANEL.
3. 3/4" AC FIRERATED PLYWOOD ON ALL WALLS, PAINTED WITH WHITE FIRE RETARDANT PAINT (DO NOT PAINT PLYWOOD LABEL). MOUNT PLYWOOD AT 22" AFF VERTICALLY WITH STAINLESS STEEL HARDWARE.
4. MOUNT QUAD OUTLETS WITH TOP OF BACKBOX AT 1" BELOW LADDER TRAY OR AT 85" AFF, WHICHEVER IS LOWER. LOCATIONS TO BE COORDINATED WITH RU.
5. BY IT/COMMUNICATIONS CONTRACTOR.
6. BY ARCHITECT/ROOM CONSTRUCTION CONTRACTOR. COORDINATE LOCATION WITH RUTGERS UNIVERSITY.
7. BY IT CONTRACTOR FOR EXISTING ROOMS. BY BUILDING CONTRACTOR FOR NEW ROOMS.

2	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS	4	ADM	DH	4/2/2020	UPDATE
1	CMW		12/11/17	GENERAL REVISION	3	ADM	DH	3/17/2020	UPDATE

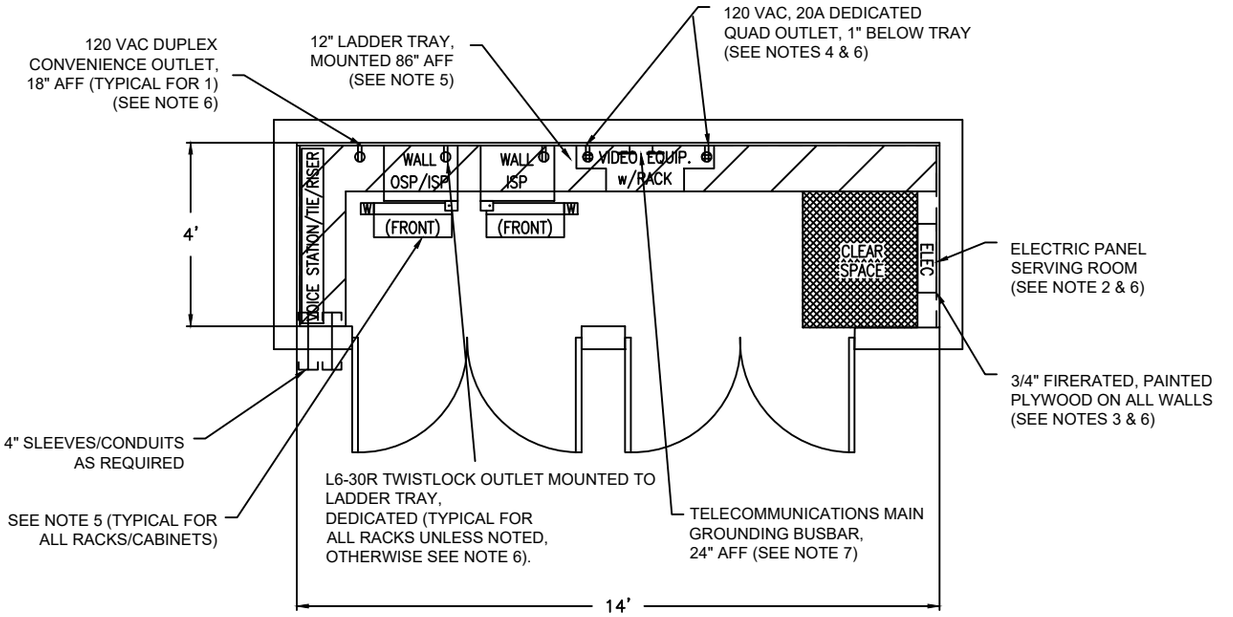
DRAWN BY: SAO
CHECKED BY: WZ
DATE: 8/27/13
DWG. # ER-03A

TITLE: TYPICAL SMALL EQUIPMENT ROOM LAYOUT

THE STATE UNIVERSITY OF NEW JERSEY RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

ER-3
3 RACKS (120 FT^2)



NOTES:

1. PROVIDE SERVICE LOOP FOR ALL HORIZONTAL VOICE, DATA, AND VIDEO CABLES NOT TO EXCEED 10 FEET. LOCATION TO BE DETERMINED BY THE RUPM.
2. PROVIDE (3) 30A SPARE CIRCUITS IN ELECTRIC PANEL.
3. 3/4" AC FIRERATED PLYWOOD ON ALL WALLS, PAINTED WITH WHITE FIRE RETARDANT PAINT (DO NOT PAINT PLYWOOD LABEL). MOUNT PLYWOOD VERTICALLY AT 22" AFF WITH STAINLESS STEEL HARDWARE.
4. MOUNT QUAD OUTLETS WITH TOP OF BACKBOX AT 1" BELOW LADDER TRAY OR AT 85" AFF, WHICHEVER IS LOWER. LOCATIONS TO BE COORDINATED WITH RUTGERS UNIVERSITY.
5. BY IT/COMMUNICATIONS CONTRACTOR.
6. BY ARCHITECT/ROOM CONSTRUCTION CONTRACTOR. COORDINATE LOCATION WITH RUTGERS UNIVERSITY.
7. BY IT CONTRACTOR FOR EXISTING ROOMS. BY BUILDING CONTRACTOR FOR NEW ROOMS.

2	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS	4	ADM	DH	4/2/2020	UPDATE
1	ADM		2/10/14	GENERAL REVISION	3	ADM	DH	3/17/2020	UPDATE

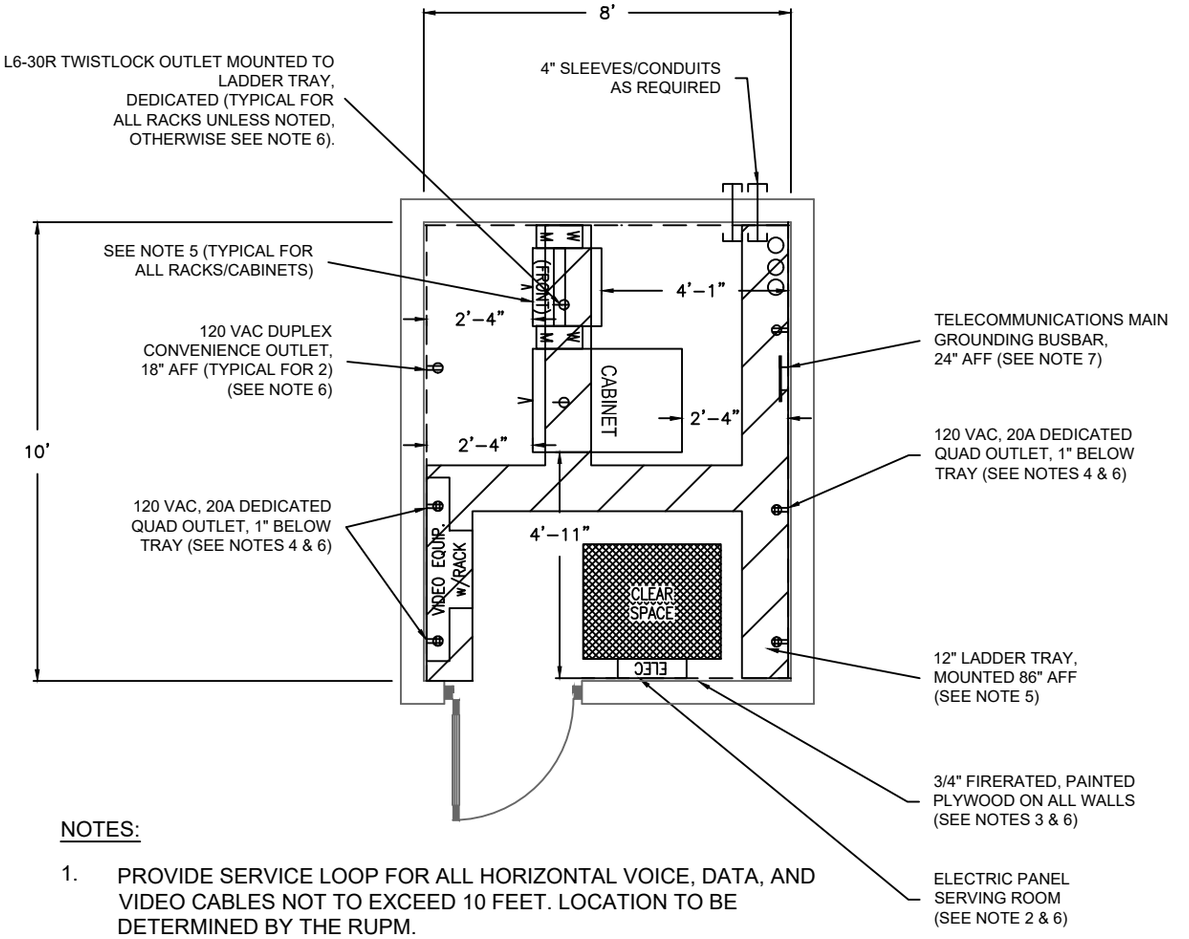
DRAWN BY: SAO
CHECKED BY: WZ
DATE: 3/26/02
DWG. # TR-02

TITLE: TYPICAL SHALLOW EQUIPMENT ROOM LAYOUT

THE STATE UNIVERSITY OF NEW JERSEY RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

ER-4



NOTES:

1. PROVIDE SERVICE LOOP FOR ALL HORIZONTAL VOICE, DATA, AND VIDEO CABLES NOT TO EXCEED 10 FEET. LOCATION TO BE DETERMINED BY THE RUPM.
2. PROVIDE (3) 30A SPARE CIRCUITS IN ELECTRIC PANEL.
3. 3/4" AC FIRERATED PLYWOOD ON ALL WALLS, PAINTED WITH WHITE FIRE RETARDANT PAINT (DO NOT PAINT PLYWOOD LABEL). MOUNT PLYWOOD VERTICALLY AT 22" AFF WITH STAINLESS STEEL HARDWARE.
4. MOUNT QUAD OUTLETS WITH TOP OF BACKBOX AT 1" BELOW LADDER TRAY OR AT 85" AFF, WHICHEVER IS LOWER. LOCATIONS TO BE COORDINATED WITH RUTGERS UNIVERSITY.
5. BY IT/COMMUNICATIONS CONTRACTOR.
6. BY ARCHITECT/ROOM CONSTRUCTION CONTRACTOR. COORDINATE LOCATION WITH RUTGERS UNIVERSITY.
7. BY IT CONTRACTOR FOR EXISTING ROOMS. BY BUILDING CONTRACTOR FOR NEW ROOMS.

2	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS					
1	ADM		2/10/14	GENERAL REVISION	3	ADM	DH	3/17/2020	UPDATE

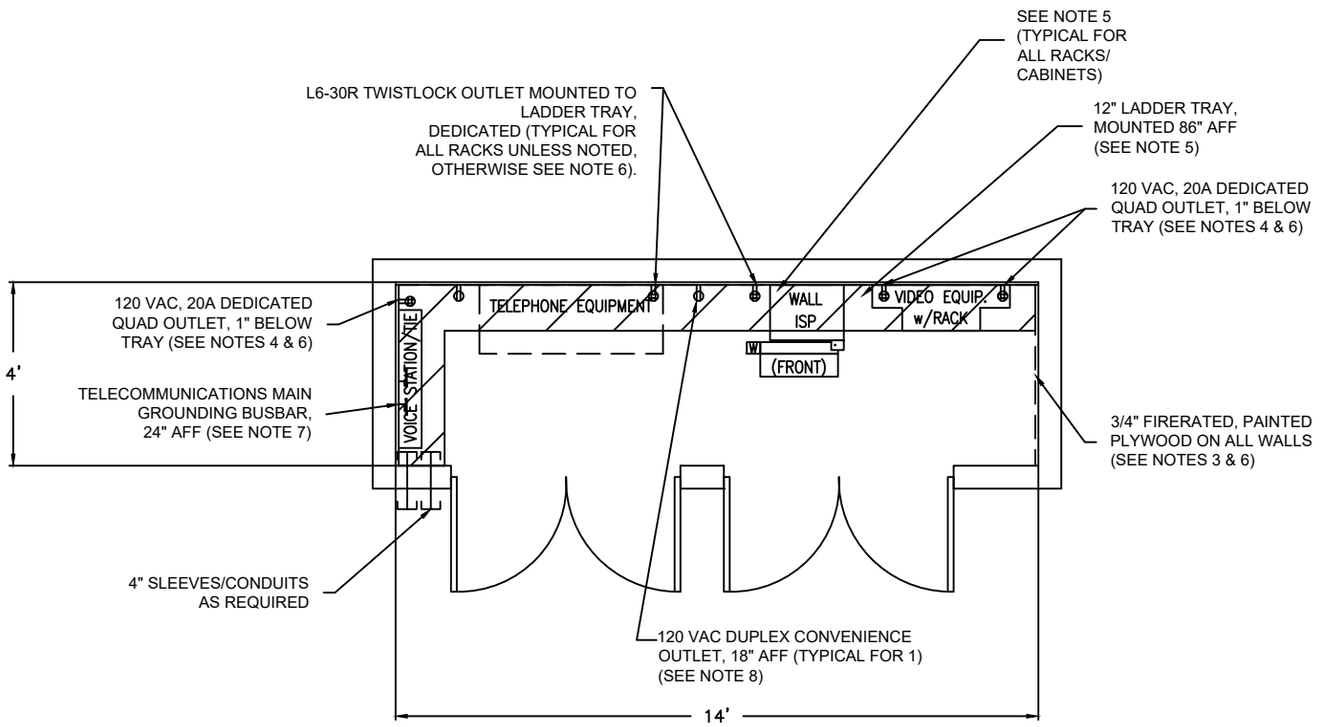
DRAWN BY: SAO
CHECKED BY: JF
DATE: 3/26/02
DWG. # TR-01

TITLE: TYPICAL TELECOMMUNICATIONS ROOM LAYOUT

THE STATE UNIVERSITY OF NEW JERSEY RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

TR-1
1 RACK (80 FT^2)



NOTES:

1. PROVIDE SERVICE LOOP FOR ALL HORIZONTAL VOICE, DATA, AND VIDEO CABLES NOT TO EXCEED 10 FEET. LOCATION TO BE DETERMINED BY THE RUPM.
2. PROVIDE (3) 30A SPARE CIRCUITS IN ELECTRIC PANEL.
3. 3/4" AC FIRERATED PLYWOOD ON ALL WALLS, PAINTED WITH WHITE FIRE RETARDANT PAINT (DO NOT PAINT PLYWOOD LABEL). MOUNT PLYWOOD VERTICALLY AT 22" AFF WITH STAINLESS STEEL HARDWARE.
4. MOUNT QUAD OUTLETS WITH TOP OF BACKBOX AT 1" BELOW LADDER TRAY OR AT 85" AFF, WHICHEVER IS LOWER. LOCATIONS TO BE COORDINATED WITH RUTGERS UNIVERSITY.
5. BY IT/COMMUNICATIONS CONTRACTOR.
6. BY ARCHITECT/ROOM CONSTRUCTION CONTRACTOR. COORDINATE LOCATION WITH RUTGERS UNIVERSITY.
7. BY IT CONTRACTOR FOR EXISTING ROOMS. BY BUILDING CONTRACTOR FOR NEW ROOMS.

2	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS					
1	ADM		2/10/14	GENERAL REVISION	3	ADM	DH	3/17/2020	UPDATE

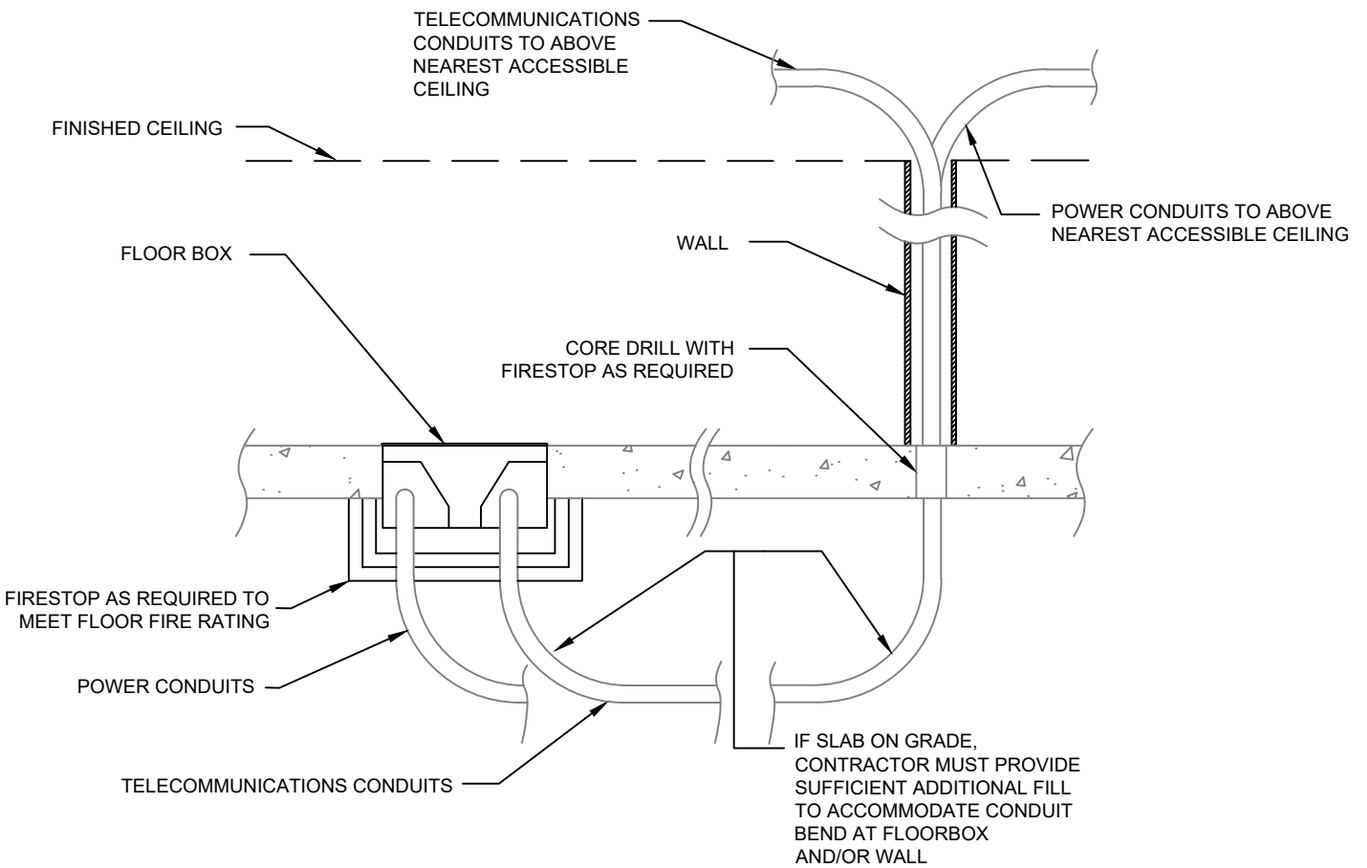
DRAWN BY: SAO
CHECKED BY: WZ
DATE: 3/26/02
DWG. # ER-4

TITLE: TYPICAL SHALLOW TELECOMMUNICATIONS ROOM LAYOUT

THE STATE UNIVERSITY OF NEW JERSEY RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

TR-2



NOTE: SEE DRAWING FB-2 FOR RELATED INFORMATION

1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

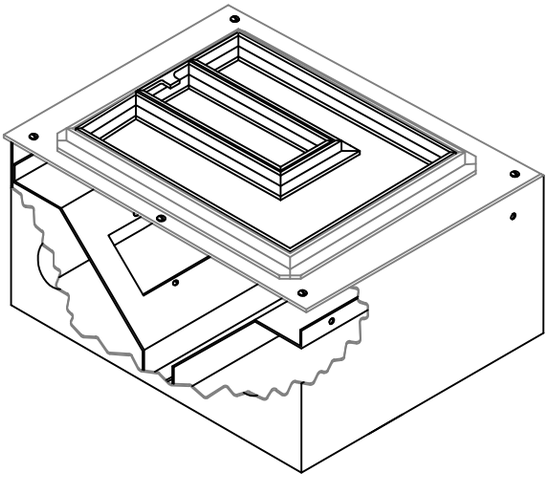
DRAWN BY: SAO
CHECKED BY: JF
DATE: 4/5/02
DWG. # FB-01

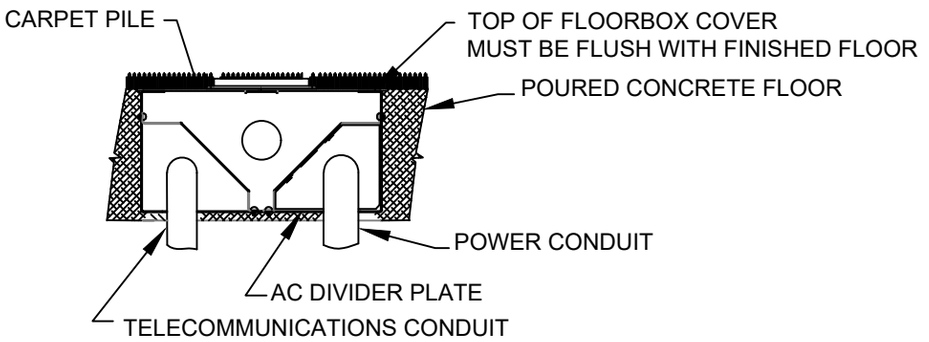
TITLE: POWER & TELCOM FLOORBOX INSTALLATION DETAIL 1

THE STATE UNIVERSITY OF NEW JERSEY RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

FB-1





NOTES:

1. CONDUIT DIRECTION UPON LEAVING FLOORBOX ARE FOR EXAMPLE ONLY AND ARE NOT FOR CONSTRUCTION.

2. SEE DRAWING FB-1 FOR RELATED DETAILS.

SPECIAL NOTES FOR SLAB ON GRADE CONDITION

1. BOX IS NOT RATED FOR CONTACT WITH GRADE - CONTRACTOR IS TO PROVIDE SUFFICIENT PAD BELOW BOX AS REQUIRED TO MEET LOCAL CODE.

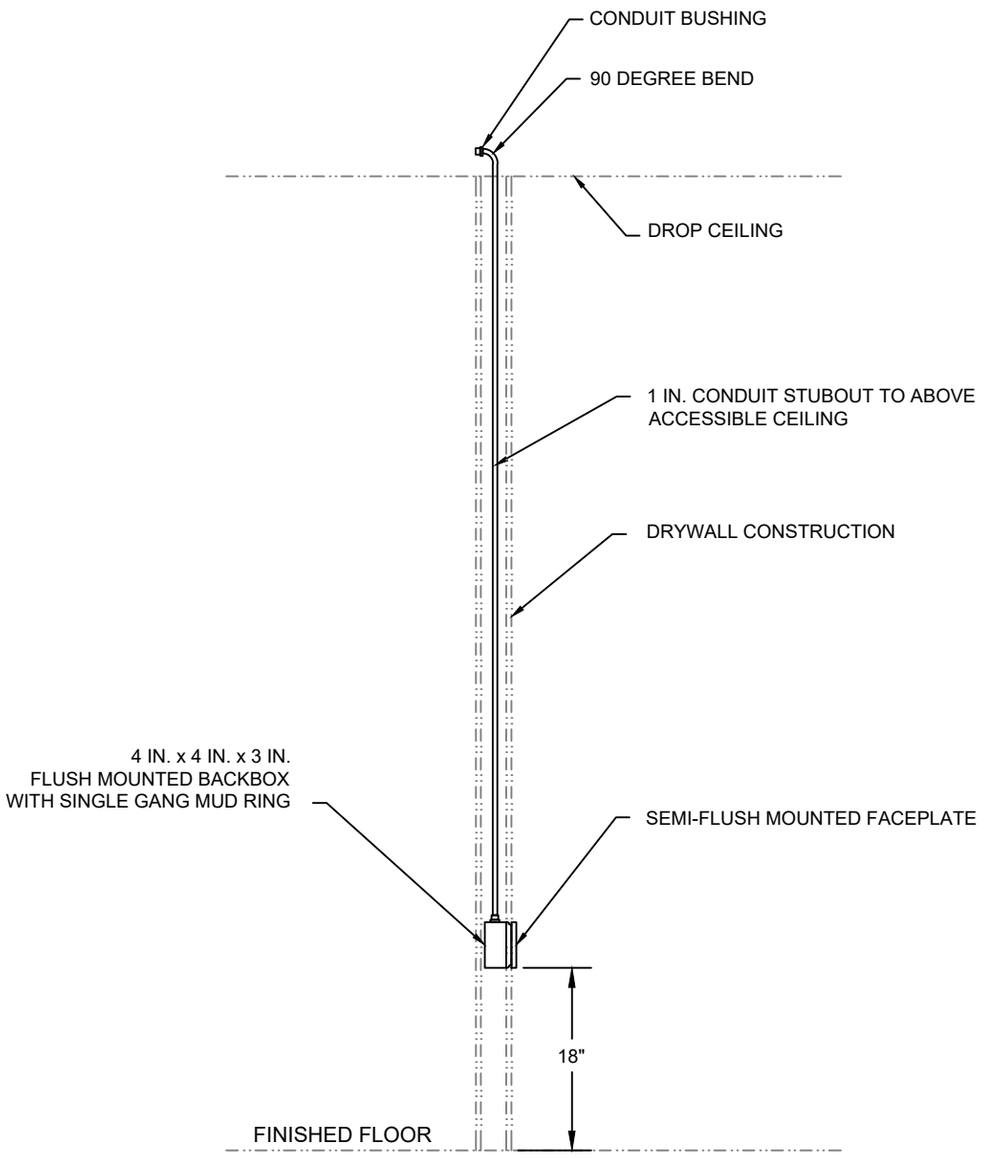
1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

DRAWN BY: SAO
CHECKED BY: JF
DATE: 4/5/02
DWG. # FB-02

TITLE: POWER & TELCOM FLOORBOX INSTALLATION DETAIL 2

THE STATE UNIVERSITY OF NEW JERSEY
RUTGERS
OFFICE OF INFORMATION TECHNOLOGY

FB-2



SECTION
N.T.S.

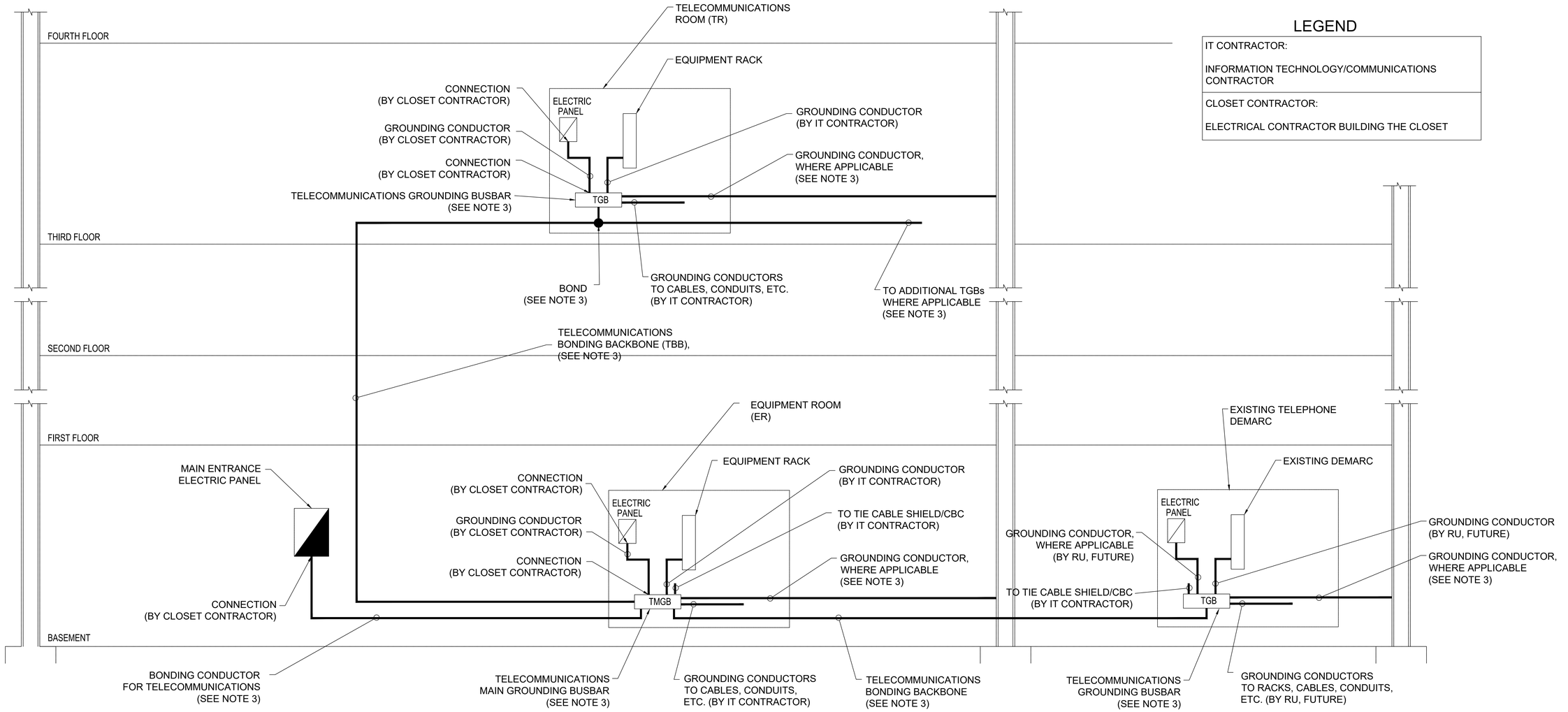
1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

DRAWN BY: SAO
CHECKED BY: JF
DATE: 3/26/02
DWG. # FM-01

TITLE: TYPICAL FLUSH MOUNTED INSTALLATION FOR NEW CONSTRUCTION

THE STATE UNIVERSITY OF NEW JERSEY
RUTGERS
OFFICE OF INFORMATION TECHNOLOGY

FM-1



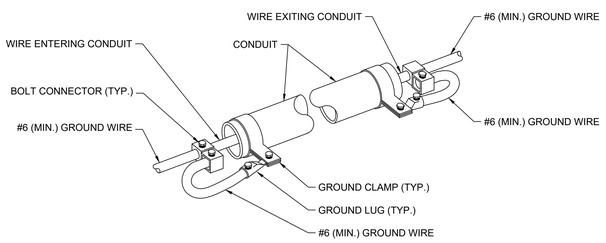
LEGEND

IT CONTRACTOR: INFORMATION TECHNOLOGY/COMMUNICATIONS CONTRACTOR
CLOSET CONTRACTOR: ELECTRICAL CONTRACTOR BUILDING THE CLOSET

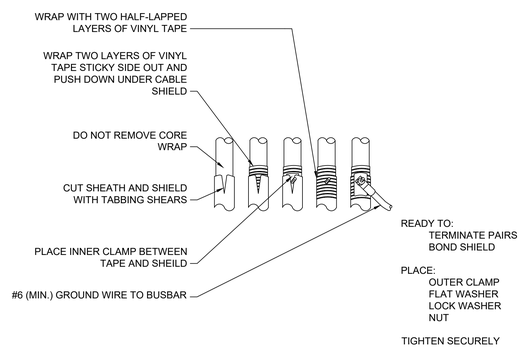
NOTES:

1. ALL CABLES AND CONNECTIONS BY IT CONTRACTOR EXCEPT WHERE INDICATED OTHERWISE.
2. ALL CABLES SHALL BE #6 AWG. OR LARGER.
3. BY IT CONTRACTOR FOR EXISTING CLOSETS.
   BY CLOSET CONTRACTOR FOR NEW CLOSETS.

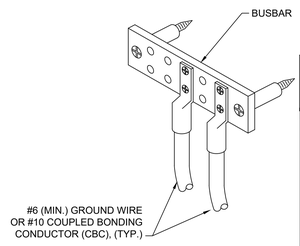
TYPICAL GROUNDING & BONDING REQUIREMENTS



TYPICAL BONDING FOR GROUND WIRE IN CONDUIT



BACKBONE CABLE SHIELDS



TMGB & TGB BUSBAR DETAIL

1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

DRAWN BY: S.A.O.
REVISED BY: ADM 2/10/14
CHECKED BY: MS
DATE: 4/03/02
DWG. # GB-01

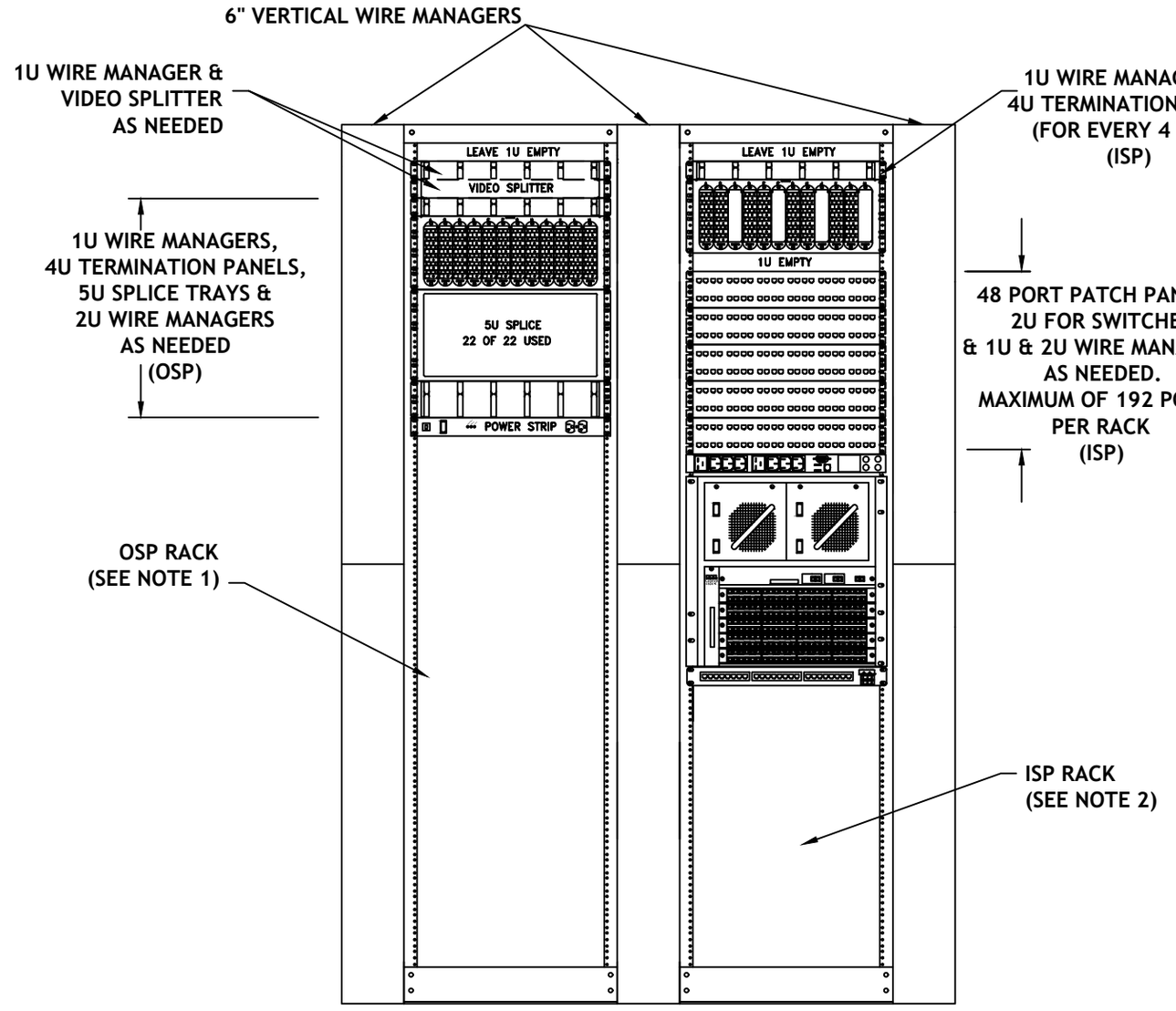
TITLE:

TYPICAL GROUNDING AND BONDING REQUIREMENTS

THE STATE UNIVERSITY OF NEW JERSEY
RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

GB-1



NOTES:

1. WIRE MANAGERS, VIDEO SPLITTERS, FIBER TERMINATION PANELS AND SPLICE TRAYS TO BE INSTALLED IN A DEDICATED OSP RACK. DETAILS OF THE EQUIPMENT NEEDED WILL VARY FROM BUILDING TO BUILDING.

2. DETAILS OF THE EQUIPMENT NEEDED WILL VARY FROM BUILDING TO BUILDING.

3	ADM	DH	2/4/2020	UPDATE	6	ADM	DH	4/2/2020	UPDATE
2	ADM	DH	1/31/2020	UPDATE	5	ADM	DH	3/17/2020	UPDATE
1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS	4	ADM	DH	2/27/2020	UPDATE

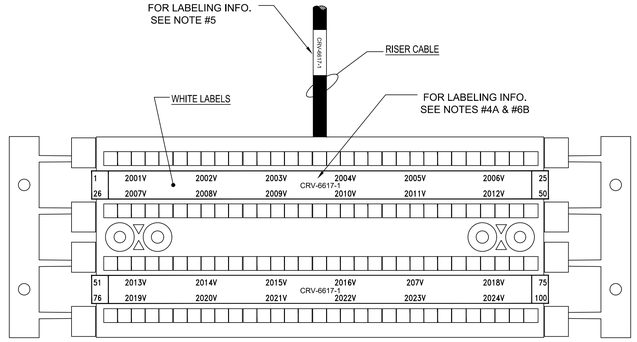
DRAWN BY: SAO
CHECKED BY: JF
DATE: 3/26/02
DWG. # RL-B2

TITLE: TYPICAL EQUIPMENT ROOM RACK LAYOUT

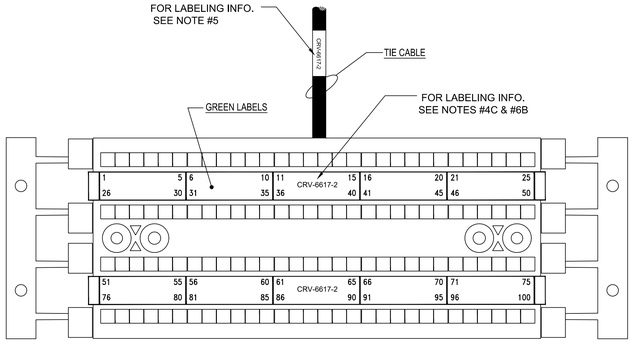
THE STATE UNIVERSITY OF NEW JERSEY RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

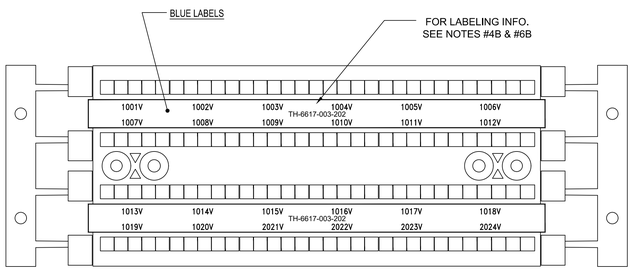
RL-ER



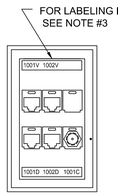
VOICE RISER CABLE
110 DETAIL



VOICE TIE CABLE
110 DETAIL

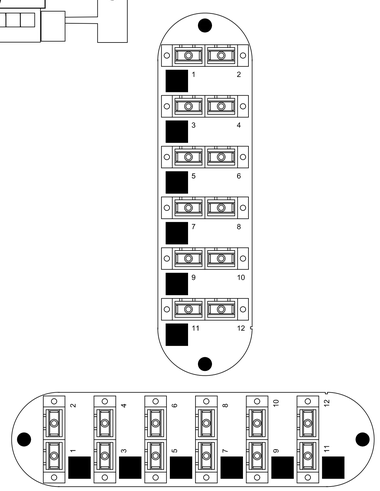


VOICE STATION CABLE
110 DETAIL

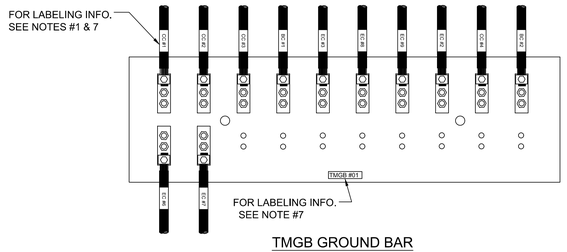


TYPICAL WP

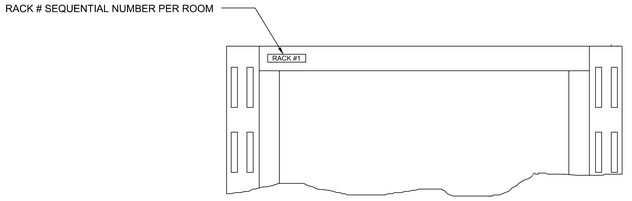
	Equipment/Cable	Label Type or Equal	Font	Size
1.	Patch Panel Port	Panduit CPPLF-5	Aerial Black-Regular	10
2.	Patch Panel	Panduit PLL22P01	Aerial Black-Regular	24
3.	Tie 110 Block	Panduit DS110-GR	Aerial-Regular	8
4.	Riser 110 Block	Panduit DS110	Aerial-Regular	8
5.	Station 110 Block	Panduit DS110-BU	Aerial-Regular	8
6.	WP Station Single Gang	P-Touch Tape TZ-231 9MM Black on White	Bold	8,27,13
7.	WP Station Double Gang	P-Touch Tape TZ-231 9MM Black on White	Bold	8,27,13
8.	WP Station Single Gang	Panduit JLEFPS-1	Aerial Black-Regular	12
9.	WP Station Double Gang	Panduit JLEFPD-1	Aerial Black-Regular	11
10.	Rack Panduit	Panduit PLL22P01	Aerial Black-Regular	24
11.	ISP Fiber Cabinet TH	Panduit PLL22P01	Aerial Black-Regular	24
12.	ISP Fiber terminations	P-Touch Tape TZ-SE4 18MM Black on White	Bold	8,27,13
13.	OSP Fiber Cabinet TH	Panduit PLL22P01	Aerial Black-Regular	24
14.	OSP Fiber terminations	P-Touch Tape TZ-SE4 18MM Black on White	Bold	8,27,13
15.	Ground Bar ID	Panduit PLL22P01	Aerial Black-Regular	24
16.	Ground Cable	Panduit PLL-11-Y3-2.5	Aerial Black-Regular	12
17.	Data Station Cable	Panduit PLL-11-Y3-2.5	Aerial Black-Regular	12
18.	Voice Station Cable	Panduit PLL-11-Y3-2.5	Aerial Black-Regular	12
19.	Coax Station Cable	Panduit PLL-12-Y3-2.5	Aerial Black-Regular	12
20.	Voice Riser Cable	SLCT-WH sleeve with P-Touch Label	Bold	13
21.	Coax Riser Cable	Panduit SLCT-WH sleeve with P-Touch Label	Bold	13
22.	Fiber Optic Riser Cable	Panduit SLCT-WH sleeve with P-Touch Label	Bold	13



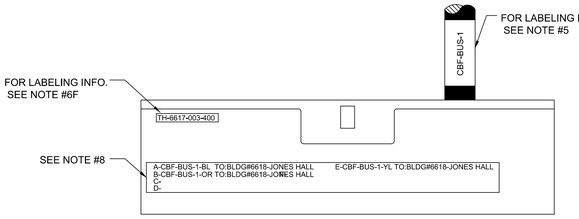
CLOSET CONNECTOR HOUSING PANELS



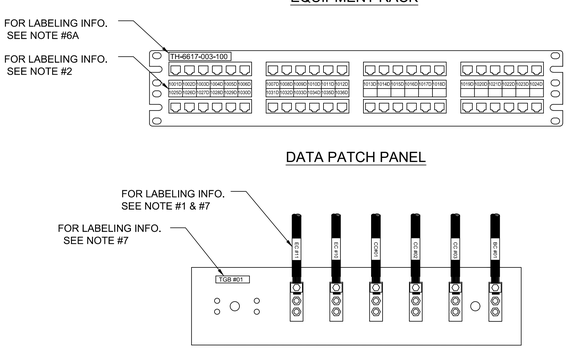
TMGB GROUND BAR



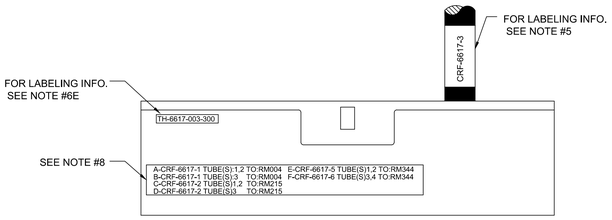
EQUIPMENT RACK



OUTSIDE PLANT FIBER CABINET

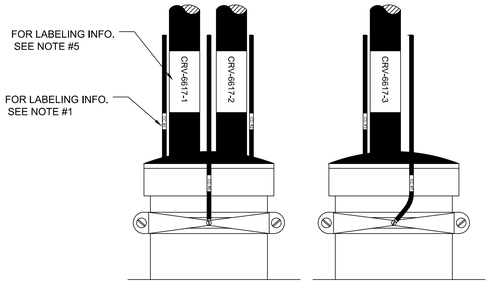


DATA PATCH PANEL

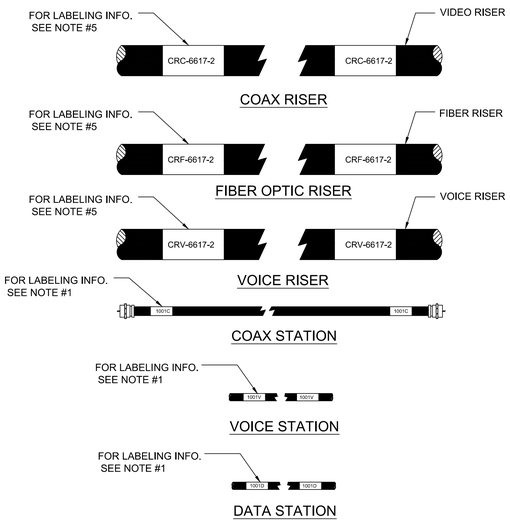


INSIDE PLANT FIBER CABINET

TGB GROUND BAR



RISER CONDUIT DETAIL



NOTES:

1. EACH VOICE, DATA, COAX CABLE AND GROUND WIRE THAT IS PULLED WILL BE LABELED ON BOTH ENDS OF THE CABLE WITH A PRINTED WRAP AROUND CABLE LABEL.

2. EACH DATA PATCH PANEL PORT WILL BE LABELED IN THE CENTER OF THE PATCH PANEL WHITE LABEL.

3. EACH FACEPLATE WILL BE LABELED BLACK ON WHITE LABELS WITH BOLD FONT.

4. EACH 110 PUNCH DOWN BLOCK WILL BE LABELED.
   A.)RISER CABLE. (WHITE)
   B.)HORIZONTAL CABLES. (BLUE)
   C.)TIE CABLE. (GREEN)

5. TIE, RISER AND BACKBONE CABLE SLEEVE.
   THE VOICE CABLE (RISER & TIE) ID WILL BE CRV-BUILDING NUMBER-SEQUENTIAL NUMBER.
   THE VOICE CABLE (BACKBONE) ID WILL BE CBV-CAMPUS NAME-SEQUNETIAL NUMBER.
   THE FIBER CABLE (RISER) ID WILL BE CRF-BUILDING NUMBER-SEQUENTIAL NUMBER.
   THE FIBER CABLE (BACKBONE) ID WILL BE CBF-CAMPUS NAME-SEQUNETIAL NUMBER.
   THE COAX CABLE (RISER) ID WILL BE CRC-BUILDING NUMBER-SEQUENTIAL NUMBER.
   THE COAX CABLE (BACKBONE) ID WILL BE CBC-CAMPUS NAME-SEQUNETIAL NUMBER.

6. TERMINATION HARDWARE NUMBERS AND RACK NUMBER.
   THE TERMINATION HARDWARE NUMBERS WILL BE DETERMINED
   THE FOLLOWING WAYS:
   A.) DATA PATCH PANEL TH-BUILDING#-ER/TR ROOM#-100 THROUGH 199.
   B.) VOICE PUNCHDOWN BLOCK (RISER) CRV-BUILDING#-SEQUENTIAL# (TO MATCH RISER CABLE NUMBER).
   C.) VOICE PUNCHDOWN BLOCK (TIE) CRV-BUILDING#-SEQUENTIAL# (TO MATCH TIE CABLE NUMBER).
   D.) VOICE PUNCHDOWN BLOCK (STATION) TH-BUILDING#-ER/TC ROOM#-200 THROUGH 299.
   E.) ISP FIBER TH-BUILDING#-ER/TR ROOM#-300 THROUGH 399.
   F.) OSP FIBER TH-BUILDING#-ER/TR ROOM#-400 THROUGH 499.
   G.) VIDEO TH-BUILDING#-ER/TR ROOM#-500 THROUGH 599.

7. GROUNDING LABELING:
   1. GROUND BARS WILL BE LABELED WITH TMGB OR TGB-THE # OF THE GND BAR AND THE BUILDING.
   2. GROUND CONDUCTORS WILL BE LABELED
      A.) BONDING CONDUCTORS (FROM TMGB TO THE TGB'S OR SERVICE GND AND BLDG STEEL) WILL BE BC AND A SEQUENTIAL NUMBER THROUGHOUT THE ROOM.
      B.) EQUIPMENT BONDING CONDUCTOR (LADDER RACKS FREESTANDING, SWING GATE, AND CONDUITS) EC AND A SEQUENTIAL NUMBER THROUGHOUT THE ROOM.
      C.) CABLE BONDING CONDUCTOR (THE BONDING CONDUCTOR THAT IS RUN WITH THE BACKBONE CABLE) CC AND THE NUMBER OF THE TIE OR RISER CABLE IT IS ASSOCIATED WITH.

8. FIBER LABELING:
   1.) INSIDE PLANT LABEL AS FOLLOWS:
   CONNECTOR PANEL LOCATION IN HOUSING-CRF-BUILDING#-SEQUENTIAL#-TUBE#(s) TO:RM# IT IS FEEDING.
   2.) OUTSIDE PLANT LABEL AS FOLLOWS:
   CONNECTOR PANEL LOCATION IN HOUSING-CBF-CAMPUS NAME-CABLE#-TUBE COLOR TO:BLDG# - BLDG. NAME IT IS FEEDING.

1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

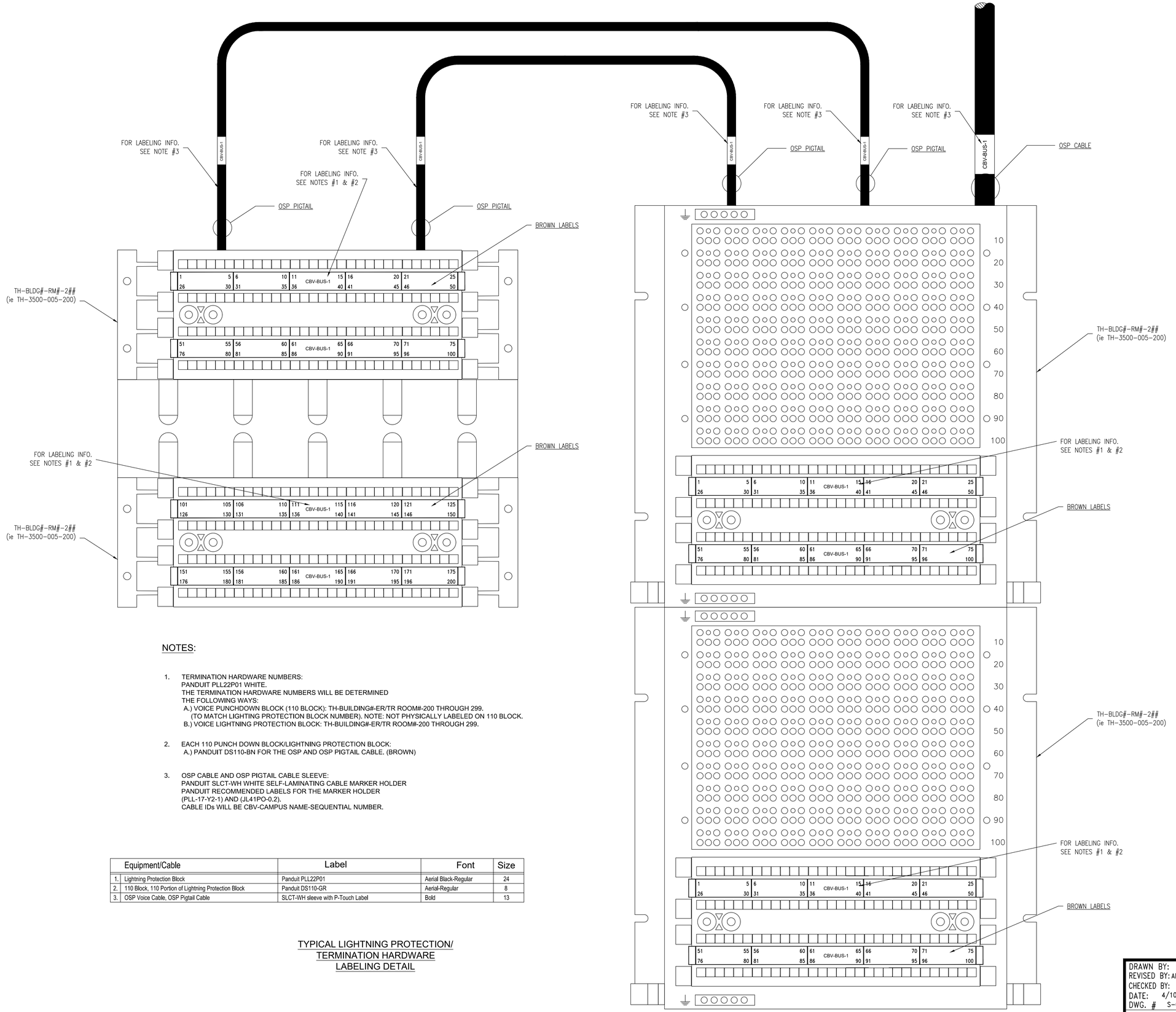
DRAWN BY: SAO
REVISED BY: ADM 2/10/14
CHECKED BY: MS
DATE: 4/10/02
DWG. # S-02

TITLE:
TYPICAL CABLE AND HARDWARE LABELING DETAILS

THE STATE UNIVERSITY OF NEW JERSEY
RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

S-2



NOTES:

1. TERMINATION HARDWARE NUMBERS:
   PANDUIT PLL22P01 WHITE.
   THE TERMINATION HARDWARE NUMBERS WILL BE DETERMINED
   THE FOLLOWING WAYS:
   A.) VOICE PUNCHDOWN BLOCK (110 BLOCK): TH-BUILDING#-ER/TR ROOM#-200 THROUGH 299.
   (TO MATCH LIGHTING PROTECTION BLOCK NUMBER). NOTE: NOT PHYSICALLY LABELED ON 110 BLOCK.
   B.) VOICE LIGHTNING PROTECTION BLOCK: TH-BUILDING#-ER/TR ROOM#-200 THROUGH 299.

2. EACH 110 PUNCH DOWN BLOCK/LIGHTNING PROTECTION BLOCK:
   A.) PANDUIT DS110-BN FOR THE OSP AND OSP PIGTAIL CABLE. (BROWN)

3. OSP CABLE AND OSP PIGTAIL CABLE SLEEVE:
   PANDUIT SLCT-WH WHITE SELF-LAMINATING CABLE MARKER HOLDER
   PANDUIT RECOMMENDED LABELS FOR THE MARKER HOLDER
   (PLL-17-Y2-1) AND (JL41PO-0.2).
   CABLE IDs WILL BE CBV-CAMPUS NAME-SEQUENTIAL NUMBER.

	Equipment/Cable	Label	Font	Size
1.	Lightning Protection Block	Panduit PLL22P01	Aerial Black-Regular	24
2.	110 Block, 110 Portion of Lightning Protection Block	Panduit DS110-GR	Aerial-Regular	8
3.	OSP Voice Cable, OSP Pigtail Cable	SLCT-WH sleeve with P-Touch Label	Bold	13

TYPICAL LIGHTNING PROTECTION/
TERMINATION HARDWARE
LABELING DETAIL

1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

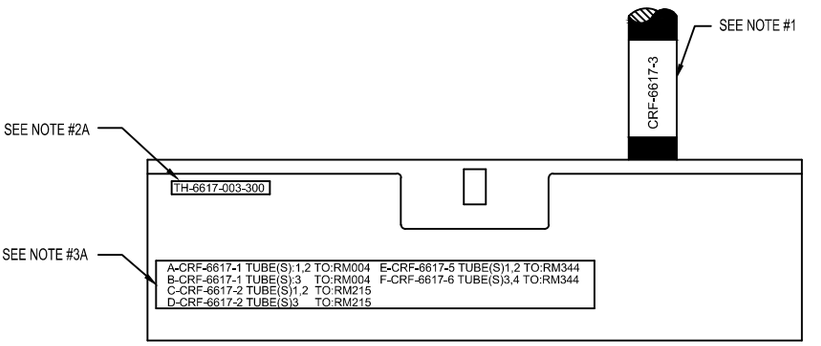
DRAWN BY: SAO
REVISED BY: ADM 2/10/14
CHECKED BY: MS
DATE: 4/10/02
DWG. # S-03

TITLE:
TYPICAL LIGHTNING PROTECTION/
TERMINATION LABELING DETAIL

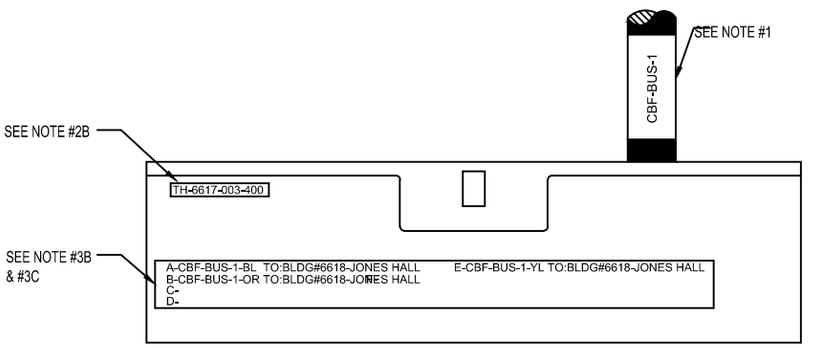
THE STATE UNIVERSITY OF NEW JERSEY
RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

S-3



INSIDE PLANT FIBER CABINET



OUTSIDE PLANT FIBER CABINET

NOTES:

1. TIE, RISER AND BACKBONE CABLE SLEEVE.
   PANDUIT SLCT-WH WHITE SELF-LAMINATING CABLE MARKER HOLDER
   PANDUIT RECOMMENDED LABELS FOR THE MARKER HOLDER
   (PLL-17-Y2-1) AND (JL41PO-0.2)
   THE FIBER CABLE (RISER) ID WILL BE CRF-BUILDING NUMBER-SEQUENTIAL NUMBER.
   THE FIBER CABLE (BACKBONE) ID WILL BE CBF-CAMPUS NAME-SEQUNETIAL NUMBER.

2. TERMINATION HARDWARE NUMBERS AND RACK NUMBER.
   PANDUIT PLL22P01 WHITE.
   THE TERMINATION HARDWARE NUMBERS WILL BE DETERMINED
   THE FOLLOWING WAYS:

   A.) ISP FIBER TH-BUILDING#-ER/TR ROOM#-300 THROUGH 399.
   B.) OSP FIBER TH-BUILDING#-ER/TR ROOM#-400 THROUGH 499.

3. FIBER LABELING:
   THE LABEL WILL BE A P-TOUCH TZ-3/4" TAPE FROM THE P-TOUCH PT330 MACHINE.

   A.) INSIDE PLANT LABEL WILL BE AS FOLLOWS:
   CONNECTOR PANEL LOCATION IN HOUSING-CRF-BUILDING#-SEQUENTIAL#-TUBE#(s)
   TO:RM# IT IS FEEDING.
   B.) OUTSIDE PLANT LABEL AS FOLLOWS:
   CONNECTOR PANEL LOCATION IN HOUSING-CBF-CAMPUS NAME-CABLE#-TUBE COLOR
   TO:BLDG# - BLDG. NAME IT IS FEEDING.
   C.) LABEL FOR OSP COMING FROM UTILITY POLE, MANHOLE OR OTHER SPLICE LOCATION.

   EXAMPLE:

   TO UTILITY POLE UP-CAC-61832
   FINAL DESTINATION:
   (B-C 6SM/6MM) CBF-CAC-00389
   BLDG. 3115 COLLEGE OF NURSING

   TO MANHOLE MH-CAM-0054
   FINAL DESTINATION:
   (A-C 12SM/12MM) CBF-CAM-00074
   BLDG. 7004 CAMDEN STUDENT CENTER

NOTES

ISSUE	BY	CHK	DATE	DESCRIPTION
3	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS
2	ADM	MS	2/10/14	UPDATED PER DESIGN SPEC REVIEW
1	ADM	DH	1/5/06	ISP/OSP LABELING

APPROVED FOR RUTGERS–THE STATE UNIVERSITY

THE STATE UNIVERSITY OF NEW JERSEY RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

PROJECT
RUTGERS UNIVERSITY

DRAWING TITLE
TYPICAL LABELING AID
ISP/OSP OPTICAL FIBER

DSR NO.	SCALE	DRAWN BY	FILE NAME
	NONE	ADM	S-5
BLDG. NO.	DATE 1/5/06	CHECKED BY JF	

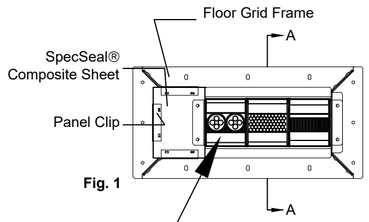


Fig. 1

COMPOSITE SHEET DIMENSIONS



Fig. 2

FLOOR OPENING DIMENSIONS

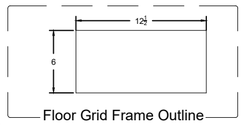


Fig. 3

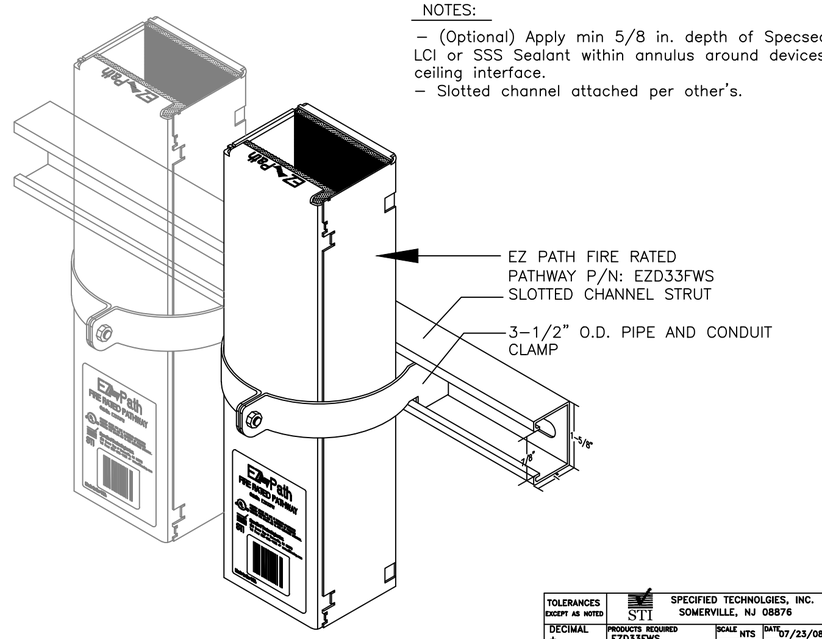
INSTALLATION INSTRUCTIONS

1. Form or cut nom 6 in. by 12-1/2 in. opening in floor. (See Fig. 3)
2. Install Floor Grid Frame (Part No. EZDG444) as specified in Series 44 Modular Grid System Installation Guide over opening.
3. Assemble Pathway Module as specified in Installation Guide and using three (3) Series 44 EZ-Path Pathways.
4. Obtain Composite Sheet Filler Panel from EZG844 or EZG1644 kits that is not in use.
5. Cut Filler Panel to 5 in. width so that it measures 6-3/4 in. by 5 in. (See Fig 2.)
6. Obtain three (3) Panel Clips from EZG844 or EZG1644 kits that are not in use.
7. Install pre-cut Composite Sheet Filler Panel and three (3) Panel Clips as shown in Fig. 1 above.
8. Insert previously assembled Pathway Module in step 3 as specified in Installation Guide. Tighten two (2) bolts and two (2) nuts (Provided) onto frame on one side. On second side, install two (2) No. 10 self tapping steel sheet metal screws using two (2) min 3/8 in. diam steel fender washers (Purchased Separately).

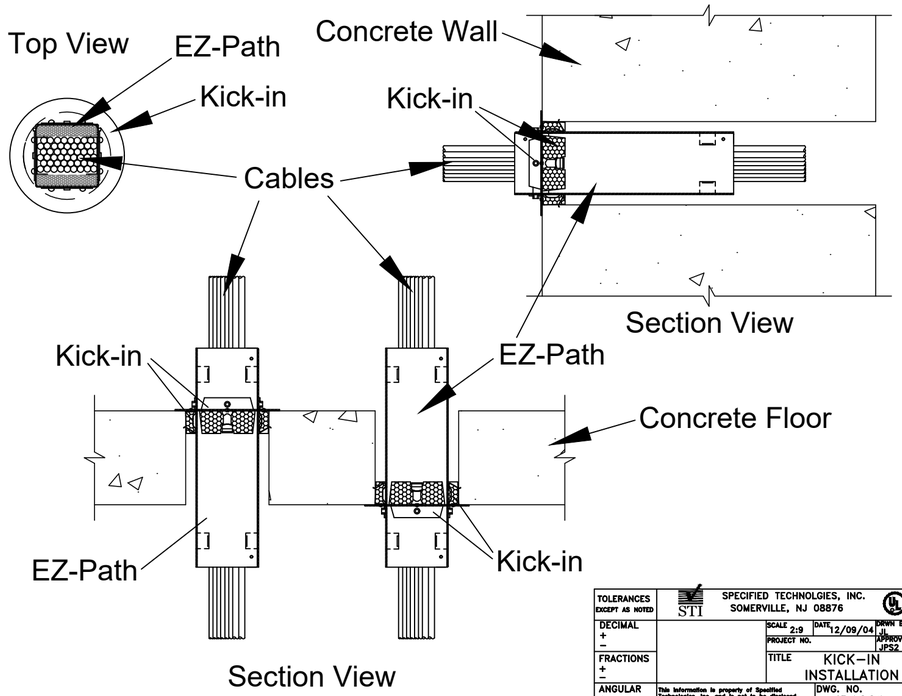
TOLERANCES EXCEPT AS NOTED	SPECIFIED TECHNOLGIES, INC. SOMERVILLE, NJ 08876		
DECIMAL ±	PRODUCTS REQUIRED EZD44(3) Filler Panel(1) Panel Clip(3) *No. 10 Screws(2) *3/8 Dia. Washer(2)	SCALE NTS	DATE 07/15/08 DRWN BY: JL
FRACTIONS ±		PROJECT INFO. MOHEGAN TC&G CENTER	APPROV. BY: JPS2
ANGULAR ±	This information is property of Specified Technologies, Inc. and is not to be disclosed, utilized or copied without written consent of Specified Technologies, Inc.	TITLE EZ-PATH Configuration for Riser Applications	
		DWG. NO. ID1006	A

NOTES:

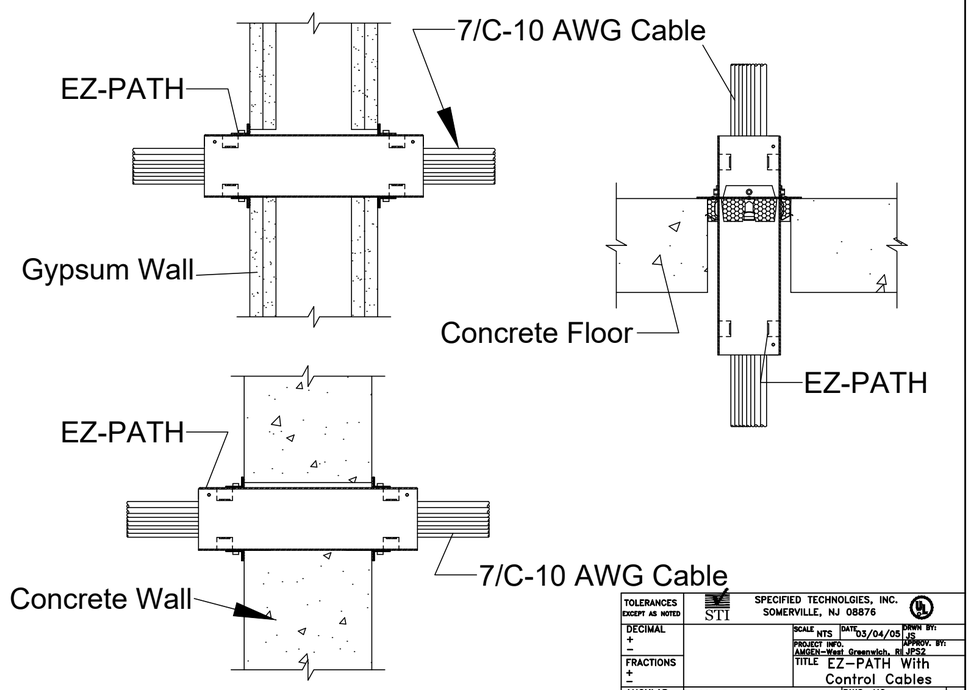
– (Optional) Apply min 5/8 in. depth of Specseal® LCI or SSS Sealant within annulus around devices at ceiling interface.
– Slotted channel attached per other's.



TOLERANCES EXCEPT AS NOTED	SPECIFIED TECHNOLGIES, INC. SOMERVILLE, NJ 08876		
DECIMAL ±	PRODUCTS REQUIRED EZD33FWS Slotted Channel Pipe or Conduit Clamp	SCALE NTS	DATE 07/25/08 DRWN BY: JL
FRACTIONS ±		PROJECT INFO. LOMA LINDA UNIVERSITY	APPROV. BY: JPS2
ANGULAR ±	This information is property of Specified Technologies, Inc. and is not to be disclosed, utilized or copied without written consent of Specified Technologies, Inc.	TITLE EZ–PATH Configuration Non–Rated Applications	
		DWG. NO. ID1008	A



TOLERANCES EXCEPT AS NOTED	SPECIFIED TECHNOLGIES, INC. SOMERVILLE, NJ 08876		
DECIMAL ±		SCALE 2:9	DATE 12/09/04 DRWN BY: JL
FRACTIONS ±		PROJECT NO.	APPROV. BY: JPS2
ANGULAR ±	This information is property of Specified Technologies, Inc. and is not to be disclosed, utilized or copied without written consent of Specified Technologies, Inc.	TITLE KICK–IN INSTALLATION	
		DWG. NO. ID1001	A



TOLERANCES EXCEPT AS NOTED	SPECIFIED TECHNOLGIES, INC. SOMERVILLE, NJ 08876		
DECIMAL ±		SCALE NTS	DATE 03/04/05 DRWN BY: JS
FRACTIONS ±		PROJECT INFO. AMGEN–West Greenwich, RI	APPROV. BY: JPS2
ANGULAR ±	This information is property of Specified Technologies, Inc. and is not to be disclosed, utilized or copied without written consent of Specified Technologies, Inc.	TITLE EZ–PATH With Control Cables	
		DWG. NO. ID2001	A

NOTES:

NOTE:
Follow UL System and Instructions to Install EZ-PATH. This is an engineering judgment that supplements UL System Nos. W-L-3218, W-J-3098, F-A-3015, and F-A-3021.

DRAWN BY: ADM REVISED BY: ADM 2/10/14 CHECKED BY: MS DATE: 2/8/10 DWG. # S–06	TITLE: TYPICAL STI - EZ PATH INSTALLATION SCENARIOS		
THE STATE UNIVERSITY OF NEW JERSEY RUTGERS	OFFICE OF INFORMATION TECHNOLOGY	S–6	

1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

1. Floor Assembly - Min 4-1/2 in. (114 mm) thick reinforced lightweight or normal weight (100-150 pcf or 1600-2400 kg/m3) concrete. Floor may also be constructed of min 6 in. thick hollow-core Precast Concrete Units* . Max diam of opening is 4-1/2 in. (114 mm).

   See Precast Concrete Units (CAZT) category in the Fire Resistance Directory for names of manufacturers.

2. Firestop Device* - Steel sleeve device centered within opening and installed in accordance with the accompanying installation instructions. Steel sleeve device secured in place by means of steel escutcheon plates installed with gasketing material supplied with product. Steel escutcheon plates installed on both sides of floor and secured to device by means of steel screws provided with device. As an option, device may be cast into floor assembly. Steel sleeve device provided in nom 1, 2 and 4 in. (25, 51 and 102 mm) sizes. The min and max opening diameters for each device are tabulated below:

Device Size	Min Opening Diam	Max Opening Diam
1 in. (25 mm)	1.12 in. (28 mm)	1.25 in. (32 mm)
2 in. (51 mm)	2.38 in. (60 mm)	2.50 in. (64 mm)
4 in. (102 mm)	4.50 in. (114 mm)	4.50 in. (114 mm)

   When device is cast in place, the gasketing and escutcheon plates are optional.

   SPECIFIED TECHNOLOGIES INC - FS100, FS200, FS201, FS400 and FS401 SpecSeal Ready-Sleeves

3. Cables - Aggregate cross-sectional area of cables in sleeve to be max 48 percent of the cross-sectional area of the sleeve. Tight bundle of cables to be installed in the steel sleeve. The annular space within the sleeve shall be a min of 0 in. (0 mm, point contact) to a max of 2 in. (51 mm). Cables to be rigidly supported on both sides of the floor assembly. Any combination of the following types and sizes of cables may be used:
   A. Max 400 pair No. 24 AWG (or smaller) copper conductor cable with polyvinyl chloride (PVC) jacketing and insulation.
   B. Max 3/C No. 2/0 AWG (or smaller) aluminum or copper conductor service entrance cable with PVC insulation and jacket.
   C. Max 3/C No. 2/0 AWG (or smaller) copper conductor PVC jacketed aluminum clad or steel clad TECK 90 cable.
   D. Max 3/C No. 8 AWG (or smaller) nonmetallic sheathed (Romex) cable with copper conductors, PVC insulation and jacket.
   E. Max 1/C 1000 kcmil (or smaller) copper conductor power cable with XLPE or PVC insulation and XLPE or PVC jacket.
   F. Max RG59/U (or smaller) coaxial cable with fluorinated ethylene insulation and jacketing.
   G. Max 62.5/48 fiber optic cable with PVC insulation and jacketing.
   H. Max 4 pair No. 24 AWG (or smaller) copper conductor data cable with PVC insulation and jacket.

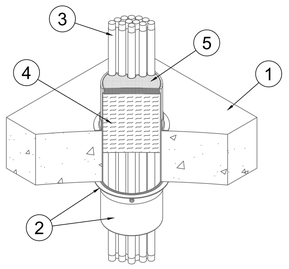
3A. Through Penetrating Product* - 3A. (Not Shown) - Max 4/C No. 2/0 AWG (or smaller) steel or aluminum Armored Cable+ or Metal Clad Cable+ with copper or aluminum conductors. Diam of cable bundle (Item 3) including armored cable not to exceed 4 in. (102 mm). Through penetrating product to be rigidly supported on both sides of a floor assembly.

   AFC CABLE SYSTEMS INC

4. Packing Material - 4. Min 4 pcf (64 kg/m3) mineral-wool batt insulation tightly packed into sleeve to fill device to min 3 in. (76 mm) depth with min 2 in. (51 mm) of the packing material depth within the confines of the floor thickness. Packing material recessed from top edge of device as required to accommodate fill material (Item 5).

   Fill, Void or Cavity Material* - Putty - 5. Min 1/2 in. (13 mm) thickness of fill material applied atop mineral wool packing material (Item 4) flush with top edge of device (Item 2). At point contact location, apply min 3/8 in. (10 mm) diam bead of fill material at grouped cable/steel sleeve interface on top surface of floor. SPECIFIED TECHNOLOGIES INC - SpecSeal Putty
   *Bearing the UL Classification Mark

F Rating - 3 Hr
T Rating - 0 Hr



TOLERANCES (UNLESS OTHERWISE SPECIFIED)	SCALE: NTS	DATE: 05/16/06
DECIMAL ±	TITLE: Ready Sleeve Applications	
FRACTIONS ±		
ANGULAR ±		
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SPECIFIED TECHNOLOGIES, INC.
SOMERVILLE, NJ 08876

1. Floor Assembly - Min 4-1/2 in. (114 mm) thick reinforced lightweight or normal weight (100-150 pcf or 1600-2400 kg/m3) concrete. Floor may also be constructed of min 6 in. thick hollow-core Precast Concrete Units *. Min distance between periphery of through opening and edge of two-piece steel plate (Item 3) is 1 in. (25 mm).

   See Precast Concrete Units (CAZT) category in the Fire Resistance Directory for names of manufacturers.

2. Cables - Aggregate cross-sectional area of cables in steel sleeve to be max 48 percent of the aggregate cross-sectional area of the sleeve. Cables to be rigidly supported on both sides of floor assembly. Any combination of the following types and sizes of copper conductor cable may be used:
   A. Max 400 pair No. AWG (or smaller) copper conductor cable with polyvinyl chloride (PVC) jacketing and insulation.
   B. Max 3/C No. 2/0 AWG (or smaller) aluminum or copper conductor service entrance cable with PVC insulation and jacket.
   C. Max 3/C No. 8 AWG (or smaller) nonmetallic sheathed (Romex) cable with copper conductors, PVC insulation and jacket.
   D. Max 7/C No. 2/0 AWG (or smaller) multiconductor power and control cables with XLPE or PVC insulation and XLPE or PVC jacket.
   E. Nom 5/8 in. (16 mm) (or smaller) coaxial cable with fluorinated ethylene insulation and jacketing.
   F. Nom 5/8 in. (16 mm) fiber optic cable with PVC insulation and jacketing.
   G. Max 4 pair No. 22 AWG (or smaller) copper conductor data cable with PVC insulation and jacket.
   H. Max 4/C No. 2/0 aluminum or copper conductor aluminum or steel Metal-Clad# or Armored-Clad# cable.
   I. Max 1/C-750 kcmil copper conductor cable with XLPE jacket and insulation.

3. Firestop Device * - Device consisting of steel sleeve halves provided with two-piece steel plates. Steel sleeve device shall be installed around cables in accordance with the accompanying installation instructions. Steel sleeve device secured in place by means of two-piece steel plates installed with gasketing material supplied with product. Steel plates and gaskets installed on both sides of floor assembly and secured to device by means of steel screws provided with device. Steel sleeve device provided in nom 1, 2 and 4 in. sizes. Steel plates fastened to floor assembly by means of nom 1/4 in. (6 mm) diam by 1-1/4 (32 mm) long steel concrete screws. As an alternate to the steel concrete screws, nom 1-1/4 in. (32 mm) long steel powder actuated fasteners may be used to secure steel plates. Steel plates shall be aligned on both sides of the floor assembly to cover opening with min 1 in. (25 mm) overlap beyond periphery of through opening. The min and max opening sizes for each sleeve device are shown in the table below:

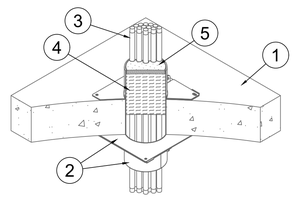
4. Packing Material - Min 4 pcf (64 kg/m3) mineral-wool batt insulation tightly packed into sleeve to fill device to min 3 in. (76 mm) depth with min 2 in. (51 mm) of the packing material depth within the confines of the floor thickness. Packing material recessed from top edge of device as required to accommodate fill material (Item 5).

5. Fill, Void or Cavity Material * - Putty - Min 1/2 in. (13 mm) thickness of fill material applied atop mineral wool packing material (Item 4) flush with top edge of device (Item 2).

   SPECIFIED TECHNOLOGIES INC - SpecSeal Putty

   *Bearing the UL Classification Mark

F Ratings - 2 hr
T Rating - 0 Hr



Device Size	Min Opening Diam	Max Opening Diam
1 in. (25 mm)	1.12 in. (28 mm)+	1.25 in. (32 mm)
2 in. (51 mm)	2.38 in. (60 mm)+	2.50 in. (64 mm)
4 in. (102 mm)	4.50 in. (114 mm)+	4.50 in. (114 mm)

+ - When opening is greater than the min opening size specified, steel fasteners are required.

SPECIFIED TECHNOLOGIES INC -
FSR100, FSR200, FSR400 Ready Split Sleeves

TOLERANCES (UNLESS OTHERWISE SPECIFIED)	SCALE: NTS	DATE: 05/16/06
DECIMAL ±	TITLE: Ready Sleeve Applications	
FRACTIONS ±		
ANGULAR ±		
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SPECIFIED TECHNOLOGIES, INC.
SOMERVILLE, NJ 08876

1. Wall Assembly - The 1 hr or 2 hr fire rated gypsum board/stud wall assembly shall be constructed of the materials and in the manner specified in the individual U300, U400 or V400 Series Wall and Partition Designs in the UL Fire Resistance Directory and shall include the following construction features:

   A. Studs - Wall framing may consist of either wood studs or steel channel studs. Wood studs to consist of nom 2 by 4 in. (51 by 102 mm) lumber spaced max 16 in. (406 mm) OC. Steel studs to be min 3-1/2 in. (89 mm) wide and spaced max 24 in. (610 mm) OC.

   B. Gypsum Board * - 5/8 in. (16 mm) thick, 4 ft (1.2 m) wide with square or tapered edges. The gypsum board type, thickness, number of layers, fastener type and sheet orientation shall be as specified in the individual U300, U400 or V400 Series Design in the UL Fire Resistance Directory.

   The hourly F rating of the firestop system is equal to the hourly fire rating of the wall assembly in which it is installed.

2. Firestop Device * - Steel sleeve device centered within opening and installed in accordance with the accompanying installation instructions. Steel sleeve device secured in place by means of steel escutcheon plates installed with gasket material supplied with product. Steel escutcheon plates installed on both sides of wall and secured to device by means of steel screws provided with device. Steel sleeve device provided in nom 2 in. and 4 in. (51 and 102 mm) sizes. The min and max opening sizes for each sleeve device are shown in the table below:

2A. Firestop Device * - (Not Shown) - As an alternate to Item 2, device consisting of steel sleeve halves provided with two-piece steel plates. Steel sleeve device shall be installed around cables in accordance with the accompanying installation instructions. Steel sleeve device secured in place by means of two-piece steel plates installed with gasket material supplied with product. Steel plates and gaskets installed on both sides of wall and secured to device by means of steel screws provided with device. Steel sleeve device provided in nom 2 in. and 4 in. (51 and 102 mm) sizes. Steel plates fastened to gypsum board layers with nom 1/8 in. (3.2 mm) diam steel hollow-wall anchors. Steel plates shall be aligned on each side of the wall and installed to lap min 1 in. (25 mm) beyond periphery of through opening. The min and max opening sizes for each sleeve device are shown in the table below:

3. Cables - Aggregate cross-sectional area of cables in steel sleeve to be max 33 percent of the aggregate cross-sectional area of the sleeve. Cables to be rigidly supported on both sides of wall assembly. Any combination of the following types and sizes of copper conductor cable may be used:

   A. Max 300 pair No. AWG (or smaller) copper conductor cable with polyvinyl chloride (PVC) jacketing and insulation.

   B. Max 3/C No. 2/0 AWG (or smaller) aluminum or copper conductor service entrance cable with PVC insulation and jacket.

   C. Max 3/C No. 8 AWG (or smaller) nonmetallic sheathed (Romex) cable with copper conductors, PVC insulation and jacket.

   D. Max 7/C No. 2/0 AWG (or smaller) multiconductor power and control cables with XLPE or PVC insulation and XLPE or PVC jacket.

   E. Max RG/U (or smaller) coaxial cable with fluorinated ethylene insulation and jacketing.

   F. Max 62.5/48 fiber optic cable with PVC insulation and jacketing.

   G. Max four pair No. 22 AWG (or smaller) copper conductor data cable with PVC insulation and jacket.

   H. Max 4/C No. 2/0 aluminum or copper conductor aluminum or steel Metal-Clad# or Armored-Clad# cable.

   I. Max 750 kcmil power cable with XLPE jacket and insulation.

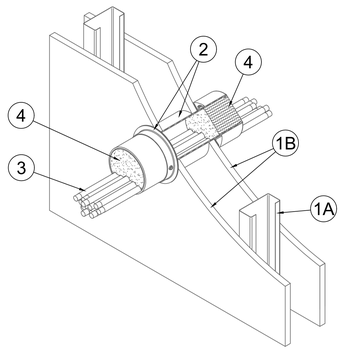
4. Fill, Void or Cavity Materials * - Plug - Nom 2 in. (51 mm) or 4 in. (102 mm) diam plug firmly installed within the sleeve or opening such that the outer circumference of the dome-shaped plug is flush with both ends of sleeve. Plug cut to fit around the cable bundle and installed tightly within the opening or sleeve.

   SPECIFIED TECHNOLOGIES INC - SpecSeal Series FP Firestop Plug

   #Bearing the UL Listing Mark

   *Bearing the UL Classification Mark

F Ratings - 1 and 2 hr (See Item 1)
T Rating - 0 Hr



Device Size	Min Opening Diam	Max Opening Diam
2 in. (51 mm)	2-3/8 in. (60 mm)+	2-1/2 in. (64 mm)
4 in. (102 mm)	4-1/2 in. (114 mm)+	4-1/2 in. (114 mm)

SPECIFIED TECHNOLOGIES INC -
FS200, FS201, FS400 and FS401 SpecSeal Ready Split Sleeves

Device Size	Min Opening Diam	Max Opening Diam
2 in. (51 mm)	2-3/8 in. (60 mm)+	4 in. (102 mm
4 in. (102 mm)	4-1/2 in. (114 mm)+	6 in. (152 mm)

+ - When opening is greater than the min opening size specified, steel fasteners are required.

TOLERANCES (UNLESS OTHERWISE SPECIFIED)	SCALE: NTS	DATE: 05/16/06
DECIMAL ±	TITLE: Ready Sleeve Applications	
FRACTIONS ±		
ANGULAR ±		
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SPECIFIED TECHNOLOGIES, INC.
SOMERVILLE, NJ 08876

1. Wall Assembly - The 1 or 2 hr fire rated gypsum board/stud wall assembly shall be constructed of the materials and in the manner specified in the individual U300, U400 or V400 Series Wall and Partition Designs in the UL Fire Resistance Directory and shall include the following construction features:

   A. Studs - Wall framing may consist of either wood studs or steel channel studs. Wood studs to consist of nom 2 by 4 in. (51 by 102 mm) lumber spaced 16 in. (406 mm) OC. Steel studs to be min 3-1/2 in. (89 mm) wide and spaced max 24 in. (610 mm) OC.

   B. Gypsum Board * - 5/8 in. (16 mm) thick, 4 ft (1.2 m) wide with square or tapered edges. The gypsum board type, thickness, number of layers, fastener type and sheet orientation shall be as specified in the individual U300, U400 or V400 Series Design in the UL Fire Resistance Directory. Min distance between periphery of through opening and edge of two-piece steel plate (Item 3) is 1 in. (25 mm).

   The hourly F rating of the firestop system is equal to the hourly fire rating of the wall assembly in which it is installed.

2. Cables - Aggregate cross-sectional area of cables in steel sleeve to be max 48 percent of the aggregate cross-sectional area of the sleeve. Cables to be rigidly supported on both sides of wall assembly. Any combination of the following types and sizes of copper conductor cable may be used:

   A. Max 400 pair No. AWG (or smaller) copper conductor cable with polyvinyl chloride (PVC) jacketing and insulation.

   B. Max 3/C No. 2/0 AWG (or smaller) aluminum or copper conductor service entrance cable with PVC insulation and jacket.

   C. Max 3/C No. 8 AWG (or smaller) nonmetallic sheathed (Romex) cable with copper conductors, PVC insulation and jacket.

   D. Max 7/C No. 2/0 AWG (or smaller) multiconductor power and control cables with XLPE or PVC insulation and XLPE or PVC jacket.

   E. Nom 5/8 in. (16 mm) (or smaller) coaxial cable with fluorinated ethylene insulation and jacketing.

   F. Nom 5/8 in. (16 mm) fiber optic cable with PVC insulation and jacketing.

   G. Max 4 pair No. 22 AWG (or smaller) copper conductor data cable with PVC insulation and jacket.

   H. Max 4/C No. 2/0 aluminum or copper conductor aluminum or steel Metal-Clad# or Armored-Clad# cable.

   I. Max 1/C-750 kcmil copper conductor cable with XLPE jacket and insulation.

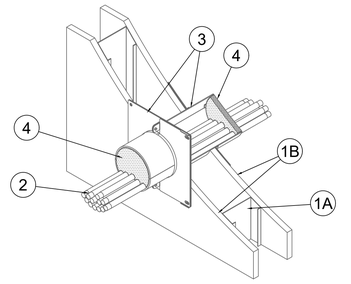
3. Firestop Device * - Device consisting of steel sleeve halves provided with two-piece steel plates. Steel sleeve device shall be installed around cables in accordance with the accompanying installation instructions. Steel sleeve device secured in place by means of two-piece steel plates installed with gasketing material supplied with product. Steel plates and gaskets installed on both sides of wall and secured to device by means of steel screws provided with device. Steel sleeve device provided in nom 1, 2 and 4 in. sizes. Steel plates fastened to gypsum board layers with nom 1/8 in. (3.2 mm) diam steel hollow-wall anchors. Steel plates shall be nom aligned on each side of the wall to cover opening with min 1 in. (25 mm) overlap beyond periphery of through opening. The min and max opening sizes for each sleeve device are shown in the table below:

4. Fill, Void or Cavity Material * - Putty - Fill material supplied with firestop device (Item 3) applied to 1 in. (25 mm) thickness flush with each end of steel sleeve device . At point contact location, min 3/8 in. (9 mm) diam bead of fill material applied at grouped cable/steel sleeve interface on each side of wall.

   SPECIFIED TECHNOLOGIES INC - SpecSeal Putty

   *Bearing the UL Classification Mark

F Ratings - 1 and 2 hr (See Item 1)
T Rating - 1 Hr



Device Size	Min. Opening Size	Max. Opening Size
1 in. (25mm)	1.12 in. (28mm)+	2.50 in. (64 mm)
2 in. (51mm)	2.38 in. (60mm)+	.00 in. (102 mm)
4 in. (102mm)	4.50 in. (114mm)+	6.00 in. (152 mm)

+ - When opening is greater than the min opening size specified, steel fasteners are required.

SPECIFIED TECHNOLOGIES INC -
FSR100, FSR200, FSR400 Ready Split Sleeves

TOLERANCES (UNLESS OTHERWISE SPECIFIED)	SCALE: NTS	DATE: 05/16/06
DECIMAL ±	TITLE: Ready Sleeve Applications	
FRACTIONS ±		
ANGULAR ±		
This information is property of Specified Technologies, Inc. and is not to be disclosed, utilized or copied without written consent of Specified Technologies, Inc.		DWG. NO. W-L-3033

SPECIFIED TECHNOLOGIES, INC.
SOMERVILLE, NJ 08876

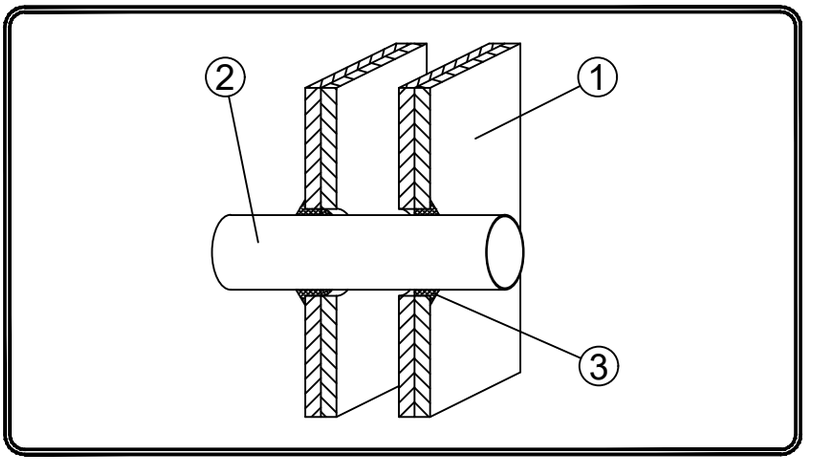
1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

DRAWN BY: ADM
REVISED BY: ADM 2/10/14
CHECKED BY: MS
DATE: 7/9/10
DWG. # S-07

TITLE:
TYPICAL STI - READY SLEEVE INSTALLATION SCENARIOS

THE STATE UNIVERSITY OF NEW JERSEY
RUTGERS
OFFICE OF INFORMATION TECHNOLOGY

S-7



1. GYPSUM WALLBOARD WALL - WALL THICKNESS MAY VARY.
2. METALLIC PIPE OR CONDUIT - NOMINAL 4 INCH DIAMETER OR SMALLER SCH. 5 OR HEAVIER STEEL PIPE, EMT OR RMC.
3. NELSON LBS SEALANT - APPLIED TO FILL THE ANNULAR SPACE AROUND PIPE TO A MINIMUM 5/8 INCH DEPTH. APPLY AN ADDITIONAL 3/8" BEAD OF LBS AROUND THE PIPE AT ITS INTERFACE WITH BOTH WALL SURFACES. NOMINAL 1/2 POLYURETHANE BACKER ROD MAY BE USED TO CONTROL DEPTH OF SEALANT.

THE SYSTEM DESCRIBED IN THIS DETAIL HAS TESTED SUCCESSFULLY TO ASTM E814 AND ANSI/UL 1479 AND HAS ACHIEVED F AND T RATINGS AS SHOWN IN THE ILLUSTRATION BOX.

GS NELSON FIRESTOP PRODUCTS
A DIVISION OF O-Z/GEDNEY
A UNIT OF GENERAL SIGNAL
P.O. BOX 726 TULSA OK 74101

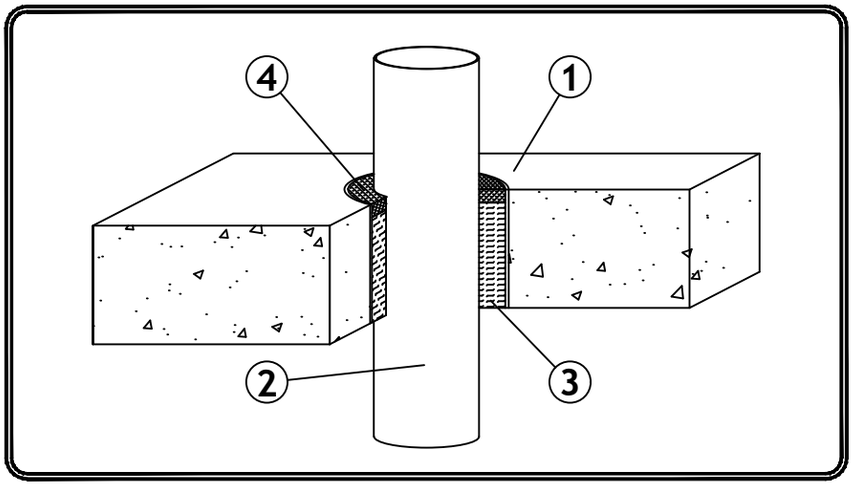
1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

DRAWN BY: SAO
CHECKED BY: JF
DATE: 3/26/02
DWG. # ED-14A

TITLE: TYPICAL FIRESTOPPING DETAIL
GYPSUM WALL SECTIONS

THE STATE UNIVERSITY OF NEW JERSEY
RUTGERS
OFFICE OF INFORMATION TECHNOLOGY

ED-14A



1. CONCRETE FLOOR OR WALL ASSEMBLY - MINIMUM 4-1/2 INCH THICK LIGHTWEIGHT OR NORMAL WEIGHT CONCRETE FLOOR OR WALL OR CMU BLOCK WALL. ANNULAR SPACE MAY RANGE FROM 1/2 INCH TO 2-1/4 INCHES.

2. METALLIC PIPE OR CONDUIT - NOMINAL 4 INCH DIAMETER OR SMALLER EMT.

3. FORMING MATERIAL - MINIMUM 4 INCH THICKNESS OF MINIMUM 4 PCF MINERAL WOOL OR CERAMIC FIBER TIGHTLY PACKED INTO OPENING AND RECESSED 1/2 INCH FROM TOP SURFACE OF FLOOR OR BOTH WALL SURFACES.

4. NELSON LBS SEALANT - APPLY LBS OVER THE FORMING MATERIAL TO A MINIMUM 1/2 INCH DEPTH, FLUSH WITH TOP SURFACE OF FLOOR OR BOTH WALL SURFACES.

THE SYSTEM DESCRIBED IN THIS DETAIL HAS TESTED SUCCESSFULLY TO ASTM E814 AND ANSI/UL 1479 AND HAS ACHIEVED F AND T RATINGS AS SHOWN IN THE ILLUSTRATION BOX.

GS NELSON FIRESTOP PRODUCTS
A DIVISION OF O-Z/GEDNEY
A UNIT OF GENERAL SIGNAL
P.O. BOX 726 TULSA OK 74101

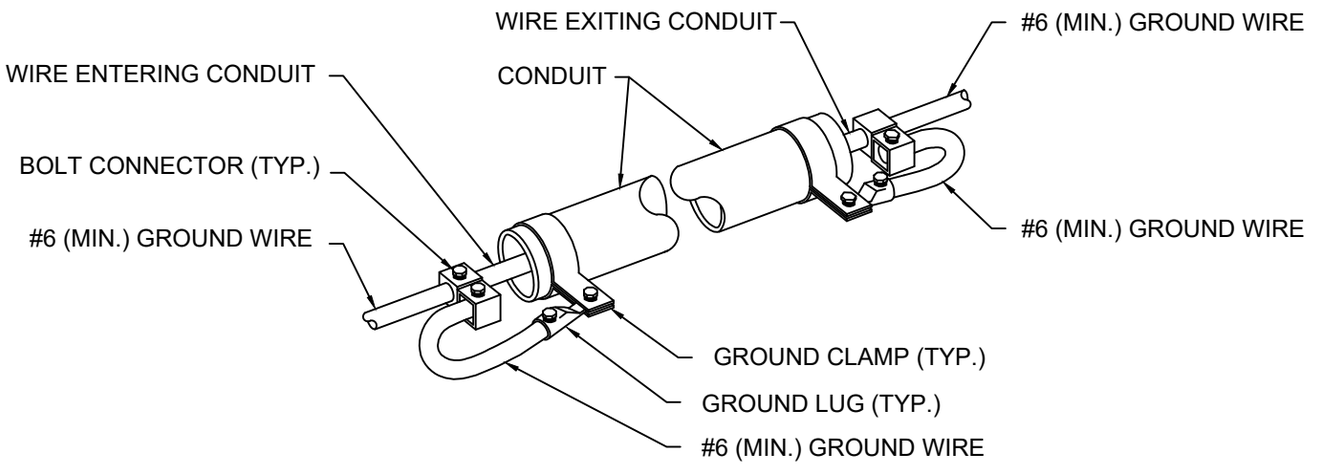
1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

DRAWN BY: SAO
CHECKED BY: JF
DATE: 3/26/02
DWG. # ED-14B

TITLE: TYPICAL FIRESTOPPING DETAIL CONCRETE OR WALL SECTIONS

THE STATE UNIVERSITY OF NEW JERSEY RUTGERS
OFFICE OF INFORMATION TECHNOLOGY

ED-14B



1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

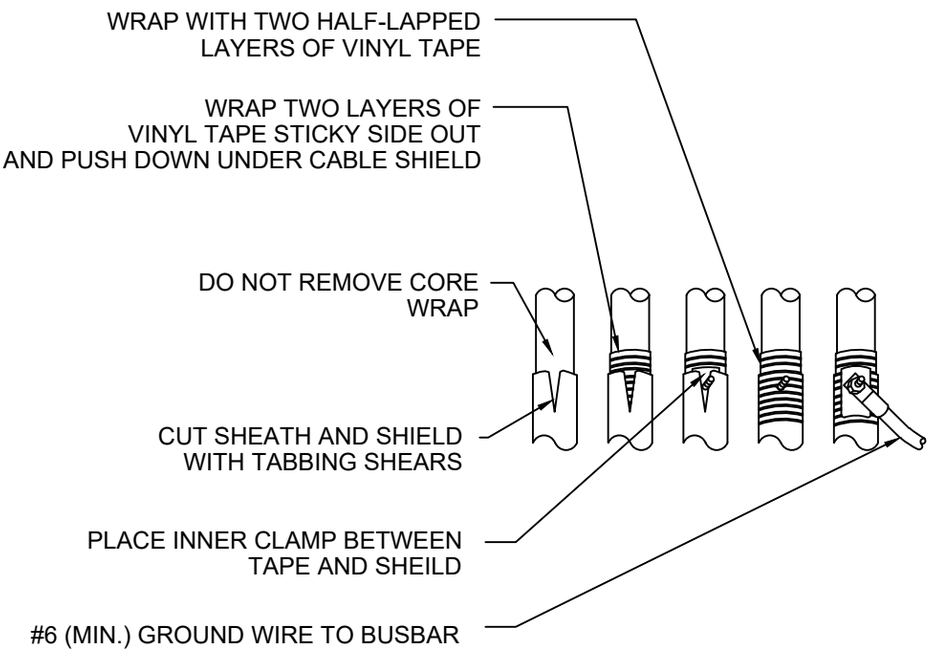
DRAWN BY: SAO
CHECKED BY: JF
DATE: 3/26/02
DWG. # ED-15A

TITLE: TYPICAL BONDING DETAIL FOR GROUND WIRE IN CONDUIT

THE STATE UNIVERSITY OF NEW JERSEY
RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

ED-15A



READY TO:
- TERMINATE PAIRS
- BOND SHIELD

PLACE:
- OUTER CLAMP
- FLAT WASHER
- LOCK WASHER
- NUT

TIGHTEN SECURELY

1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

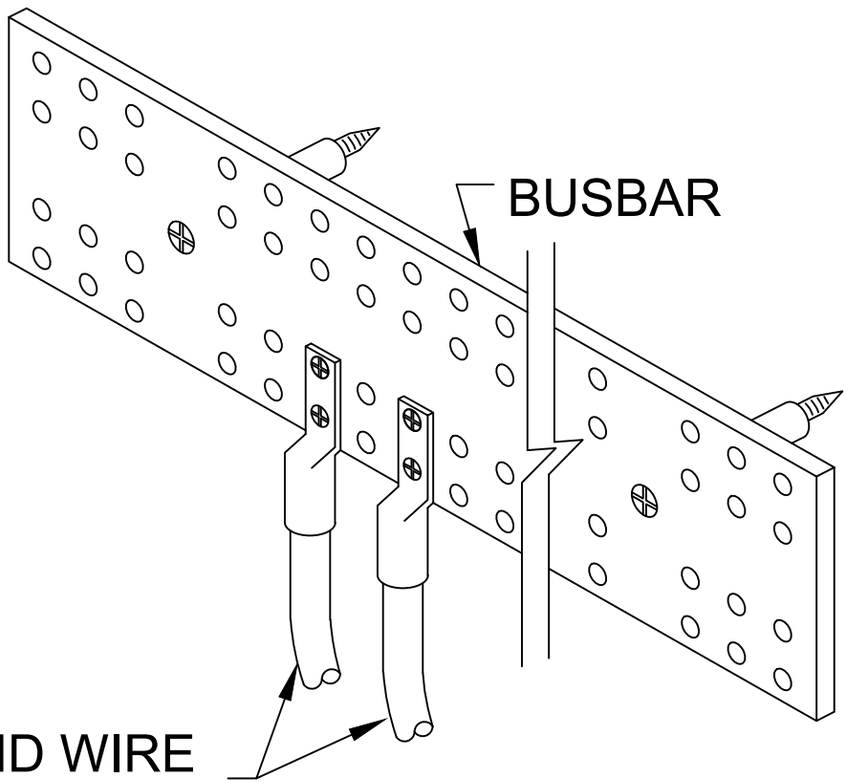
DRAWN BY: SAO
CHECKED BY: JF
DATE: 3/26/02
DWG. # ED-15B

TITLE: BACKBONE CABLE SHIELDS DETAIL

THE STATE UNIVERSITY OF NEW JERSEY RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

ED-15B



#6 (MIN.) GROUND WIRE OR #10 COUPLED BONDING CONDUCTOR (CBC), (TYP.)

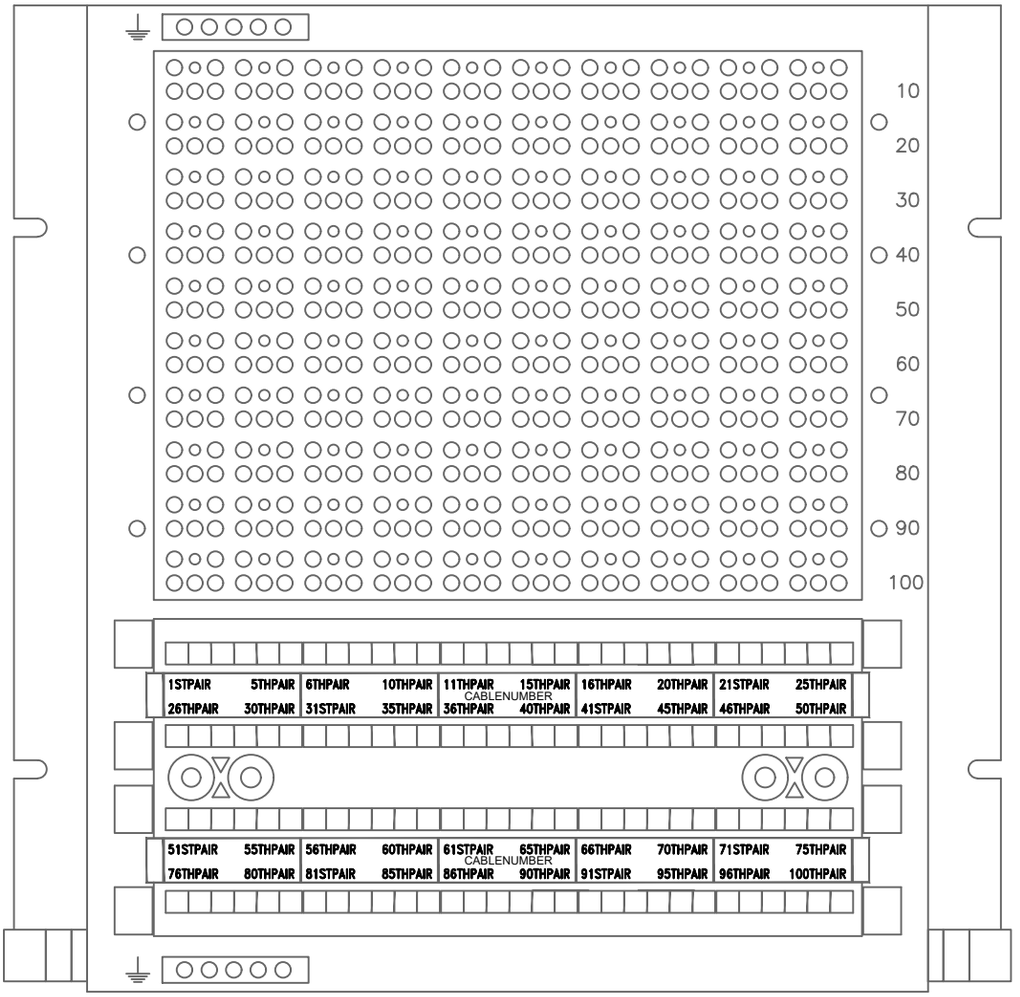
1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

DRAWN BY: SAO
CHECKED BY: JF
DATE: 3/26/02
DWG. # ED-15C

TITLE: GROUNDING AND BONDING
TMGB AND TGB BUSBAR DETAIL

THE STATE UNIVERSITY OF NEW JERSEY
RUTGERS
OFFICE OF INFORMATION TECHNOLOGY

ED-15C



1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

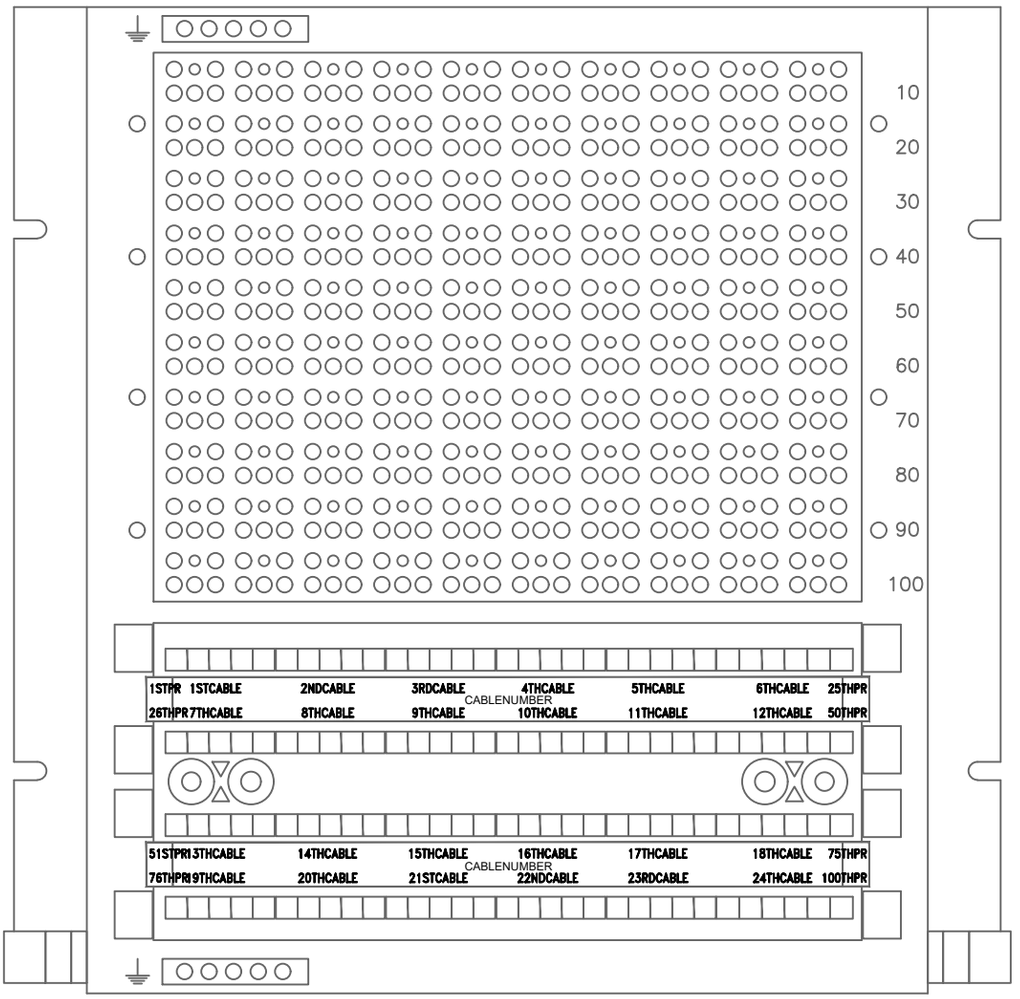
DRAWN BY: SAO
CHECKED BY: JF
DATE: 4/10/02
DWG. # ED-16A

TITLE: ELEVATION HARDWARE DETAIL
CIRCA 1880B1-100 TIE

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ED-16A



1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

DRAWN BY: SAO
CHECKED BY: JF
DATE: 4/10/02
DWG. # ED-16B

TITLE: ELEVATION HARDWARE DETAIL
CIRCA 1880B1-100 TIE/RISER

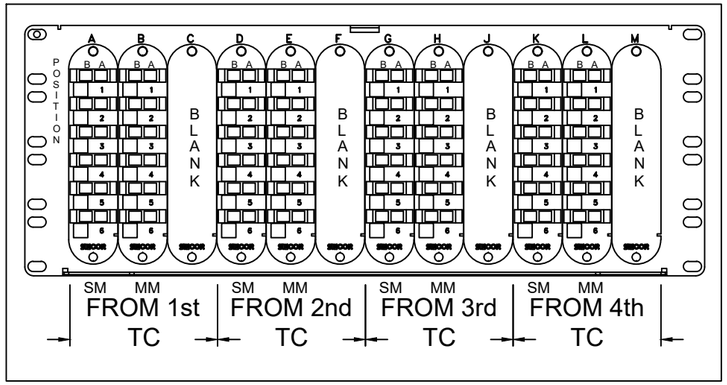
THE STATE UNIVERSITY OF NEW JERSEY
RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

ED-16B

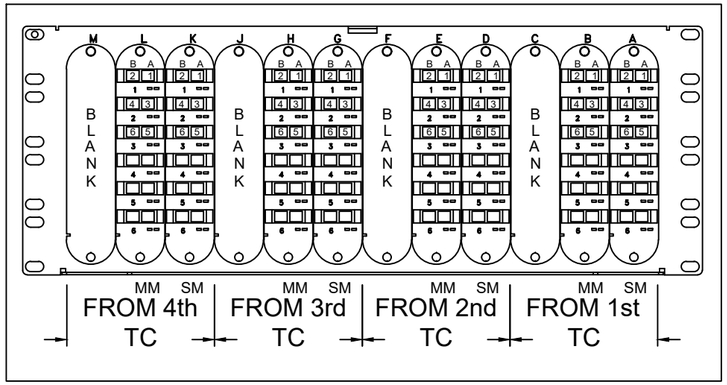
STRAND COLOR	
1	BLUE
2	ORANGE
3	GREEN
4	BROWN
5	SLATE
6	WHITE

EQUIPMENT ROOM



FRONT CONNECTORS (FRONT VIEW)

EQUIPMENT ROOM



REAR TERMINATIONS (REAR VIEW)

1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

DRAWN BY: SAO
CHECKED BY: PR
DATE: 4/03/02
DWG. # FTL-1

TITLE: TYPICAL ISP BACKBONE FIBER TERMINATION LAYOUT - ER's

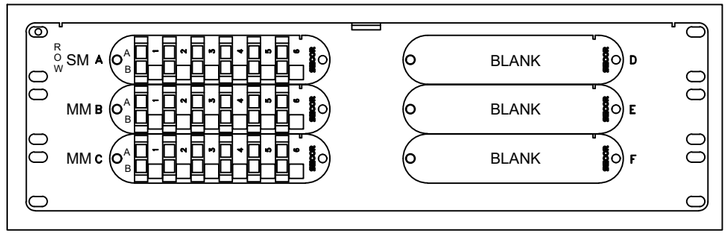
THE STATE UNIVERSITY OF NEW JERSEY RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

FTL-1

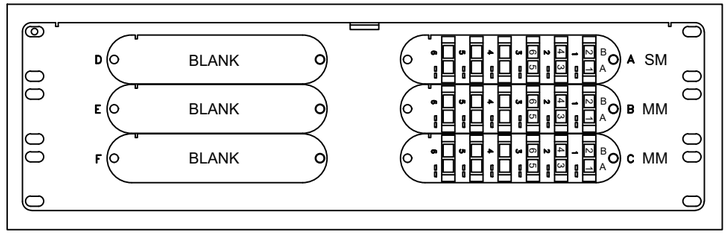
STRAND COLOR	
1	BLUE
2	ORANGE
3	GREEN
4	BROWN
5	SLATE
6	WHITE

TELECOMMUNICATIONS ROOM



FRONT CONNECTORS (FRONT VIEW)

TELECOMMUNICATIONS ROOM



REAR TERMINATIONS (REAR VIEW)

1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

DRAWN BY: SAO
CHECKED BY: PR
DATE: 4/03/02
DWG. # FTL-2

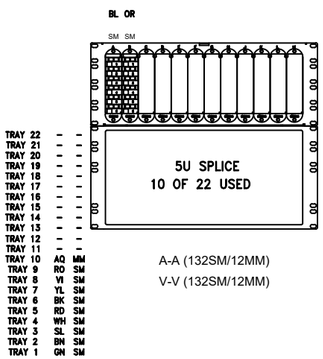
TITLE: TYPICAL ISP BACKBONE FIBER TERMINATION LAYOUT - TR's

THE STATE UNIVERSITY OF NEW JERSEY RUTGERS

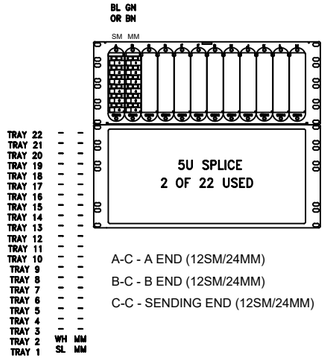
OFFICE OF INFORMATION TECHNOLOGY

FTL-2

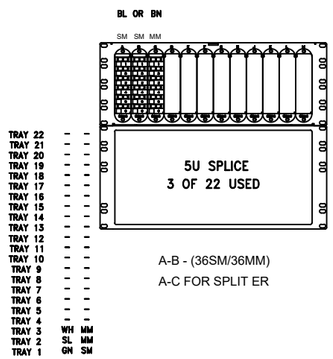
SM / MM



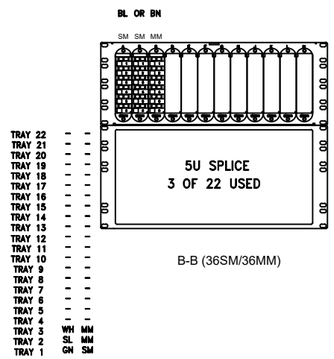
A-A (132SM/12MM)
V-V (132SM/12MM)



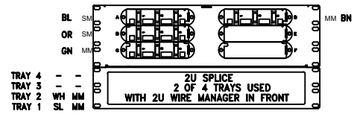
A-C - A END (12SM/24MM)
B-C - B END (12SM/24MM)
C-C - SENDING END (12SM/24MM)



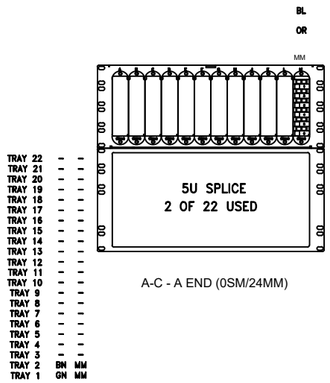
A-B - (36SM/36MM)
A-C FOR SPLIT ER



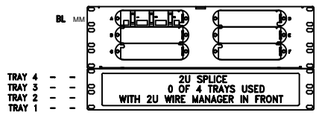
B-B (36SM/36MM)



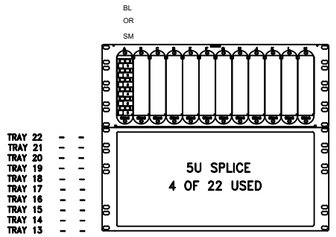
A-C - C END (12SM/24MM)
B-C - C END (12SM/24MM)
C-C - RECEIVING END (12SM/24MM)



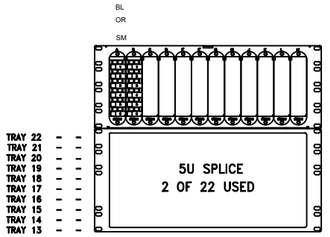
A-C - A END (0SM/24MM)



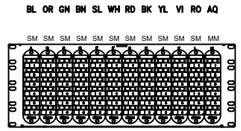
A-C - C END (0SM/6MM)
B-C - C END (0SM/6MM)
C-C - RECEIVING END (0SM/6MM)



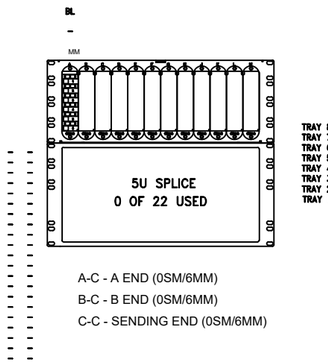
A-B - (36SM/0MM)



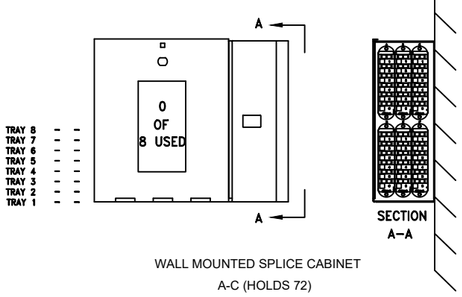
A-B (24SM/0MM)



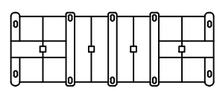
V-V (132SM/12MM) AT DAVIDSON
A-A (132SM/12MM) AT DAVIDSON



A-C - A END (0SM/6MM)
B-C - B END (0SM/6MM)
C-C - SENDING END (0SM/6MM)



WALL MOUNTED SPLICE CABINET
A-C (HOLDS 72)
B-C (HOLDS 72)
C-C (HOLDS 72)



FIBEROPTIC SPLICE CASE
B-C (0SM/6MM)

TUBE BREAKDOWN (SINGLE MODE LISTED FIRST)

6SM/6MM	2 TUBES OF 6 (BL/OR)
6SM/12MM	2 TUBES, 1 W/6, 1 W/12 (BL/OR)
12SM/12MM	2 TUBES OF 12 (BL/OR)
12SM/24MM	3 TUBES OF 6 (BL/OR/GN)
36SM/36MM	6 TUBES OF 12 (BL/OR/GN/BN/SL/WH)
12SM/132MM	12 TUBES OF 12 (BL/OR/GN/BN/SL/WH/RD/BK/YL/VI/RO/AQ)
0SM/6MM	1 TUBE OF 6 (BL)
0SM/12MM	2 TUBES OF 6 (BL/OR)
0SM/24MM	4 TUBES OF 6 (BL/OR/GN/BN)
0SM/36MM	3 TUBES OF 12 (BL/OR/GN)
12SM/0MM	2 TUBES OF 6 (BL/OR)
36SM/0MM	3 TUBES OF 12 (BL/OR/GN)

1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

DRAWN BY: S.A.O.
REVISED BY: ADM 2/10/14
CHECKED BY: MS
DATE: 4/03/02
DWG. # FTL-03

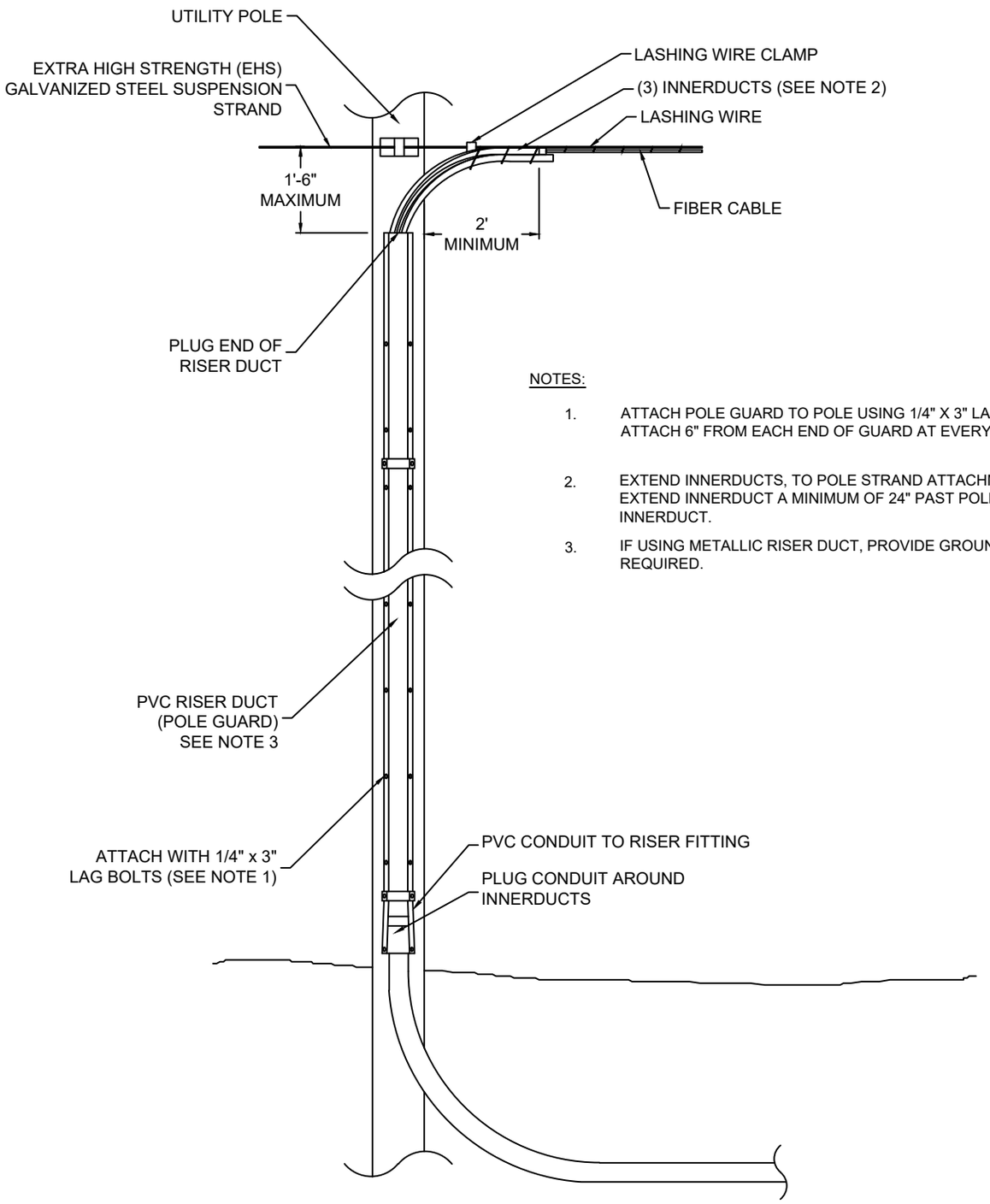
TITLE:

TYPICAL OSP FIBER TERMINATION LAYOUTS

THE STATE UNIVERSITY OF NEW JERSEY
RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

FTL-3



NOTES:

1. ATTACH POLE GUARD TO POLE USING 1/4" X 3" LAG BOLTS. ATTACH 6" FROM EACH END OF GUARD AT EVERY 18".
2. EXTEND INNERDUCTS, TO POLE STRAND ATTACHMENT POINT. EXTEND INNERDUCT A MINIMUM OF 24" PAST POLE. PLUG INNERDUCT.
3. IF USING METALLIC RISER DUCT, PROVIDE GROUND WIRE, AS REQUIRED.

1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

DRAWN BY: SAO
CHECKED BY: PR
DATE: 5/09/02
DWG. # RP-01

TITLE: TYPICAL RISER POLE GUARD INSTALLATION DETAIL

THE STATE UNIVERSITY OF NEW JERSEY RUTGERS

OFFICE OF INFORMATION TECHNOLOGY

RP-1

THE STATE UNIVERSITY OF NEW JERSEY
RUTGERS

Building Name
Telecommunications Cable Plant Drawings
(Building #)
Name of Campus

NOTES:

NOTE TO BIDDERS:
AS-BUILTS TO BE HAND DRAWN MARK-UPS OF SHOP DRAWINGS.

CONTRACTORS NAME ADDRESS AND TELEPHONE NUMBER

ISSUE	BY	CHK	DATE	DESCRIPTION
2	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS
1	M.P.	GT	4/10/02	RESPONSE TO PATP - ###

APPROVED FOR RUTGERS–THE STATE UNIVERSITY

THE STATE UNIVERSITY OF NEW JERSEY RUTGERS — OFFICE OF INFORMATION TECHNOLOGY

PROJECT
NAME OF BUILDING – NAME OF CAMPUS
ADDRESS OF BUILDING

DRAWING TITLE
COVER SHEET
TELECOMMUNICATIONS PLAN

DSR NO.	SCALE	DRAWN BY	FILE NO.	DRAWING NO.
BLDG. NO. 8500	X/XX"=1'-0"	M	SD8500-T01	T0-1
	DATE 4/10/02	CHECKED BY XX		

SYMBOLS LIST

CONDUIT AND/OR RACEWAY UP/DN

▲ WP#
OUTLET TERMINATED IN CEILING

▲ WP#
OUTLET TERMINATED IN FLOOR POKE-THRU

DETAIL LABEL
X = DETAIL LABEL
Y-Y = DRAWING NUMBER

ABBREVIATIONS LIST

AFF	=	ABOVE FINISHED FLOOR
C	=	CONDUIT
CORR	=	CORRIDOR
CAT	=	CATEGORY
EF	=	ENTRANCE FACILITY
ER	=	EQUIPMENT ROOM
F.O.	=	FIBER OPTIC
PR	=	PAIR
RM	=	ROOM
TR	=	TELECOMMUNICATIONS ROOM
TEL	=	TELEPHONE
TV	=	TELEVISION
RU	=	RUTGERS UNIVERSITY
RUCS	=	RUTGERS UNIVERSITY COMPUTING SERVICES
UTP	=	UNSHIELDED TWISTED PAIR
WP	=	WALL PLATE/WIRING PLAN
TYP	=	TYPICAL

DRAWINGS LIST

T0-1	NAME OF BUILDING – COVER SHEET TELECOMMUNICATIONS PLAN
T0-2	NAME OF BUILDING – ABBREVIATIONS & SYMBOLS LIST TELECOMMUNICATIONS PLAN
T1-B	NAME OF BUILDING – BASEMENT FLOOR TELECOMMUNICATIONS PLAN
T1-1	NAME OF BUILDING – FIRST FLOOR TELECOMMUNICATIONS PLAN
T1-2	NAME OF BUILDING – SECOND FLOOR TELECOMMUNICATIONS PLAN
T1-3	NAME OF BUILDING – THIRD FLOOR TELECOMMUNICATIONS PLAN
T2-1	NAME OF BUILDING – TELECOMMUNICATIONS RISER DIAGRAM TELECOMMUNICATIONS PLAN
T2-2	NAME OF BUILDING – GROUNDING AND BONDING RISER AND DETAILS TELECOMMUNICATIONS PLAN
T2-3	NAME OF BUILDING – EQUIPMENT ROOM DETAILS & ELEVATIONS TELECOMMUNICATIONS PLAN
T2-4	NAME OF BUILDING – TELECOMMUNICATIONS ROOM DETAILS & ELEVATIONS TELECOMMUNICATIONS PLAN
T2-5	NAME OF BUILDING – TYPICAL 110 BLOCK TERMINATIONS TELECOMMUNICATIONS PLAN
T2-6	NAME OF BUILDING – TYPICAL CABLE HARDWARE LABELING DETAILS TELECOMMUNICATIONS PLAN
T2-7	NAME OF BUILDING – LABELING DETAILS TELECOMMUNICATIONS PLAN

WP DESCRIPTION (ALL DATA: MINIMUM CAT6 RATING)

◀ WP1 WIRING PLAN 1:
- VOICE: 1 x 4-UTP CATEGORY 5e PLENUM CABLE TERMINATED ON A RJ45 JACK.
- DATA: 1 x 4-UTP CATEGORY 5e PLENUM CABLE TERMINATED ON A RJ45 JACK.

◀ WP2 WIRING PLAN 2:
- VOICE: 1 x 4-UTP CATEGORY 5e PLENUM CABLE TERMINATED ON A RJ45 JACK.
- DATA: 1 x 4-UTP CATEGORY 5e PLENUM CABLE TERMINATED ON A RJ45 JACK.
- FTV: 1 x RG6 QUAD SHIELD RISER RATED CABLE TERMINATED ON A F TYPE COUPLER.

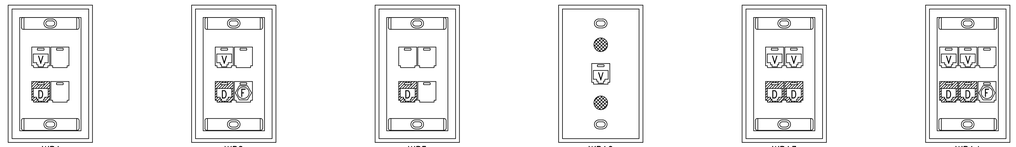
◀ WP5 WIRING PLAN 5:
- DATA: 1 x 4-UTP CATEGORY 5e PLENUM CABLE TERMINATED ON A RJ45 JACK.

◀ WP10 WIRING PLAN 10: WALL PHONE OUTLET, MOUNTED AT 48" AFF
- VOICE: 1 x 4-UTP CATEGORY 5e PLENUM CABLE TERMINATED ON A RJ45 JACK.

◀ WP13 WIRING PLAN 13:
- VOICE: 2 x 4-UTP CATEGORY 5e PLENUM CABLES TERMINATED ON TWO RJ45 JACKS.
- DATA: 2 x 4-UTP CATEGORY 5e PLENUM CABLES TERMINATED ON TWO RJ45 JACKS.

◀ WP14 WIRING PLAN 14:
- VOICE: 2 x 4-UTP CATEGORY 5e PLENUM CABLES TERMINATED ON TWO RJ45 JACKS.
- DATA: 2 x 4-UTP CATEGORY 5e PLENUM CABLES TERMINATED ON TWO RJ45 JACKS.
- FTV: 1 x RG6 QUAD SHIELD RISER RATED CABLE TERMINATED ON A F TYPE COUPLER.

WP SYMBOLS



WP1 WP2 WP5 WP10 (STAINLESS STEEL FACE PLATE WITH WALL PHONE SUPPORTS) WP13 WP14

NOTES:

NOTE TO BIDDERS: AS-BUILTS TO BE HAND DRAWN MARK-UPS OF SHOP DRAWINGS.

CONTRACTORS NAME ADDRESS AND TELEPHONE NUMBER

ISSUE	BY	CHK	DATE	DESCRIPTION
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1	M.P.	GT	4/10/02	RESPONSE TO PATP - ###

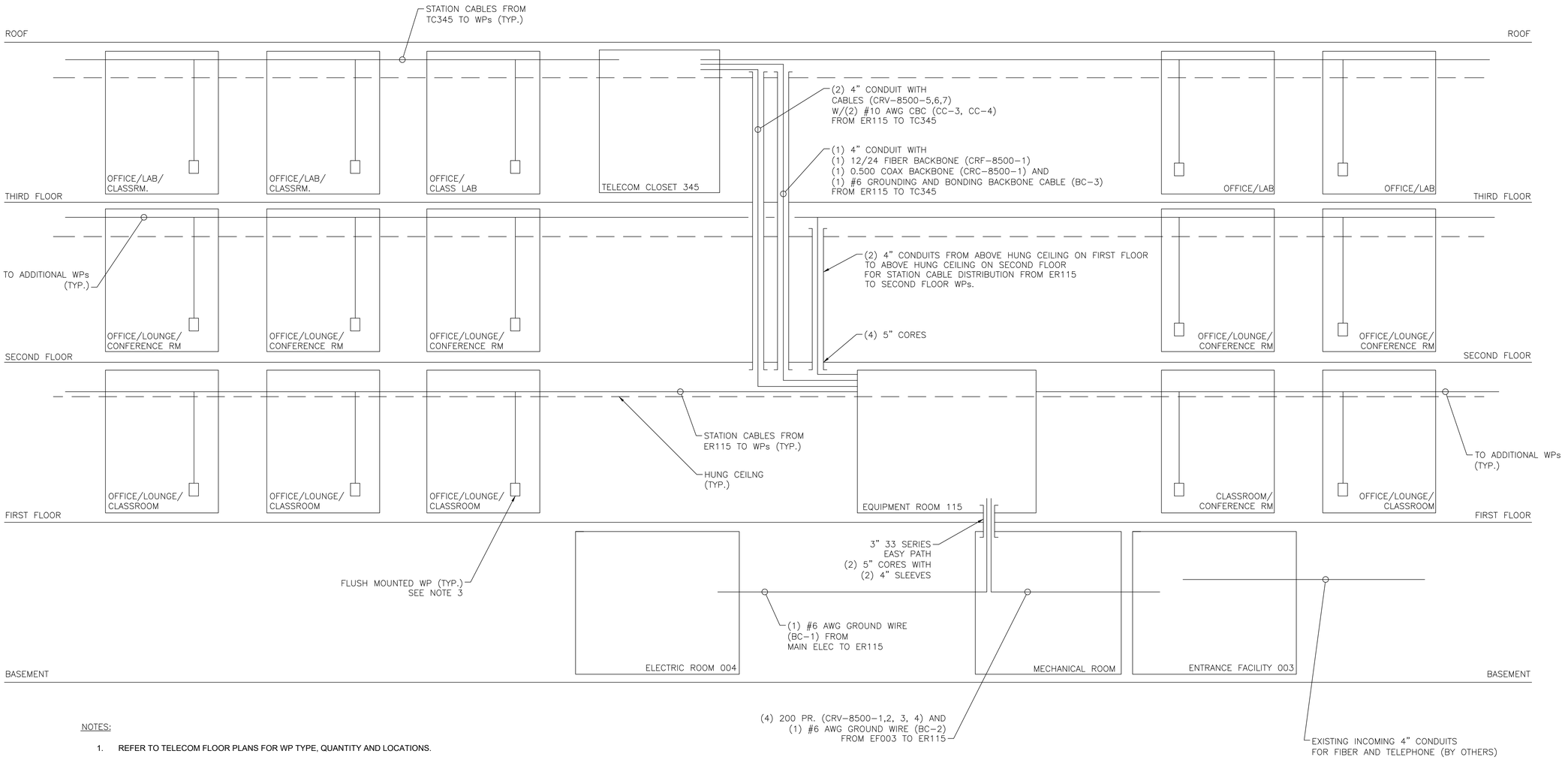
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PROJECT
NAME OF BUILDING – NAME OF CAMPUS
ADDRESS OF BUILDING

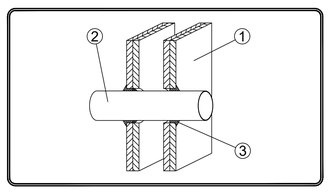
DRAWING TITLE
ABBREVIATIONS & SYMBOLS LIST
TELECOMMUNICATIONS PLAN

DSR NO.	SCALE	DRAWN BY	FILE NO.	DRAWING NO.
BLDG. NO. 8500	X/XX"-X"-X" DATE 4/10/02	XX CHECKED BY XX	SD8500-T02	T0-2



NOTES:

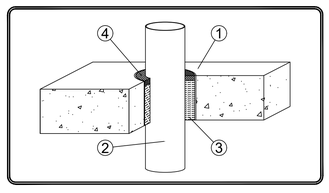
1. REFER TO TELECOM FLOOR PLANS FOR WP TYPE, QUANTITY AND LOCATIONS.
2. REFER TO TELECOM FLOOR PLANS FOR STATION AND BACKBONE CABLE ROUTES.
3. ALL PENETRATIONS, EXCEPT FOR CONDUITS, SHALL REQUIRE A SLEEVES. PROVIDE FIRESTOPPING FOR ALL CONDUIT AND SLEEVE PENETRATIONS.



1. Gypsum Wallboard Wall – Wall thickness may vary.
2. Metallic Pipe or Conduit – Nominal 4 inch diameter or smaller Sch. 5 or heavier steel pipe, EMT or RMC.
3. Nelson LBS Sealant – Applied to fill the annular space around pipe to a minimum 5/8 inch depth. Apply an additional 3/8 inch bead of LBS around the pipe at its interface with both wall surfaces. Nominal 1/2 polyurethane backer rod may be used to control depth of sealant.

The system described in this detail has tested successfully to ASTM E814 and ANSI/UL 1479 and has achieved F and T ratings as shown in the illustration box.

NELSON FIRESTOP PRODUCTS
A DIVISION OF O-Z/GEDNEY
A UNIT OF GENERAL SIGNAL
P.O. BOX 726 TULSA OK 74101



1. Concrete Floor or Wall Assembly – Minimum 4-1/2 inch thick lightweight or normal weight concrete floor or wall or CMU block wall. Annular space may range from 1/2 inch to 2-1/4 inches.
2. Metallic Pipe or Conduit – Nominal 4 inch diameter or smaller EMT.
3. Forming Material – Minimum 4 inch thickness of minimum 4 pcf mineral wool or ceramic fiber tightly packed into opening and recessed 1/2 inch from top surface of floor or both wall surfaces.
4. Nelson LBS Sealant – Apply LBS over the forming material to a minimum 1/2 inch depth, flush with top surface of floor or both wall surfaces.

The system described in this detail has tested successfully to ASTM E814 and ANSI/UL 1479 and has achieved F and T ratings as shown in the illustration box.

NELSON FIRESTOP PRODUCTS
A DIVISION OF O-Z/GEDNEY
A UNIT OF GENERAL SIGNAL
P.O. BOX 726 TULSA OK 74101

NOTES:

NOTE TO BIDDERS: AS-BUILTS TO BE HAND DRAWN MARK-UPS OF SHOP DRAWINGS.

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ISSUE	BY	CHK	DATE	DESCRIPTION
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1	M.P.	P.R.	4/10/02	RESPONSE TO PATP - 234A

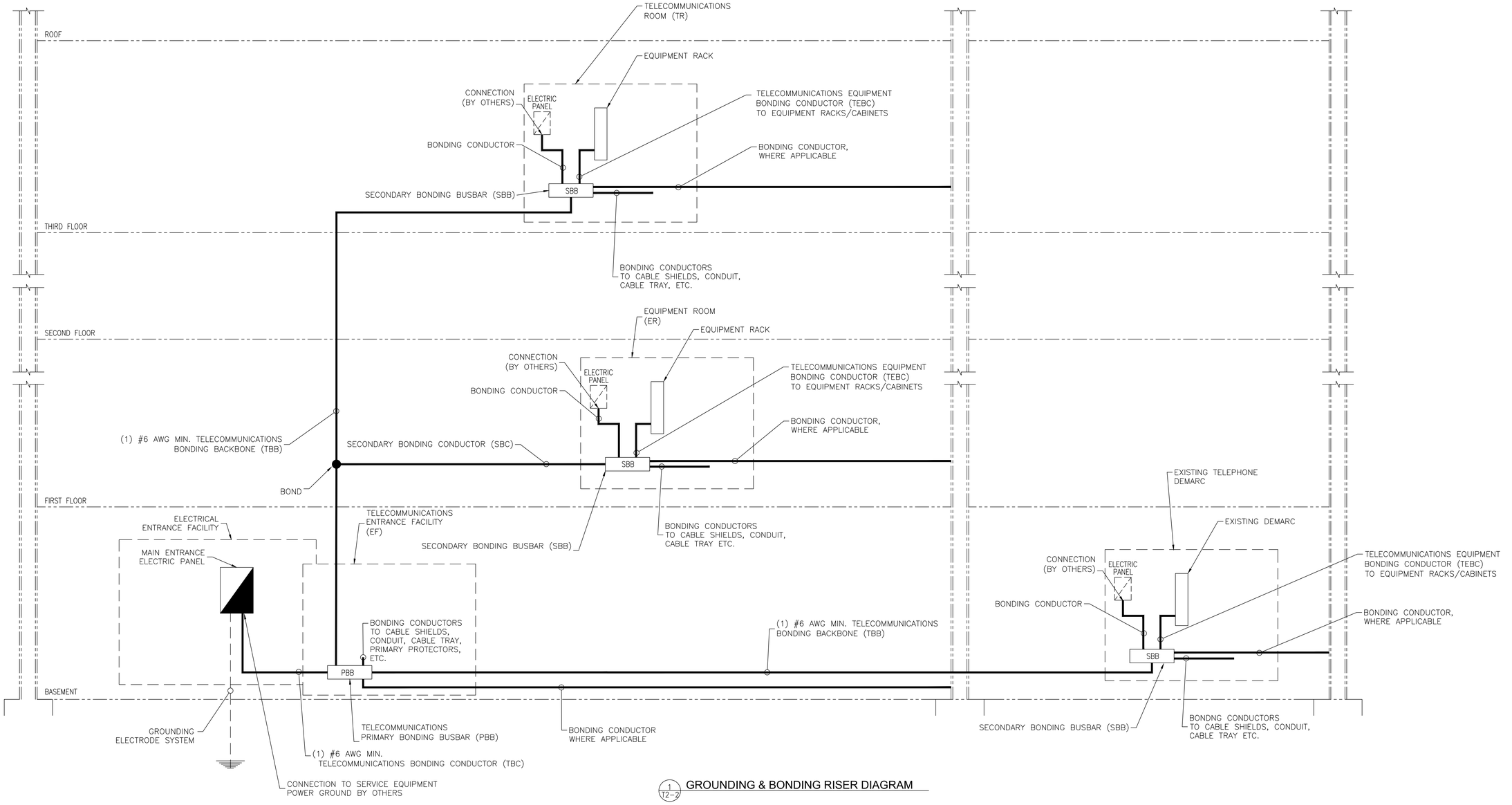
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PROJECT
NAME OF BUILDING – NAME OF CAMPUS
ADDRESS OF BUILDING

DRAWING TITLE
TELECOMMUNICATIONS RISER DIAGRAM
TELECOMMUNICATIONS PLAN

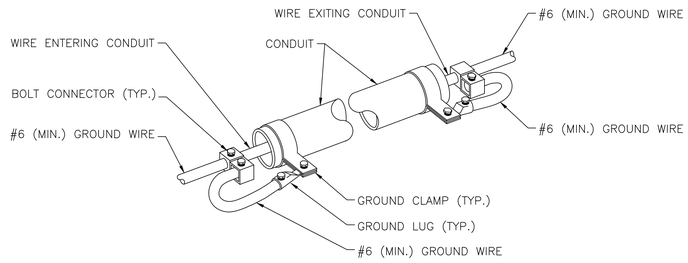
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BLDG. NO. 8500	X/0"-X"-X" DATE 4/10/02	DH CHECKED BY XX	SD8500-T21	T2-1



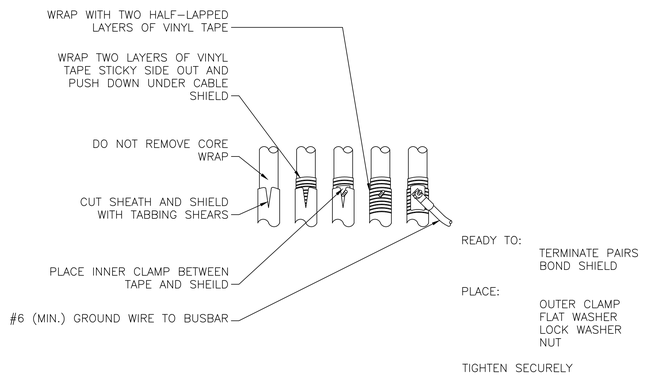
1 T2-2 GROUNDING & BONDING RISER DIAGRAM

NOTES:

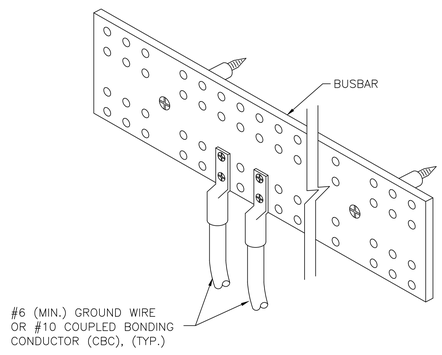
1. ALL CABLES AND CONNECTIONS BY TELECOMMUNICATIONS CONTRACTOR EXCEPT WHERE INDICATED OTHERWISE.
2. ALL CABLES SHALL BE #6 AWG. OR LARGER.



2 T2-2 TYPICAL BONDING FOR GROUND WIRE IN CONDUIT



3 T2-2 BACKBONE CABLE SHIELDS



4 T2-2 PBB & SBB BUSBAR DETAIL

NOTES:

NOTE TO BIDDERS: AS-BUILTS TO BE HAND DRAWN MARK-UPS OF SHOP DRAWINGS.

CONTRACTORS NAME ADDRESS AND TELEPHONE NUMBER

ISSUE	BY	CHK	DATE	DESCRIPTION
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1	M.P.	GT	4/10/02	RESPONSE TO PATP - ###

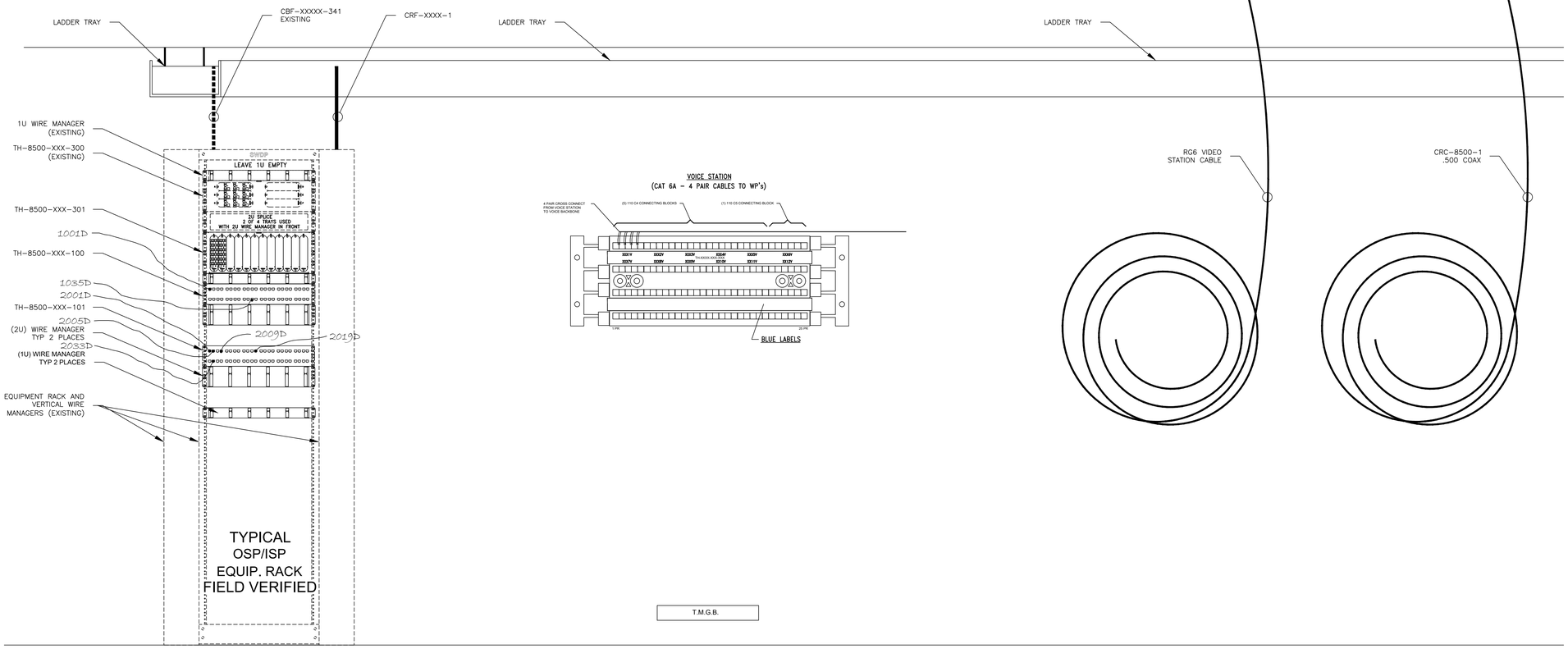
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PROJECT

DRAWING TITLE: TYPICAL TELECOMMUNICATIONS GROUNDING AND BONDING DETAILS

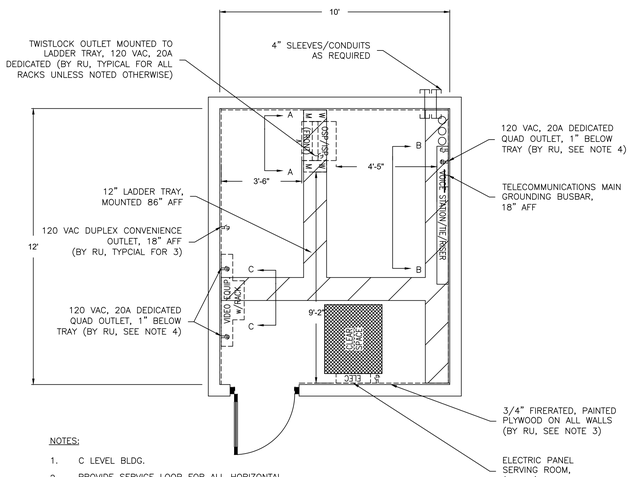
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BLDG. NO.	DATE: 4/10/02	CHECKED BY: XX		



VIEW A-A
N.T.S.

VIEW B-B
N.T.S.

VIEW C-C
N.T.S.



NOTES:

1. C LEVEL BLDG.
2. PROVIDE SERVICE LOOP FOR ALL HORIZONTAL VOICE, DATA, AND VIDEO CABLES NOT TO EXCEED 10 FEET. LOCATION TO BE DETERMINED BY THE RUPM.
3. 3/4" AC FIRERATED PLYWOOD ON ALL WALLS, PAINTED WITH WHITE FIRE RETARDANT PAINT (DO NOT PAINT PLYWOOD LABEL). MOUNT PLYWOOD WITH STAINLESS STEEL HARDWARE (BY RU).
4. MOUNT QUAD OUTLETS WITH TOP OF BACKBOX AT 1" BELOW LADDER TRAY OR AT 85" AFF, WHICHEVER IS LOWER (BY RU).

TYPICAL LAYOUT - FIELD VERIFIED

1 / T2-3 EQUIPMENT ROOM 115 LAYOUT

SCALE: 3/8" = 1'-0"

NOTES:

NOTE TO BIDDERS: AS-BUILTS TO BE HAND DRAWN MARK-UPS OF SHOP DRAWINGS.

CONTRACTORS NAME ADDRESS AND TELEPHONE NUMBER

ISSUE	BY	CHK	DATE	DESCRIPTION
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1	M.P.	G.T.	4/5/02	RESPONSE TO PATP - ###

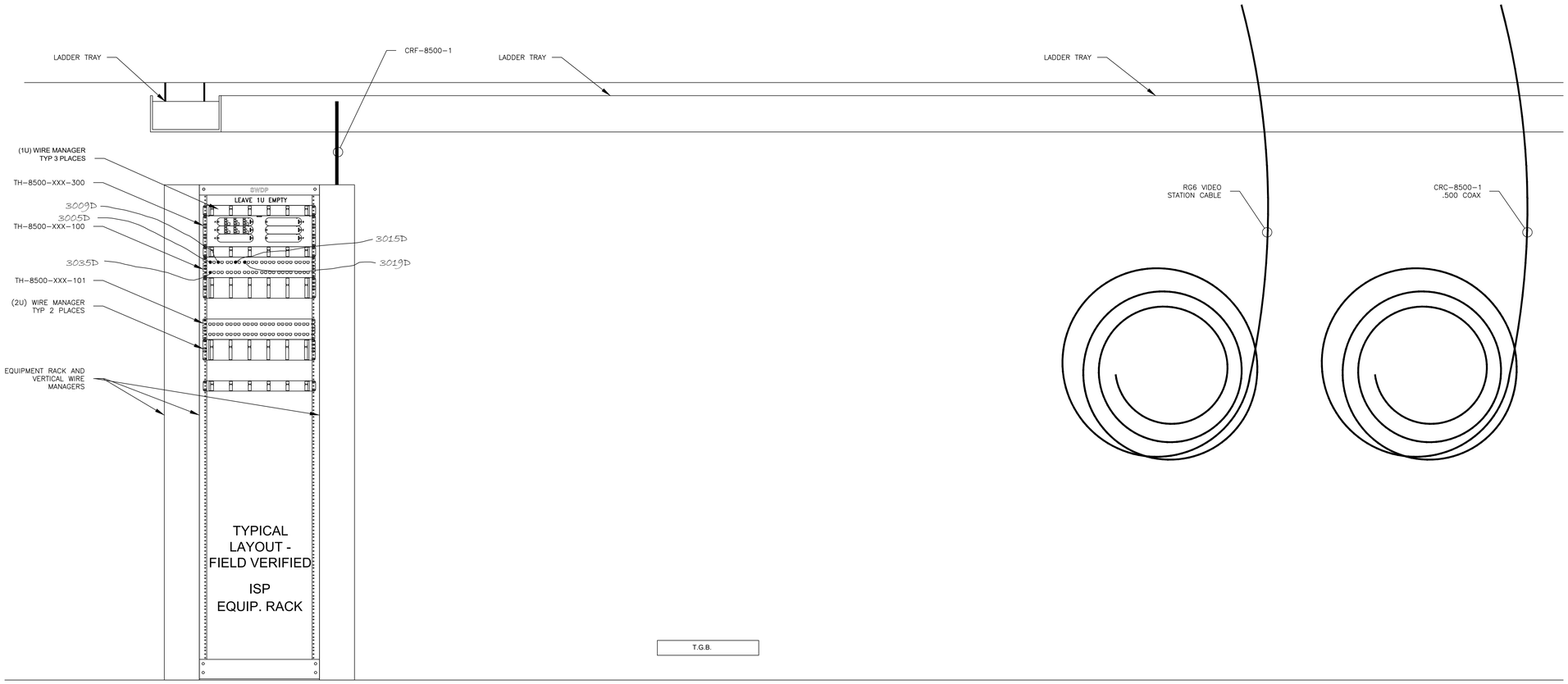
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PROJECT
NAME OF BUILDING - NAME OF CAMPUS
ADDRESS OF BUILDING

DRAWING TITLE
EQUIPMENT ROOM DETAILS & ELEVATIONS
TELECOMMUNICATIONS PLAN

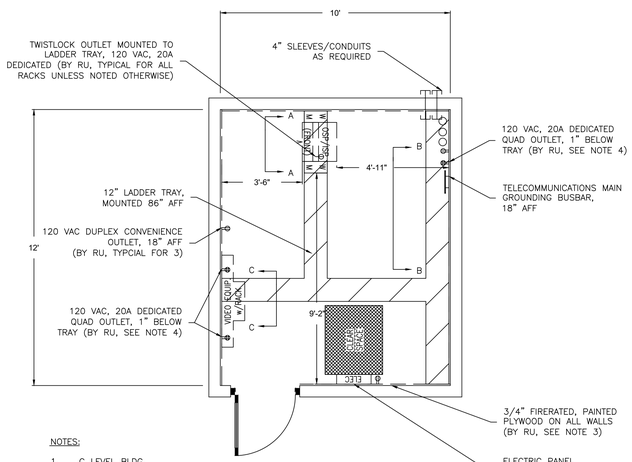
DSR NO.	SCALE 1/XX"=X'-X"	DRAWN BY M	FILE NO. SD8500-T23	DRAWING NO. T2-3
BLDG. NO. 8500	DATE 4/10/02	CHECKED BY XX		



VIEW A-A
N.T.S.

VIEW B-B
N.T.S.

VIEW C-C
N.T.S.



NOTES:

1. C LEVEL BLDG.
2. PROVIDE SERVICE LOOP FOR ALL HORIZONTAL VOICE, DATA, AND VIDEO CABLES NOT TO EXCEED 10 FEET. LOCATION TO BE DETERMINED BY THE RUPM.
3. 3/4" AC FIRERATED PLYWOOD ON ALL WALLS, PAINTED WITH WHITE FIRE RETARDANT PAINT (DO NOT PAINT PLYWOOD LABEL). MOUNT PLYWOOD WITH STAINLESS STEEL HARDWARE (BY RU).
4. MOUNT QUAD OUTLETS WITH TOP OF BACKBOX AT 1" BELOW LADDER TRAY OR AT 85" AFF, WHICHEVER IS LOWER (BY RU).

TYPICAL LAYOUT - FIELD VERIFIED
TELECOMMUNICATIONS ROOM XXX LAYOUT
1 / T2-4
SCALE: 3/8" = 1'-0"

NOTES:

NOTE TO BIDDERS: AS-BUILTS TO BE HAND DRAWN MARK-UPS OF SHOP DRAWINGS.

CONTRACTORS NAME ADDRESS AND TELEPHONE NUMBER

ISSUE	BY	CHK	DATE	DESCRIPTION
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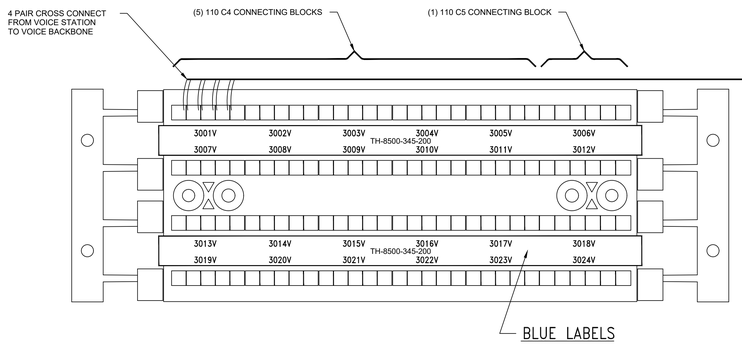
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PROJECT
NAME OF BUILDING - NAME OF CAMPUS
ADDRESS OF BUILDING

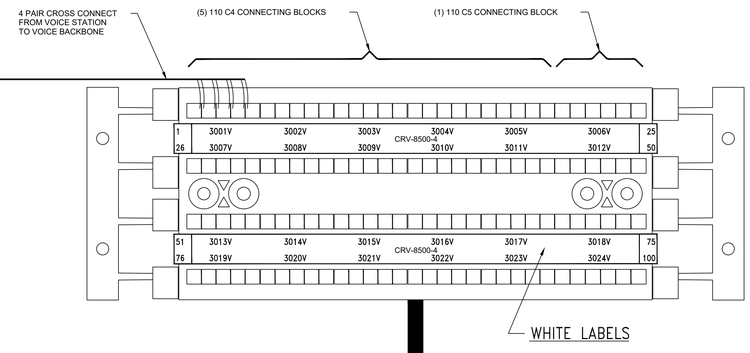
DRAWING TITLE
TELECOMMUNICATIONS ROOM DETAILS & ELEVATIONS
TELECOMMUNICATIONS PLAN

DSR NO.	SCALE	DRAWN BY	FILE NO.	DRAWING NO.
	X/XX"-X'-X"	M	SD8500-T24	T2-4
BLDG. NO. 8500	DATE 4/10/02	CHECKED BY XX		

VOICE STATION
(CAT 6 – 4 PAIR CABLES TO WP's)



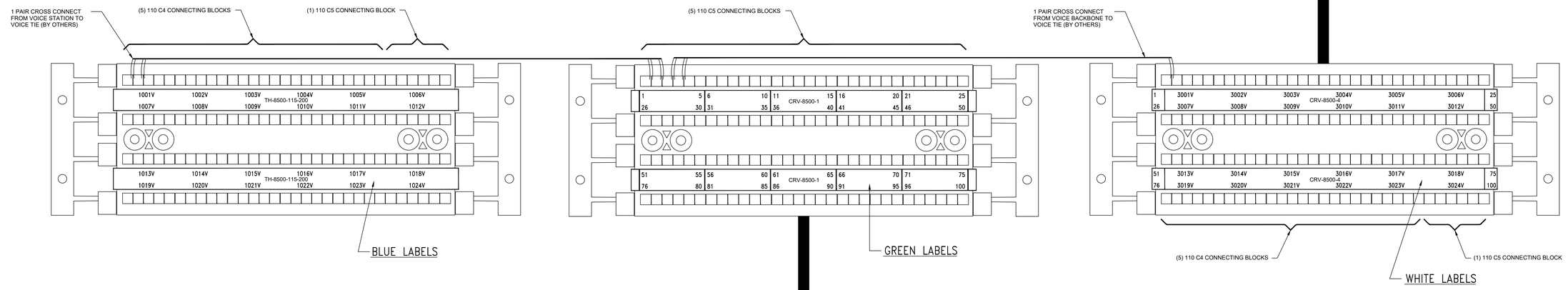
VOICE BACKBONE
(MULTI–PAIR COPPER CABLE – TO E.R.)



T.R.

VOICE BACKBONE CABLE

VOICE TIE
(MULTI–PAIR COPPER CABLE – TO E.F.)



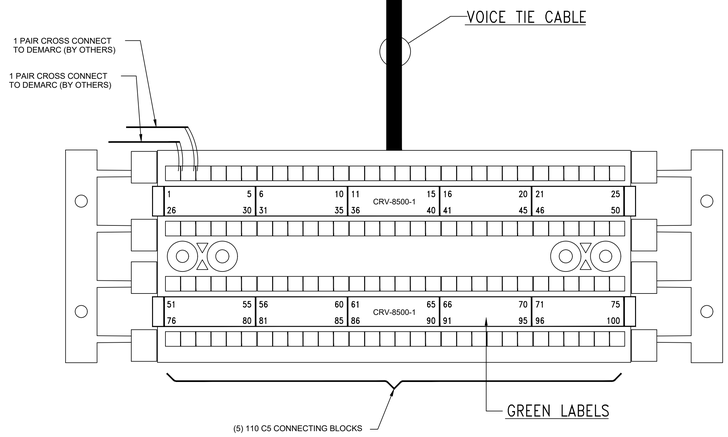
VOICE STATION
(CAT 6 – 4 PAIR CABLES TO WP's)

VOICE BACKBONE
(MULTI–PAIR COPPER CABLE – TO T.R.)

E.R.

E.F.

VOICE TIE CABLE



VOICE TIE
(MULTI–PAIR COPPER CABLE – TO E.R.)

NOTES:

NOTE TO BIDDERS: AS-BUILTS TO BE HAND DRAWN MARK-UPS OF SHOP DRAWINGS.

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1	M.P.	P.R.	4/10/02	RESPONSE TO PATP - 234A

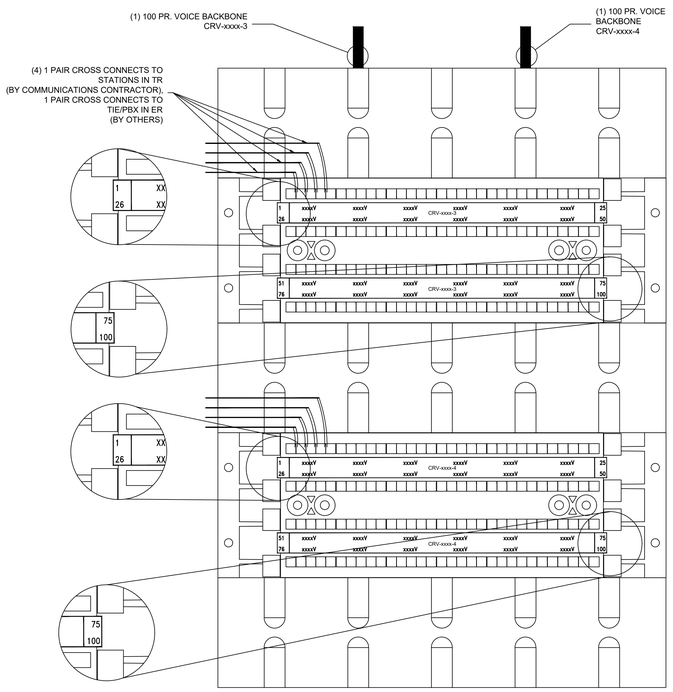
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PROJECT
NAME OF BUILDING – NAME OF CAMPUS
ADDRESS OF BUILDING

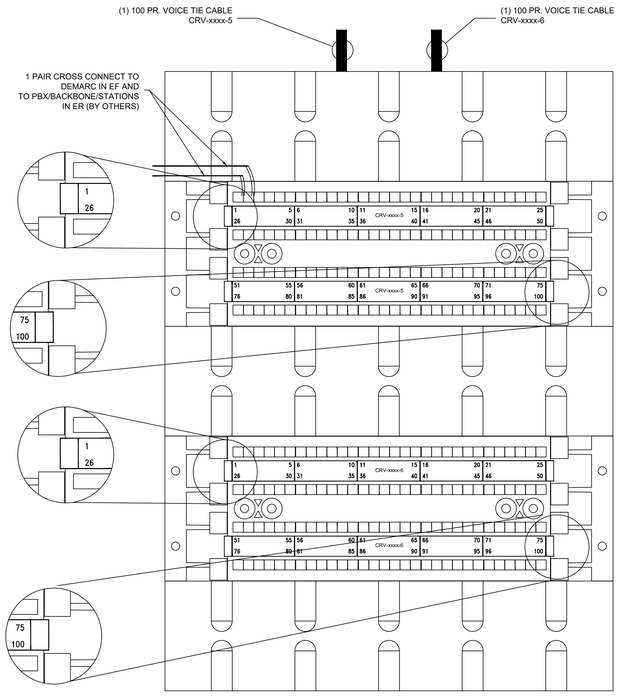
DRAWING TITLE
TYPICAL 110 BLOCK TERMINATIONS
TELECOMMUNICATIONS PLAN

DSR NO.	SCALE	DRAWN BY	FILE NO.	DRAWING NO.
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BLDG. NO. 8500	DATE 4/10/02	CHECKED BY XX		



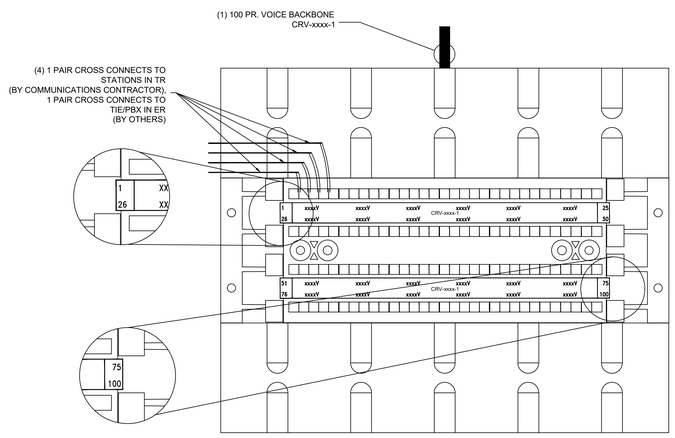
1. 110 BLOCK TO BE LABELED WITH BACKBONE CABLE ID (CRV-xxxx-x), STATION CABLE IDENTIFIERS (xxxV) AND WITH PAIR COUNT OF BACKBONE CABLE (IN GROUPS OF 25: 1, 25, 26, 50, etc.).
2. FOR MULTIPLE BACKBONE CABLES, LABELS ON 110 BLOCKS SHALL MATCH PAIR COUNT OF THE INDIVIDUAL BACKBONE CABLES (START PAIR COUNTS AT 1 FOR EACH CABLE).

1 MULTIPLE VOICE BACKBONE CABLES
T2-6 TYPICAL LABELING SCENARIO



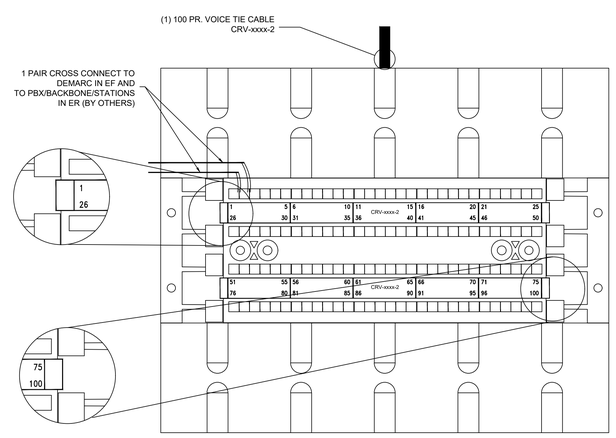
1. 110 BLOCK TO BE LABELED WITH TIE CABLE ID (CRV-xxxx-x) AND WITH PAIR COUNT OF TIE CABLE (IN GROUPS OF 5 PAIRS: 1, 5, 6, 10, 11, 15, 16, etc.).
2. FOR MULTIPLE TIE CABLES, LABELS ON 110 BLOCKS SHALL MATCH PAIR COUNT OF THE INDIVIDUAL TIE CABLES (START PAIR COUNTS AT 1 FOR EACH CABLE).

2 MULTIPLE VOICE TIE CABLES
T2-6 TYPICAL LABELING SCENARIO



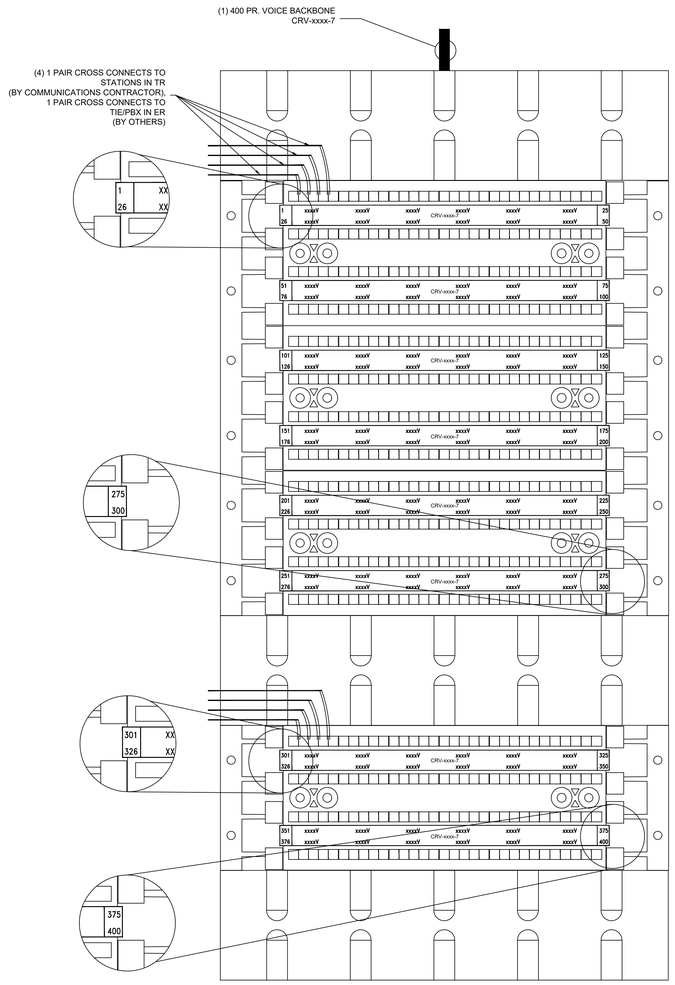
1. 110 BLOCK TO BE LABELED WITH BACKBONE CABLE ID (CRV-xxxx-x), STATION CABLE IDENTIFIERS (xxxV) AND WITH PAIR COUNT OF BACKBONE CABLE (IN GROUPS OF 25: 1, 25, 26, 50, etc.).

3 SINGLE SMALL COUNT VOICE BACKBONE CABLE
T2-6 TYPICAL LABELING SCENARIO



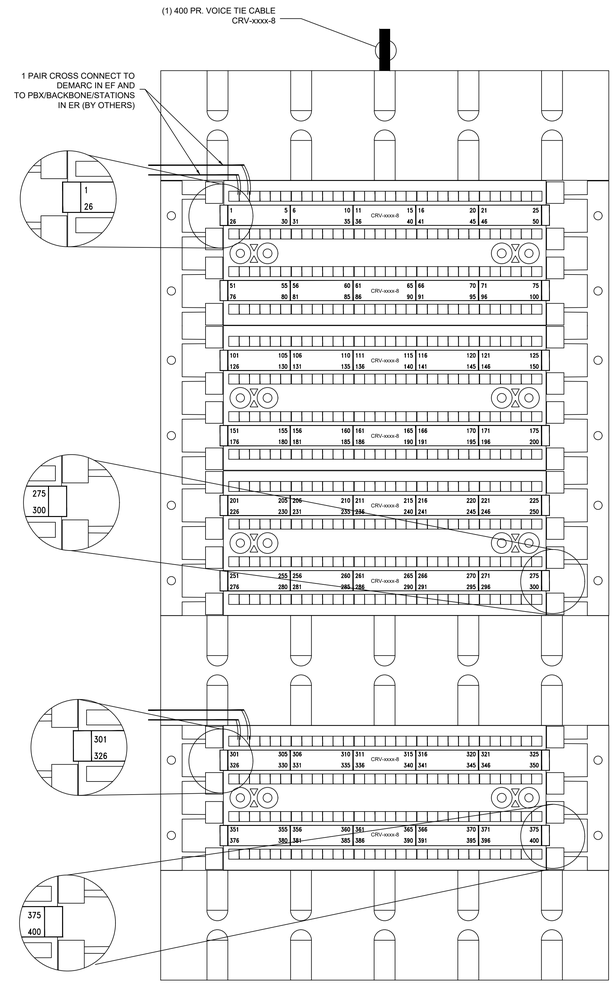
1. 110 BLOCK TO BE LABELED WITH TIE CABLE ID (CRV-xxxx-x) AND WITH PAIR COUNT OF TIE CABLE (IN GROUPS OF 5 PAIRS: 1, 5, 6, 10, 11, 15, 16, etc.).

4 SINGLE SMALL COUNT VOICE TIE CABLE
T2-6 TYPICAL LABELING SCENARIO



1. 110 BLOCK TO BE LABELED WITH BACKBONE CABLE ID (CRV-xxxx-x), STATION CABLE IDENTIFIERS (xxxV) AND WITH PAIR COUNT OF BACKBONE CABLE (IN GROUPS OF: 25: 1, 25, 26, 50, etc.).
2. WHEN A BACKBONE CABLE SPANS MULTIPLE 110 BLOCKS, LABELS ON 110 BLOCKS SHALL MATCH PAIR COUNT OF THE BACKBONE CABLE (PAIR COUNT ON FIRST BLOCK STARTS AT 1, PAIR COUNT ON SUBSEQUENT BLOCKS STARTS WHERE THE PREVIOUS BLOCK LEFT OFF: 1 THRU xxx ON FIRST BLOCK, xxx+1 THRU yyy ON SECOND BLOCK, etc. UP TO FINAL PAIR OF THE BACKBONE CABLE).

5 LARGE COUNT VOICE BACKBONE CABLE
T2-6 TYPICAL LABELING SCENARIO



1. 110 BLOCK TO BE LABELED WITH TIE CABLE ID (CRV-xxxx-x) AND WITH PAIR COUNT OF TIE CABLE (IN GROUPS OF 5 PAIRS: 1, 5, 6, 10, 11, 15, 16, etc.).
2. WHEN A TIE CABLE SPANS MULTIPLE 110 BLOCKS, LABELS ON 110 BLOCKS SHALL MATCH PAIR COUNT OF THE TIE CABLE (PAIR COUNT ON FIRST BLOCK STARTS AT 1, PAIR COUNT ON SUBSEQUENT BLOCKS STARTS WHERE THE PREVIOUS BLOCK LEFT OFF: 1 THRU xxx ON FIRST BLOCK, xxx+1 THRU yyy ON SECOND BLOCK, etc. UP TO FINAL PAIR OF THE TIE CABLE).

6 LARGE COUNT VOICE TIE CABLE
T2-6 TYPICAL LABELING SCENARIO

NOTES:

NOTE TO BIDDERS: AS-BUILTS TO BE HAND DRAWN MARK-UPS OF SHOP DRAWINGS.

CONTRACTORS NAME ADDRESS AND TELEPHONE NUMBER

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1	M.P.	P.R.	4/10/02	RESPONSE TO PATP - 234A

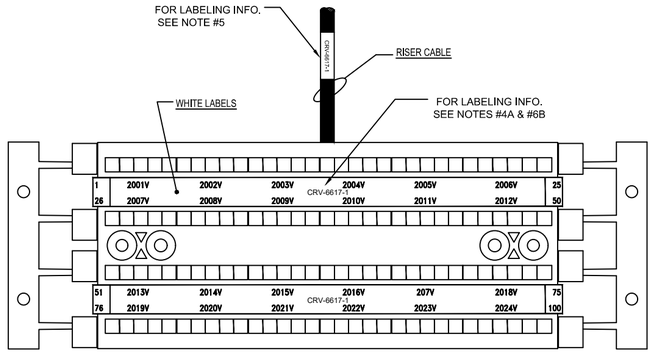
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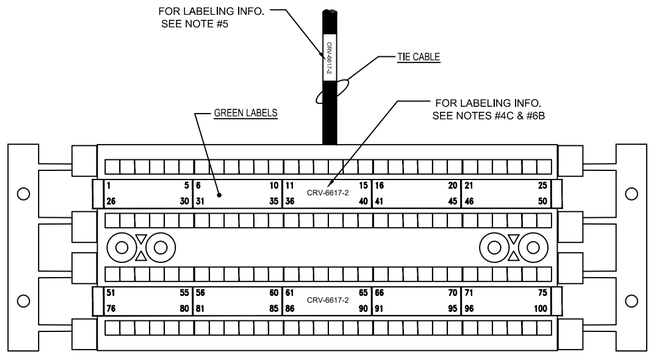
PROJECT
NAME OF BUILDING – NAME OF CAMPUS
ADDRESS OF BUILDING

DRAWING TITLE
TYPICAL CABLE HARDWARE LABELING DETAILS
TELECOMMUNICATIONS PLAN

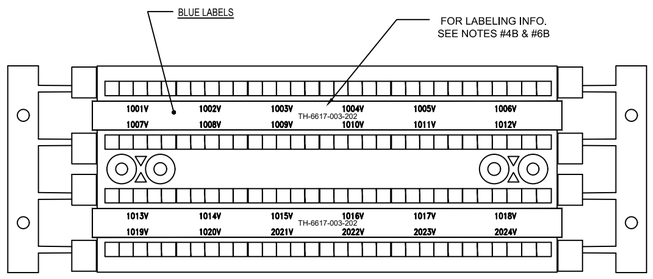
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BLDG. NO. 8500	X/XX"=X"-X"	DH	SD8500-T26	T2-6
	DATE 4/10/02	CHECKED BY XX		



VOICE RISER CABLE
110 DETAIL

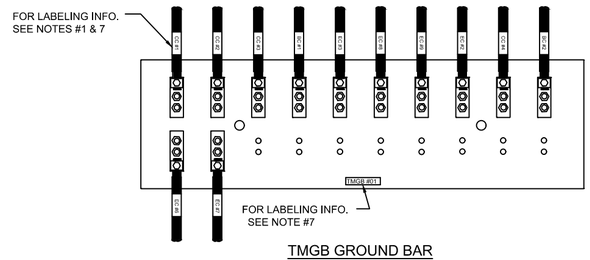


VOICE TIE CABLE
110 DETAIL

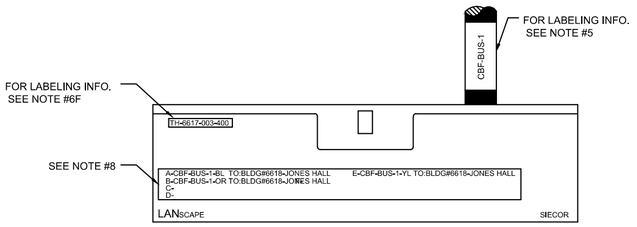


VOICE STATION CABLE
110 DETAIL

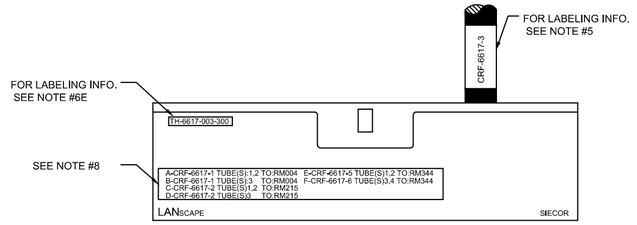
	Equipment/Cable	Label	Font	Size
1.	Patch Panel Port	Panduit CPPLF-5	Aerial Black-Regular	10
2.	Patch Panel	Panduit PLL22P01	Aerial Black-Regular	24
3.	Tie 110 Block	Panduit DS110-GR	Aerial-Regular	8
4.	Riser 110 Block	Panduit DS110	Aerial-Regular	8
5.	Station 110 Block	Panduit DS110-BU	Aerial-Regular	8
6.	WP Station Single Gang	P-Touch Tape TZ-231 9MM Black on White	Bold	13
7.	WP Station Double Gang	P-Touch Tape TZ-231 9MM Black on White	Bold	13
8.	WP Station Single Gang	Panduit JLEFPS-1	Aerial Black-Regular	12
9.	WP Station Double Gang	Panduit JLEFPD-1	Aerial Black-Regular	11
10.	Rack Panduit	Panduit PLL22P01	Aerial Black-Regular	24
11.	ISP Fiber Cabinet TH	Panduit PLL22P01	Aerial Black-Regular	24
12.	ISP Fiber terminations	P-Touch Tape TZ-SE4 18MM Black on White	Bold	10
13.	OSP Fiber Cabinet TH	Panduit PLL22P01	Aerial Black-Regular	24
14.	OSP Fiber terminations	P-Touch Tape TZ-SE4 18MM Black on White	Bold	10
15.	Ground Bar ID	Panduit PLL22P01	Aerial Black-Regular	24
16.	Ground Cable	Panduit PLL-11-Y3-2.5	Aerial Black-Regular	12
17.	Data Station Cable	Panduit PLL-11-Y3-2.5	Aerial Black-Regular	12
18.	Voice Station Cable	Panduit PLL-11-Y3-2.5	Aerial Black-Regular	12
19.	Coax Station Cable	Panduit PLL-12-Y3-2.5	Aerial Black-Regular	12
20.	Voice Riser Cable	SLCT-WH sleeve with P-Touch Label	Bold	13
21.	Coax Riser Cable	Panduit SLCT-WH sleeve with P-Touch Label	Bold	13
22.	Fiber Optic Riser Cable	Panduit SLCT-WH sleeve with P-Touch Label	Bold	13



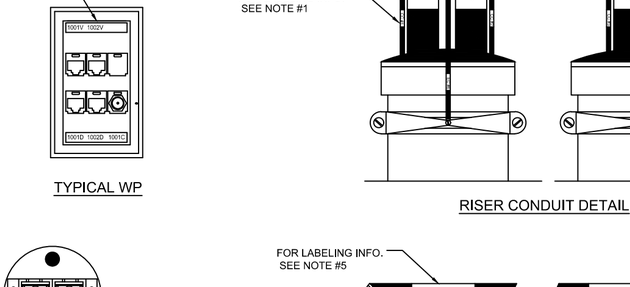
TMGB GROUND BAR



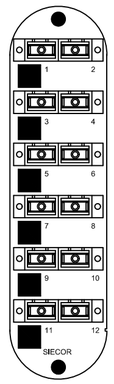
OUTSIDE PLANT FIBER CABINET

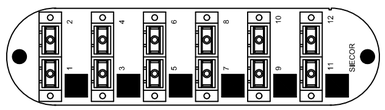


INSIDE PLANT FIBER CABINET

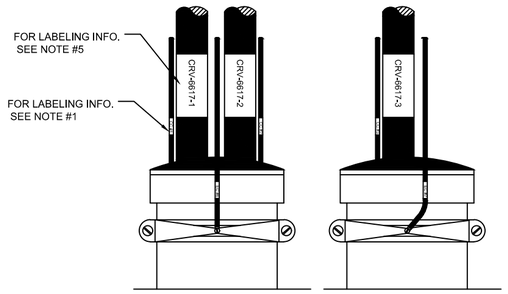


TYPICAL WP

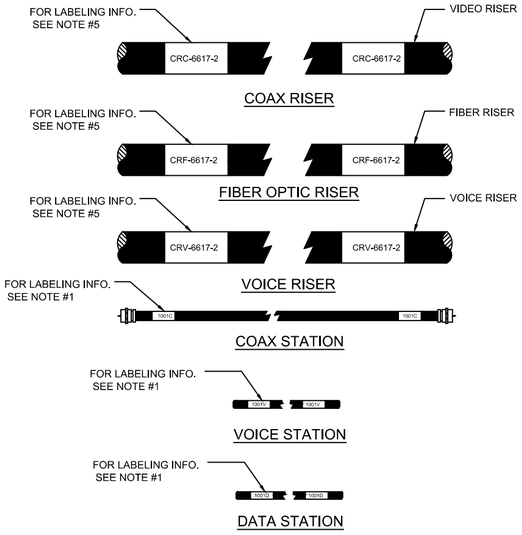


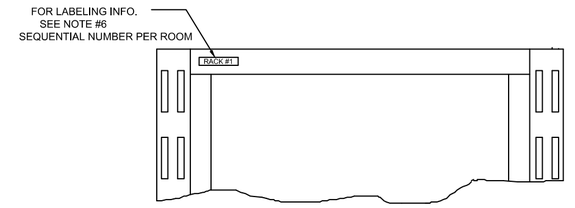


CLOSET CONNECTOR HOUSING PANELS

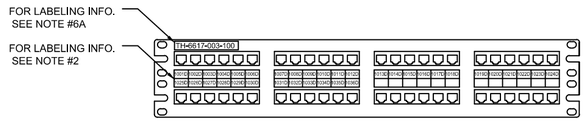


RISER CONDUIT DETAIL

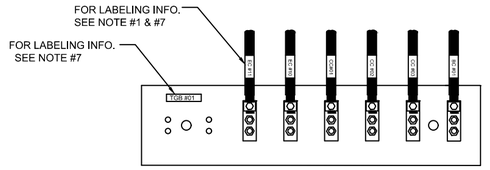




EQUIPMENT RACK



DATA PATCH PANEL



TGB GROUND BAR

NOTES:

1. EACH VOICE, DATA, COAX CABLE AND GROUND WIRE THAT IS PULLED WILL BE LABELED ON BOTH ENDS OF THE CABLE WITH A COMPUTER GENERATED WRAP AROUND CABLE PANDUIT LABEL #PLL-11-Y3-2.5

2. EACH DATA PATCH PANEL PORT WILL BE LABELED IN THE CENTER OF THE PATCH PANEL PANDUIT TYPE CPPLF-5 WHITE.

3. EACH FACEPLATE WILL BE LABELED WITH THE FOLLOWING LABEL:
   PANDUIT TYPE JLEFPS-1 FOR A SINGLE GANG PLATE
   PANDUIT TYPE JLEFPD-1 FOR A DOUBLE GANG PLATE
   A) FACEPLATE LABELS CAN BE SHOWN AS FOLLOWS.
      P-TOUCH PT-330 OR EQUAL
      P-TOUCH TAPES TZ 9MM BLACK ON WHITE.
      FONT SIZE IS 13 BOLD.

4. EACH 110 PUNCH DOWN BLOCK.
   A.)PANDUIT DS110 FOR THE RISER CABLE. (WHITE)
   B.)PANDUIT DS110-BU FOR THE HORIZONTAL CABLES. (BLUE)
   C.)PANDUIT DS110-GR FOR THE TIE CABLE. (GREEN)

5. TIE, RISER AND BACKBONE CABLE SLEEVE.
   PANDUIT SLCT-WH WHITE SELF-LAMINATING CABLE MARKER HOLDER
   PANDUIT RECOMMENDED LABELS FOR THE MARKER HOLDER
   (PLL-17-Y2-1) AND (JL41PO-0.2)
   THE VOICE CABLE (RISER & TIE) ID WILL BE CRV-BUILDING NUMBER-SEQUENTIAL NUMBER.
   THE VOICE CABLE (BACKBONE) ID WILL BE CBV-CAMPUS NAME-SEQUNETIAL NUMBER.
   THE FIBER CABLE (RISER) ID WILL BE CRF-BUILDING NUMBER-SEQUENTIAL NUMBER.
   THE FIBER CABLE (BACKBONE) ID WILL BE CBF-CAMPUS NAME-SEQUNETIAL NUMBER.
   THE COAX CABLE (RISER) ID WILL BE CRC-BUILDING NUMBER-SEQUENTIAL NUMBER.
   THE COAX CABLE (BACKBONE) ID WILL BE CBC-CAMPUS NAME-SEQUNETIAL NUMBER.

6. TERMINATION HARDWARE NUMBERS AND RACK NUMBER.
   PANDUIT PLL22P01 WHITE.
   THE TERMINATION HARDWARE NUMBERS WILL BE DETERMINED
   THE FOLLOWING WAYS:
   A.) DATA PATCH PANEL TH-BUILDING#-ER/TR ROOM#-100 THROUGH 199.
   B.) VOICE PUNCHDOWN BLOCK (RISER) CRV-BUILDING#-SEQUENTIAL# (TO MATCH RISER CABLE NUMBER).
   C.) VOICE PUNCHDOWN BLOCK (TIE) CRV-BUILDING#-SEQUENTIAL# (TO MATCH TIE CABLE NUMBER).
   D.) VOICE PUNCHDOWN BLOCK (STATION) TH-BUILDING#-ER/TR ROOM#-200 THROUGH 299.
   E.) ISP FIBER TH-BUILDING#-ER/TR ROOM#-300 THROUGH 399.
   F.) OSP FIBER TH-BUILDING#-ER/TR ROOM#-400 THROUGH 499.
   G.) VIDEO TH-BUILDING#-ER/TR ROOM#-500 THROUGH 599.

7. GROUNDING LABELING:
   1. GROUND BARS WILL BE LABELED WITH A PANDUIT PLL22P01.
      TMGB OR TGB-THE # OF THE GND BAR AND THE BUILDING.
   2. GROUND CONDUCTORS WILL BE LABELED WITH A PANDUIT PLL-11-Y3-2.5
      A.) BONDING CONDUCTORS (FROM TMGB TO THE TGB'S OR SERVICE GND AND BLDG STEEL) WILL BE BC AND A SEQUENTIAL NUMBER THROUGHOUT THE ROOM.
      B.) EQUIPMENT BONDING CONDUCTOR (LADDER RACKS FREESTANDING, SWING GATE, AND CONDUITS) EC AND A SEQUENTIAL NUMBER THROUGHOUT THE ROOM.
      C.) CABLE BONDING CONDUCTOR (THE BONDING CONDUCTOR THAT IS RUN WITH THE BACKBONE CABLE) CC AND THE NUMBER OF THE TIE OR RISER CABLE IT IS ASSOCIATED WITH.

8. FIBER LABELING:
   THE LABEL WILL BE A P-TOUCH TZ-3/4" TAPE FROM THE P-TOUCH PT330 MACHINE.
   1.) INSIDE PLANT LABEL WILL BE AS FOLLOWS:
      CONNECTOR PANEL LOCATION IN HOUSING-CRF-BUILDING#-SEQUENTIAL#-TUBE#(s) TO:RM# IT IS FEEDING.
   2.) OUTSIDE PLANT LABEL AS FOLLOWS:
      CONNECTOR PANEL LOCATION IN HOUSING-CBF-CAMPUS NAME-CABLE#-TUBE COLOR TO:BLDG# - BLDG. NAME IT IS FEEDING.

NOTES:

NOTE TO BIDDERS: AS-BUILTS TO BE HAND DRAWN MARK-UPS OF SHOP DRAWINGS.

CONTRACTORS NAME ADDRESS AND TELEPHONE NUMBER

ISSUE	BY	CHK	DATE	DESCRIPTION
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1	M.P.	P.R.	4/10/02	RESPONSE TO PATP - 234A

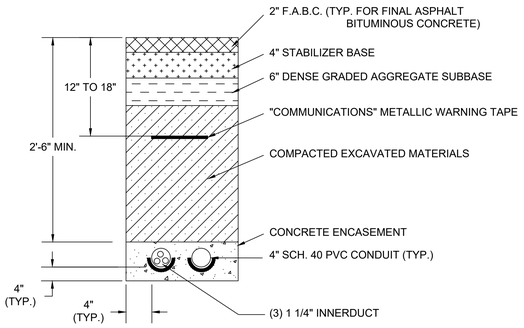
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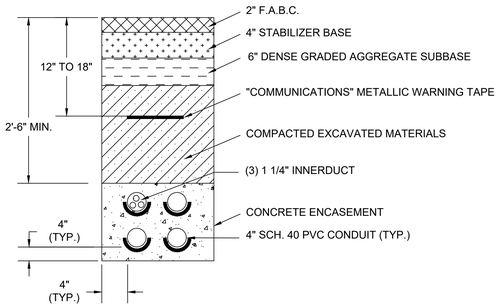
THE STATE UNIVERSITY OF NEW JERSEY
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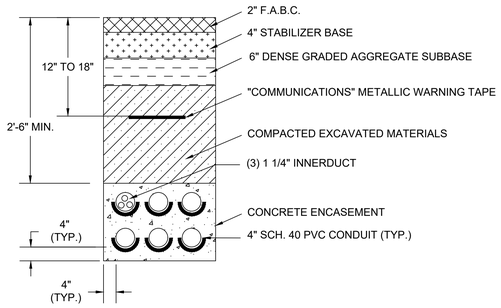
PROJECT
NAME OF BUILDING – NAME OF CAMPUS
ADDRESS OF BUILDING

DRAWING TITLE
LABELING DETAILS
TELECOMMUNICATIONS PLAN

JOB NO.	SCALE	DRAWN BY	FILE NO.	DRAWING NO.
BLDG. NO. 8500	AS-BUILT	XX	SD8500-T27	T2-7
	DATE 4/10/02	CHECKED BY XX		

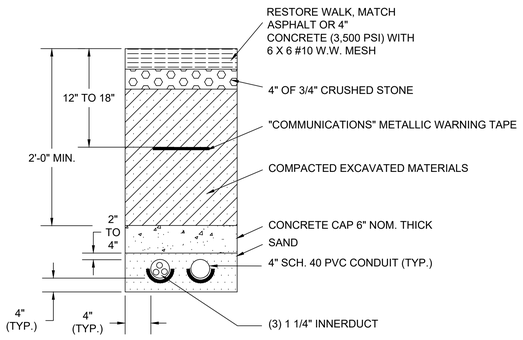


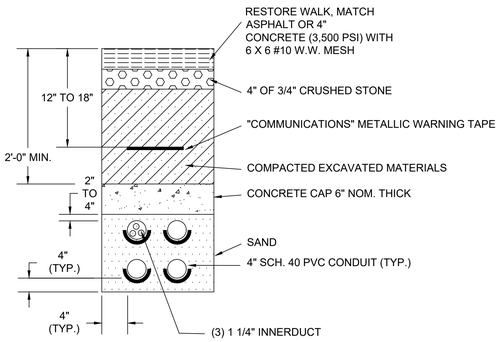


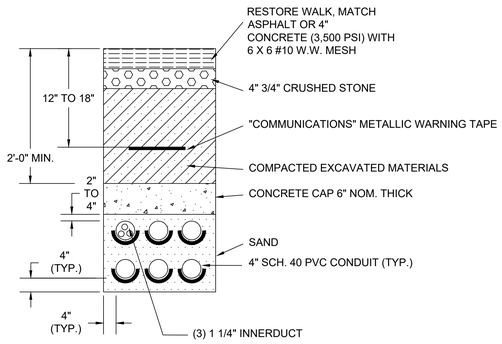


TYPICAL ASPHALT DETAILS (1 HOUR SETUP)
NOTES:

1. ALL ELEVATIONS/DEPTHS ARE FROM O.G.
2. CONDUIT INSTALLATION PROCEDURE:
   A. SAWCUT AND REMOVE EXISTING ASPHALT.
   B. EXCAVATE TO REQUIRED DEPTH.
   C. PLACE CONDUIT(S).
   D. ENCASE CONDUIT(S) WITH CONCRETE.
   E. INSTALL COMPACTED LIFTS OF SUITABLE BACKFILL AFTER 'D'.
   F. PLACE WARNING TAPE AT PROPER DEPTH.
   G. CONTINUE 'E', RESTORE ASPHALT WITH SUB-BASE.
   H. INSTALL (3) 1 1/4" FIBER OPTIC CABLE INNERDUCT.

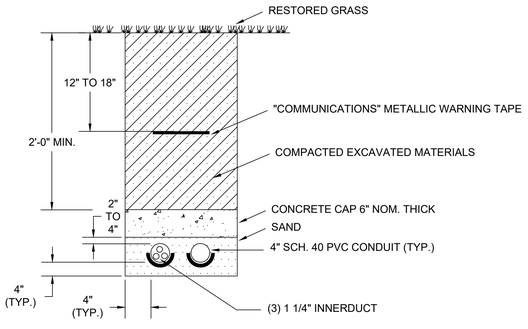


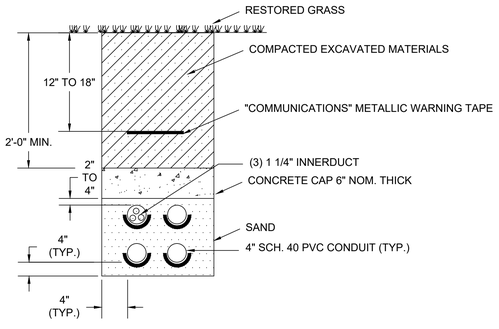


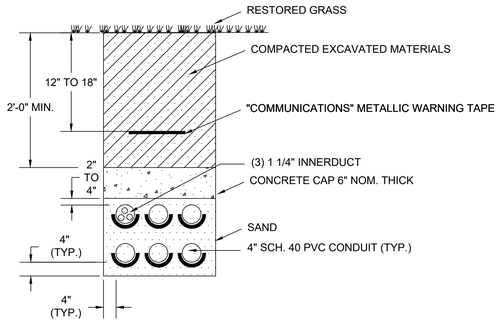


TYPICAL WALKWAY DETAILS (1 HOUR SETUP)
NOTES:

1. ALL ELEVATIONS/DEPTHS ARE FROM O.G.
2. CONDUIT INSTALLATION PROCEDURE:
   A. REMOVE SIDEWALK/ASPHALT.
   B. EXCAVATE TO REQUIRED DEPTH.
   C. PLACE CONDUIT(S) WITH SAND BED AND COVER.
   D. CAP OVER CONDUIT(S) WITH CONCRETE.
   E. INSTALL COMPACTED LIFTS OF SUITABLE BACKFILL AFTER 'D'.
   F. PLACE WARNING TAPE AT PROPER DEPTH.
   G. CONTINUE 'E', RESTORE ASPHALT OR CONCRETE WALK.
   H. INSTALL (3) 1 1/4" FIBER OPTIC CABLE INNERDUCT.







TYPICAL GRASS DETAILS (1 HOUR SETUP)
NOTES:

1. ALL ELEVATIONS/DEPTHS ARE FROM O.G.
2. CONDUIT INSTALLATION PROCEDURE:
   A. REMOVE GRASS AND TOPSOIL.
   B. EXCAVATE TO REQUIRED DEPTH.
   C. PLACE CONDUIT(S) WITH SAND BED AND COVER.
   D. CAP OVER CONDUIT(S) WITH CONCRETE.
   E. INSTALL COMPACTED LIFTS OF SUITABLE BACKFILL AFTER 'D'.
   F. PLACE WARNING TAPE AT PROPER DEPTH.
   G. CONTINUE 'E', RESTORE GRASS TURF.
   H. INSTALL (3) 1 1/4" FIBER OPTIC CABLE INNERDUCT.

2-4" CONDUIT TRENCH

4-4" CONDUIT TRENCH

6-4" CONDUIT TRENCH

1-4" CONDUIT TRENCH

SAME AS ABOVE DETAILS
LESS ONE (1) 4" PVC CONDUIT.

3-4" CONDUIT TRENCH

SAME AS ABOVE DETAILS
LESS ONE (1) 4" PVC CONDUIT.

5-4" CONDUIT TRENCH

SAME AS ABOVE DETAILS
LESS ONE (1) 4" PVC CONDUIT.

NOTE:

1. **PLEASE REPORT ANY FLOOR PLAN CHANGES TO THE OFFICE OF DOCUMENT SERVICES - LIVINGSTON CAMPUS TELEPHONE 1-848-445-2451.**
2. **DO NOT SCALE DRAWING FOR SIZES OTHER THAN ('E1') (30X42).**

NOTES:

ISSUE	BY	CHK	DATE	DESCRIPTION
1	ADM	DH	11/19/19	UPDATED PER LATEST STANDARDS

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RUTGERS
OFFICE OF INFORMATION TECHNOLOGY

PROJECT
NAME OF BUILDING – NAME OF CAMPUS
ADDRESS OF BUILDING

DRAWING TITLE
DUCTBANK
TRENCH DETAILS

DSR NO.	SCALE	DRAWN BY	FILE NO.	DRAWING NO.
	3/16"=1'-0"	ADM	DUCTBANK	DB–1
BLDG. NO. TRENCH DETAILS	DATE 3/6/03	CHECKED BY DH		